Yandex Research

University of Amsterdam

Edinburgh University of Edinburgh Natural Language Processing NLP

Word Embeddings

SEPTEMBER 12, 2018

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https://lena-voita.github.io

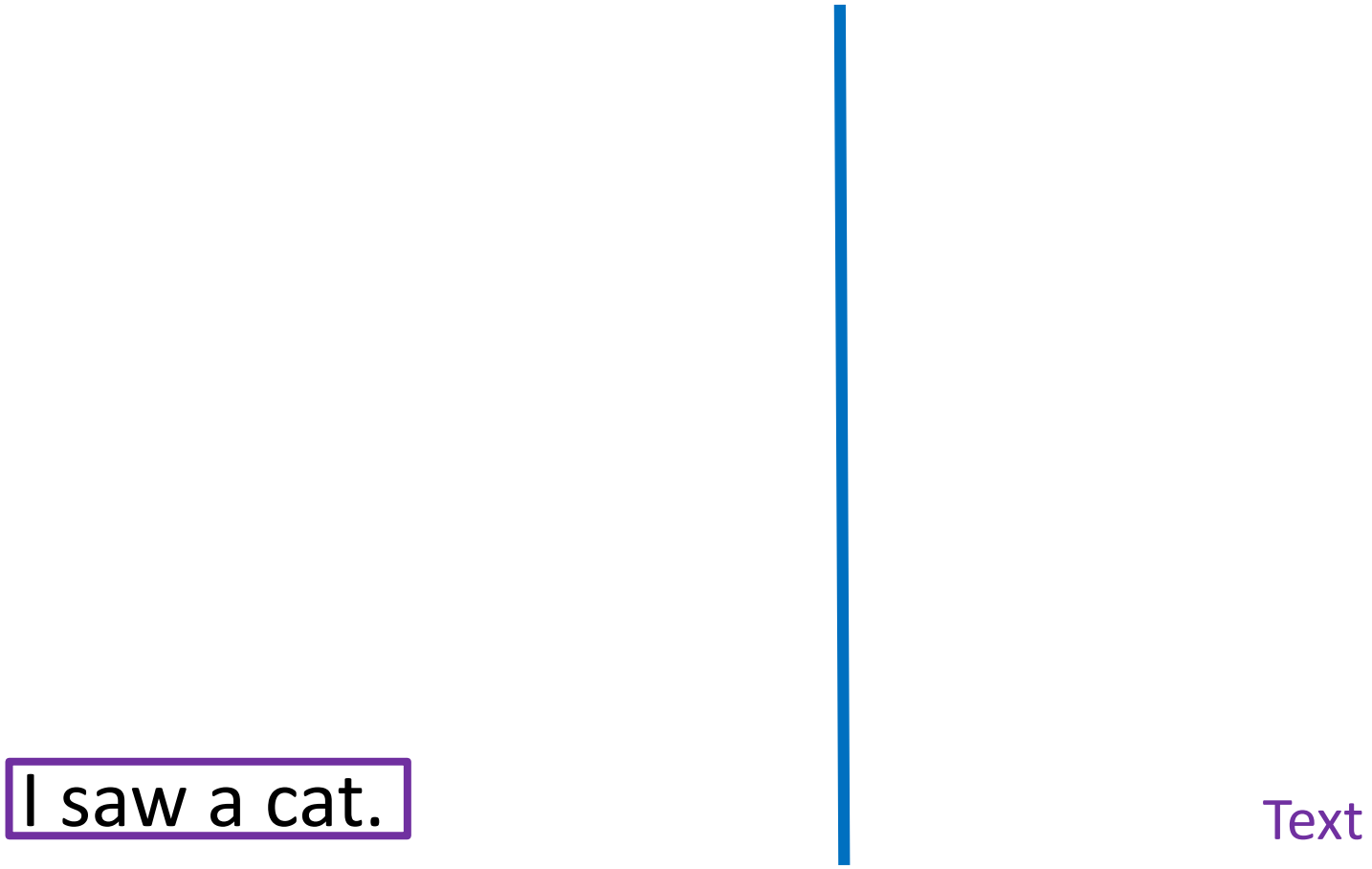

@lena_voita


lena-voita

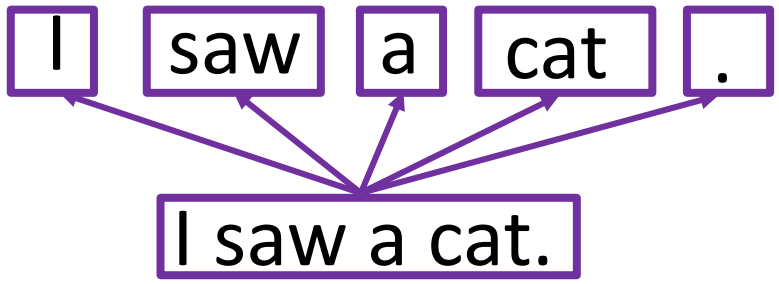
Plan

- Why do we need word representations?
- Distributional semantics
- Word2Vec in detail
- Glove overview
- Let’s take a walk!
- Further directions and recent advances
- Hack of the day!

Why do we need word representation?



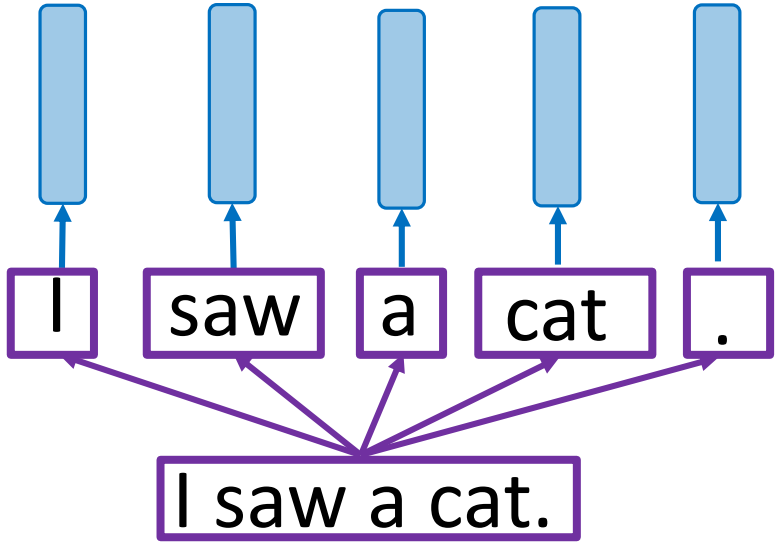
Why do we need word representation?



Sequence of tokens

Text

Why do we need word representation?

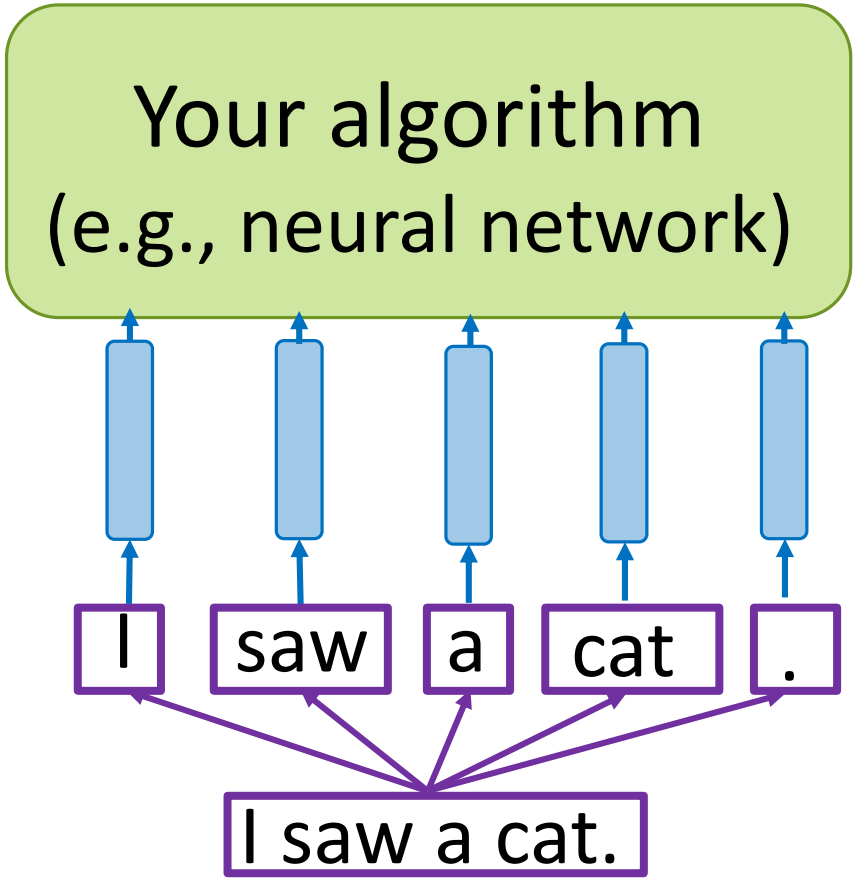


Word representation - vector (word embedding)

Sequence of tokens

Text

Why do we need word representation?



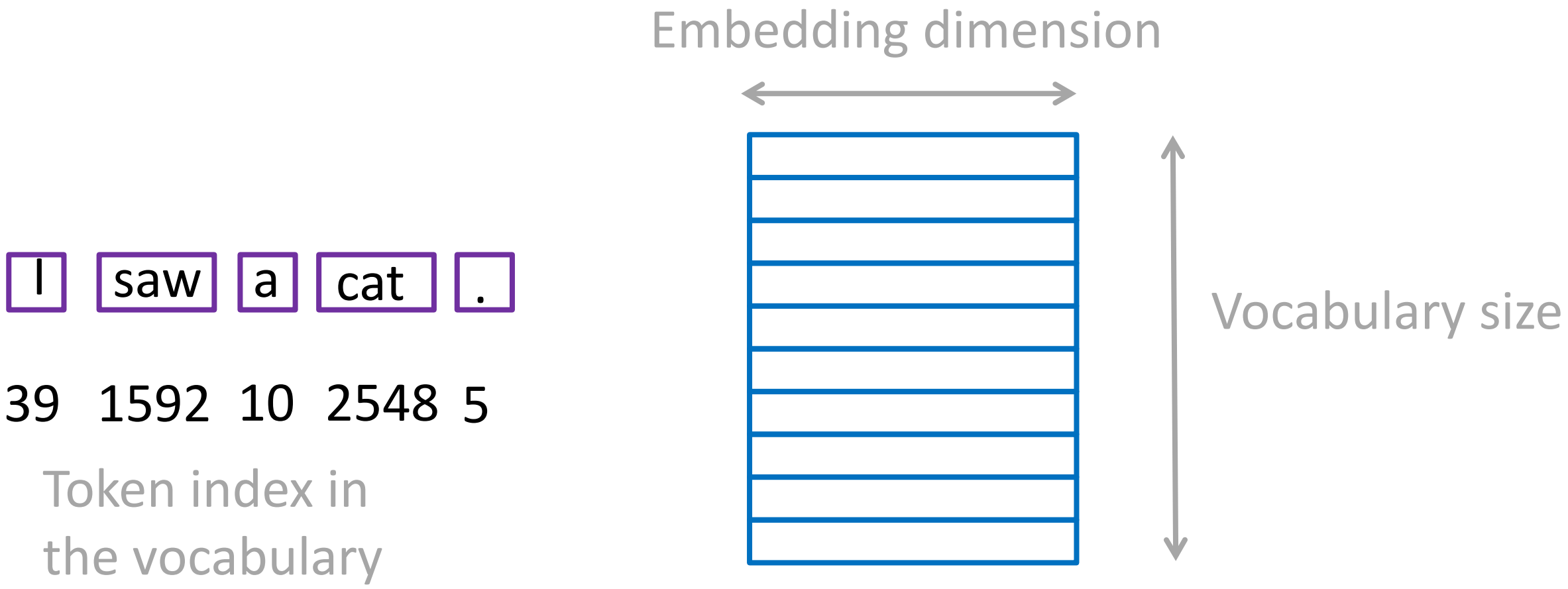
Any algorithm for solving any task

Word representation - vector
(word embedding)

Sequence of tokens

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How it works: Look-up Table (Vocabulary)

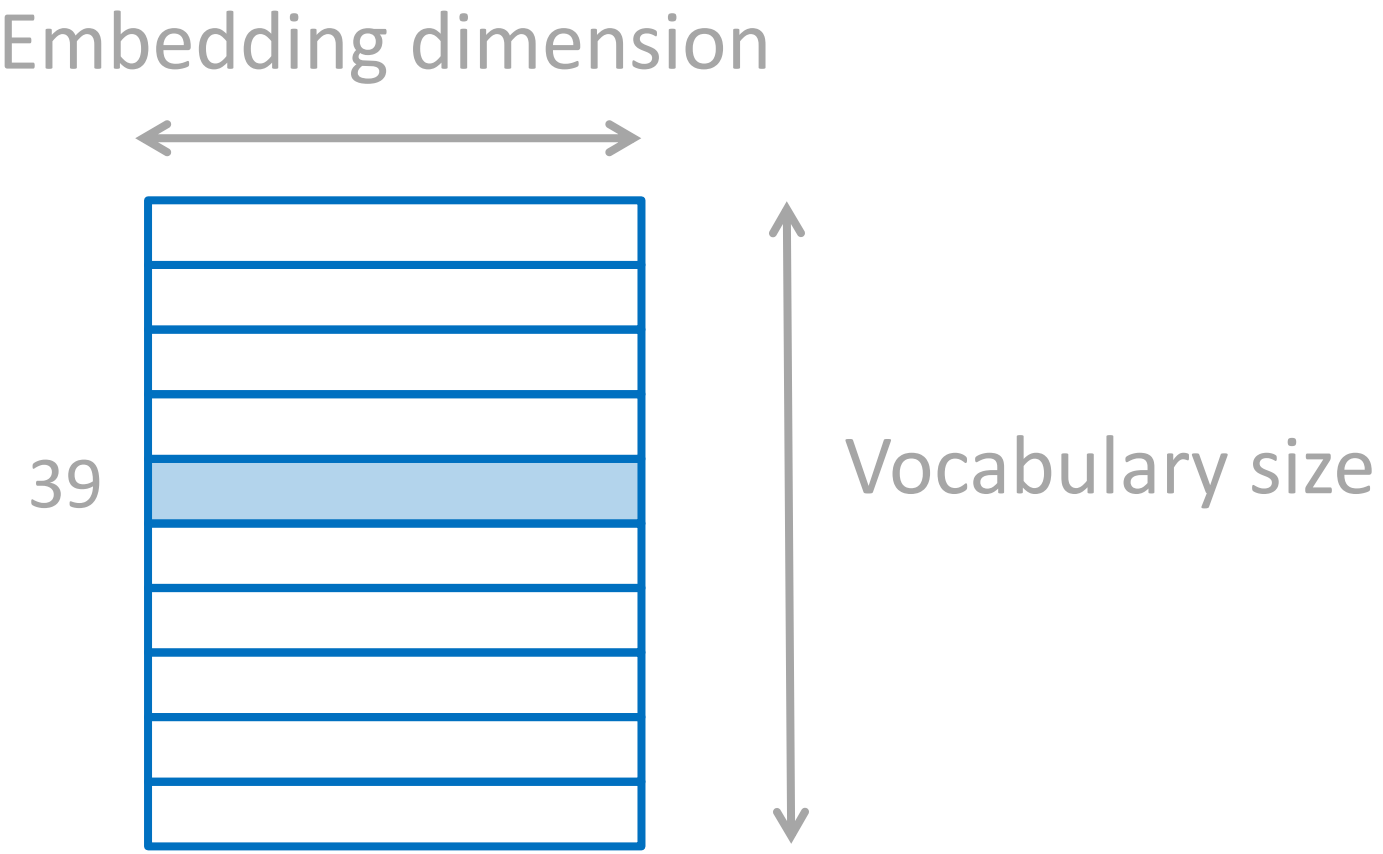


How it works: Look-up Table (Vocabulary)

I saw a cat .

39 1592 10 2548 5

Token index in the vocabulary

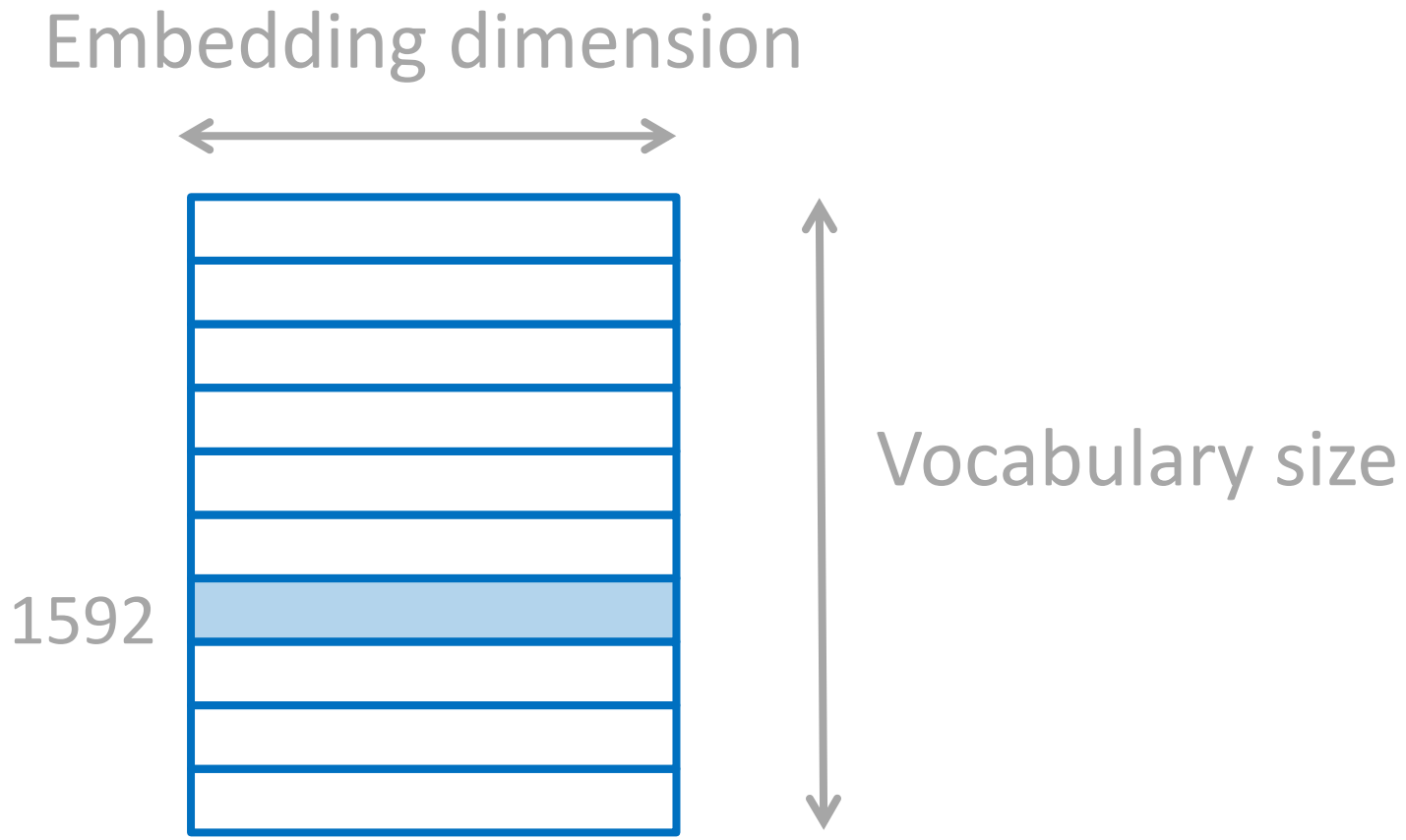


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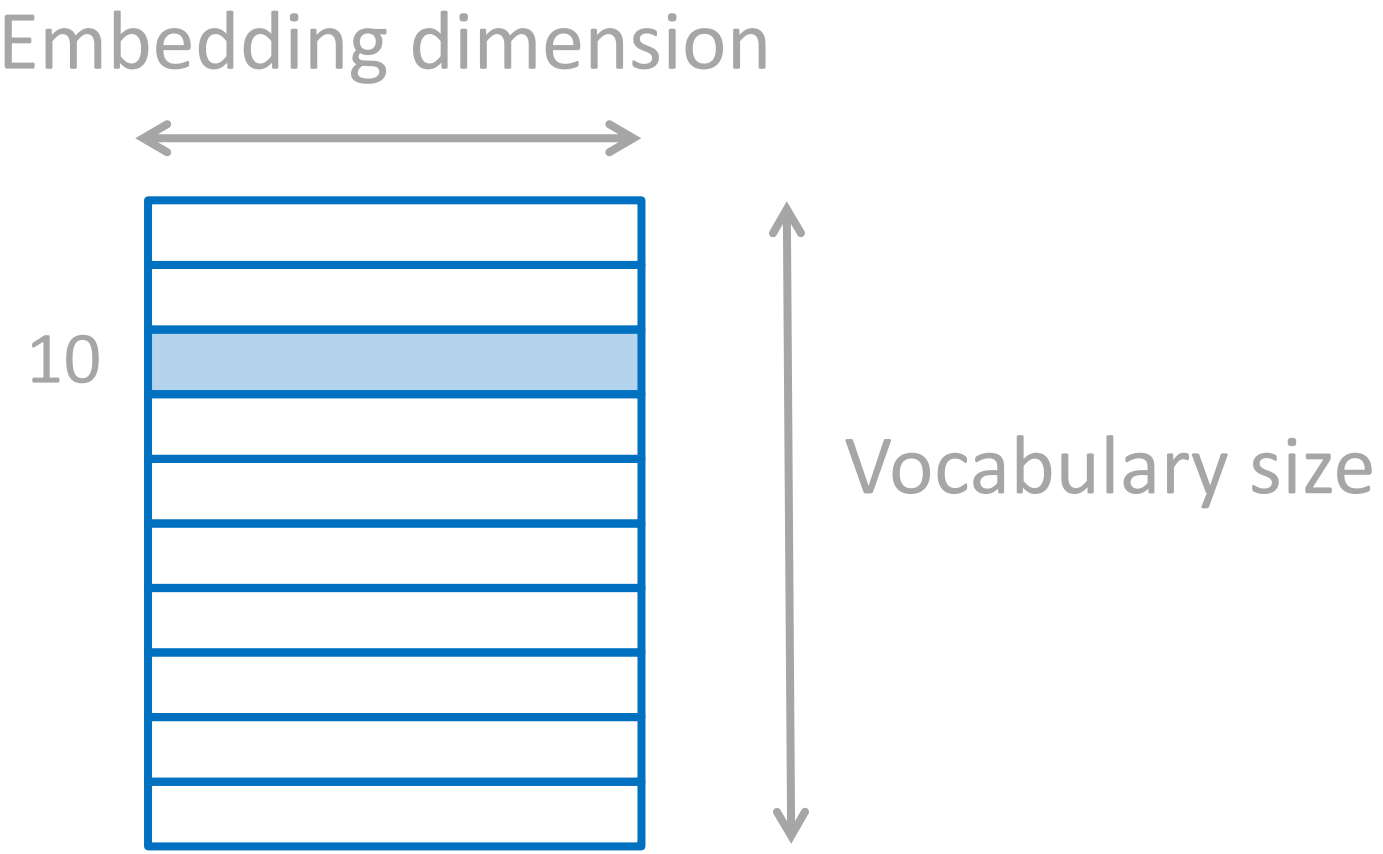


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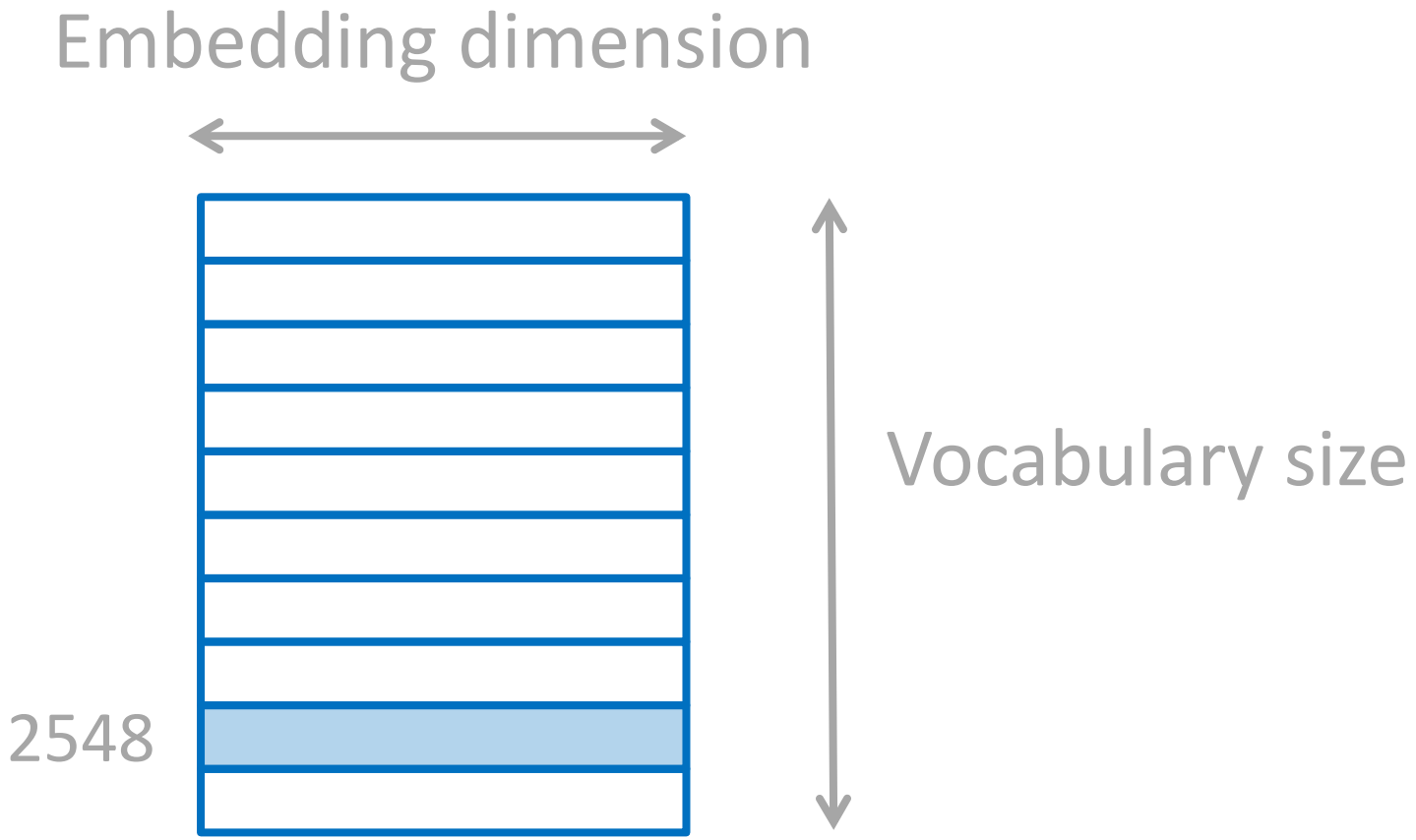


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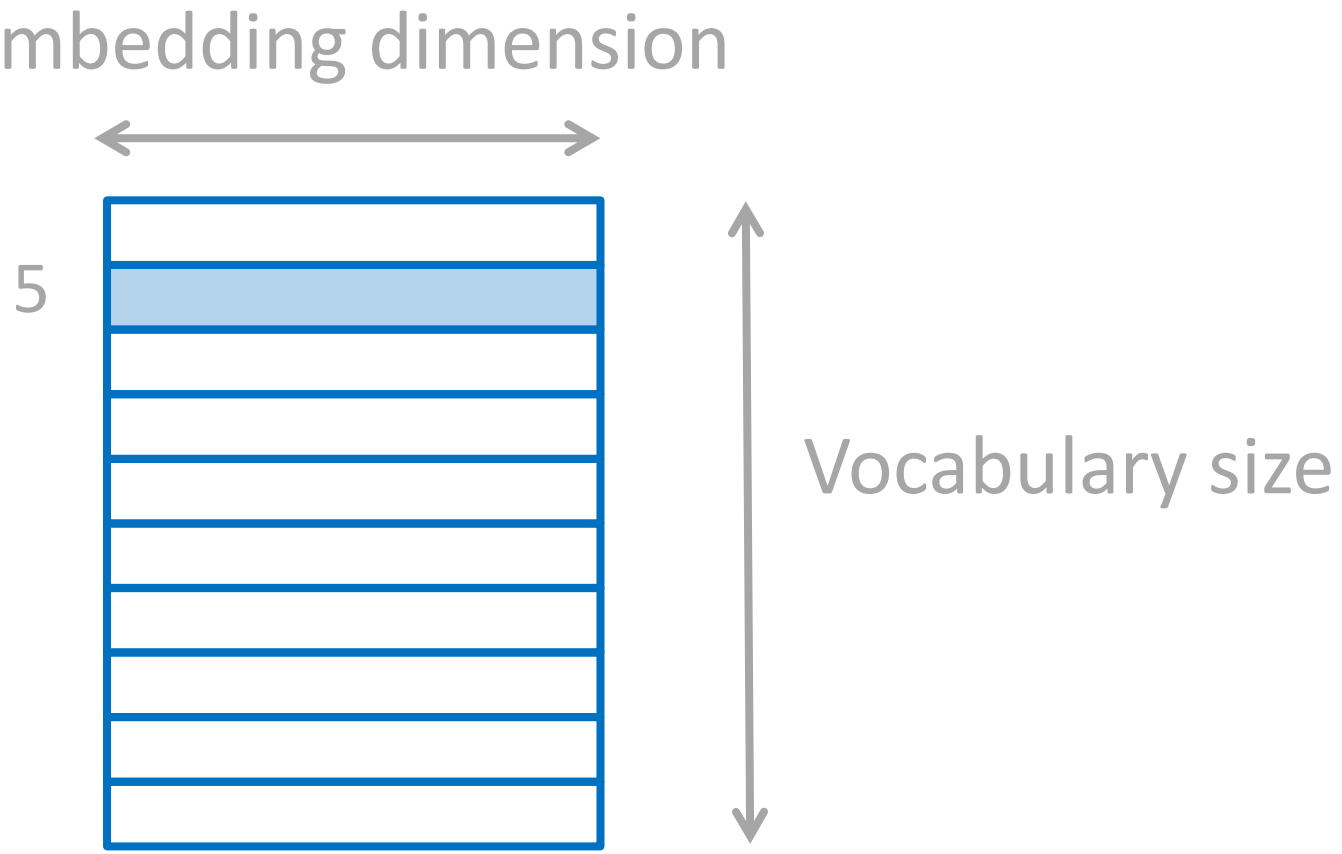


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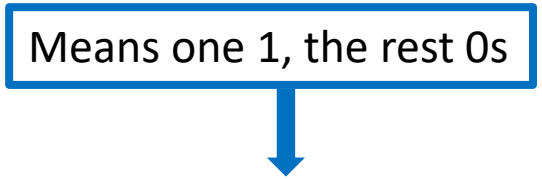
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Token index in the vocabulary



Representing words as discrete symbols

In traditional NLP, we regard words as discrete symbols: **hotel, conference, motel**



Words can be represented by **one-hot** vectors:

motel = [0 0 0 0 0 0 0 0 0 0 1 0 0 0 0]
hotel = [0 0 0 0 0 0 0 1 0 0 0 0 0 0 0]

Vector dimension = number of words in vocabulary (e.g. 500,000)

Problem with words as discrete symbols

Example: in web search, if user searches for "**Seattle motel**", we would like to match documents containing "**Seattle hotel**".

But:

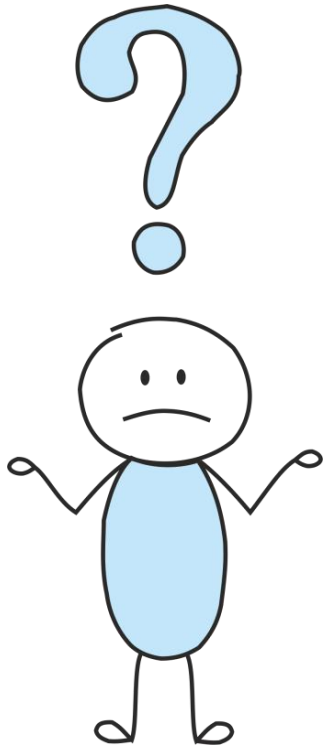
motel = [0 0 0 0 0 0 0 0 0 0 1 0 0 0 0]
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These two vectors are orthogonal.

There is no natural notion of **similarity** for one-hot vectors!

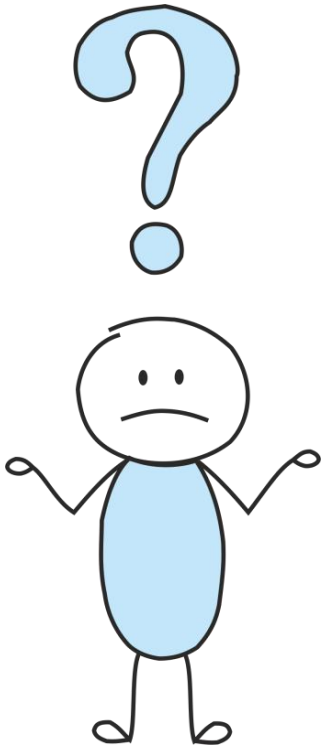
These vectors do not contain information about a **meaning** of a word.

But what is meaning?



But what is meaning?

What is bardiwac?



But what is meaning?

What is bardiwac?

- He handed her a glass of bardiwac.
- Beef dishes are made to complement the bardiwac.
- Nigel staggered to his feet, face flushed from too much bardiwac.
- Malbec, one of the lesser-known bardiwac grapes, responds well to Australia's sunshine.
- I dined off bread and cheese and this excellent bardiwac.
- The drinks were delicious: blood-red bardiwac as well as light, sweet Rhenish.

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Bardiwac is a red alcoholic beverage made from grapes

Distributional semantics

- A bottle of _bardiwac_ is on the table.
- Everybody likes _bardiwac_.
- Don’t have _bardiwac_ before you drive.
- We make _bardiwac_ out of corn.

Distributional semantics

- A bottle of _____ is on the table.
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What other words fit into these contexts?

Distributional semantics

- A bottle of _____ is on the table. (1)
- Everybody likes _____. (2)
- Don't have _____ before you drive. (3)
- We make _____ out of corn. (4)

What other words fit into these contexts?

	(1)	(2)	(3)	(4)	…
bardiwac	1	1	1	1	
loud	0	0	0	0	
motor oil	1	0	0	1	
tortillas	0	1	0	1	
wine	1	1	1	0	
choices	0	1	0	0	

Distributional semantics

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Distributional semantics

Does vector similarity imply semantic similarity?

Distributional semantics

Does vector similarity imply semantic similarity?



The **distributional hypothesis**, stated by Firth (1957):

"You shall know a word by the company it keeps."

Idea: co-occurrence counts

Corpus sentences

He also found five fish swimming in murky water in an old **bathtub**.
We do abhor dust and dirt, and stains on the **bathtub**, and any kind of filth.
Above At the far end of the garden room a **bathtub** has been planted with herbs for the winter.
They had been drinking Cisco, a fruity, wine-based fluid that smells and tastes like a mixture of cough syrup and **bathtub** gin.
Science finds that a surface tension on the water can draw the boats together, like toy boats in a **bathtub**.
In fact, the godfather of gloom comes up with a plot that takes in Windsor Davies (the ghost of sitcoms past), a **bathtub** and a big box of concentrated jelly.
'I'll tell him,' said the Dean from the bathroom above the sound of bathwater falling from a great height into the ample Edwardian **bathtub**.

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Co-occurrence counts

the	12
a	9
of	7
and	6
in	5
…	…
like	2
water	2
boat	2
from	2
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vector

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Dimensionality reduction

small vector

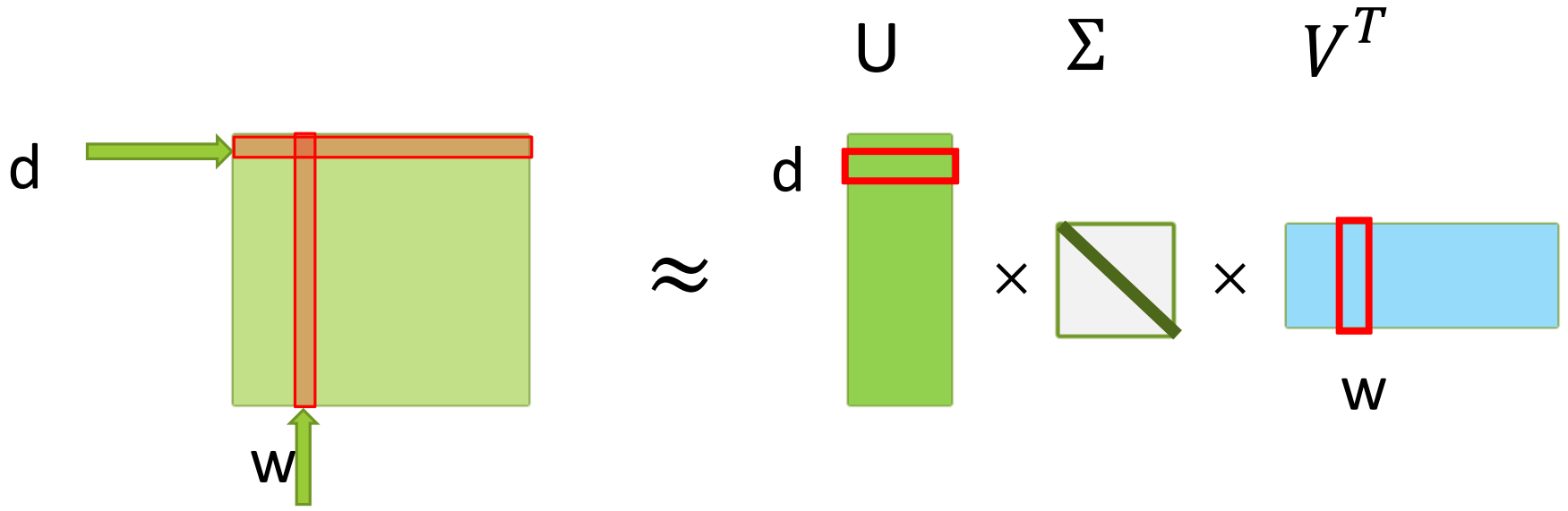
$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}$$

Latent semantic analysis (LSA)

X - document-term co-occurrence matrix

$$X \approx \hat{X} = U \Sigma V^T$$

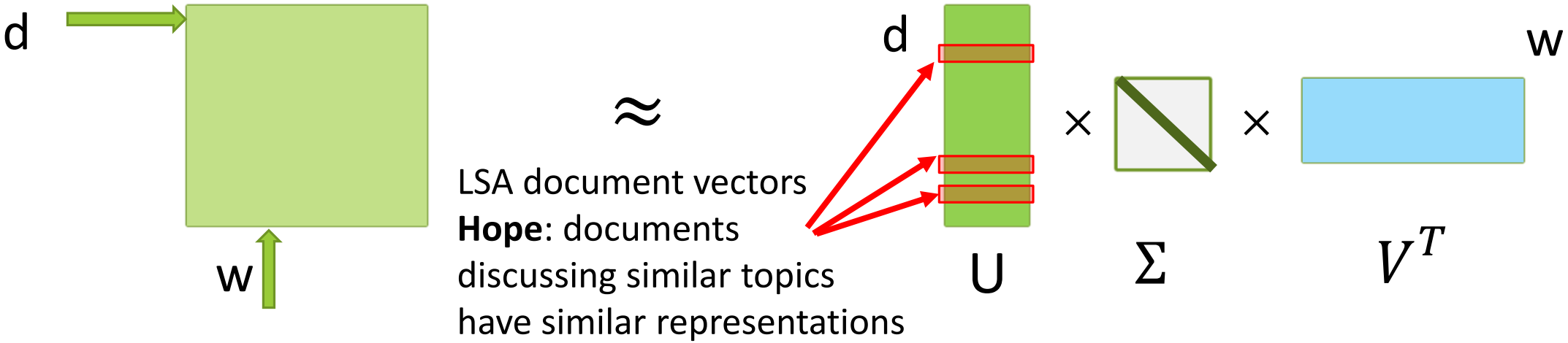
- co-occurrence counts
- tf-idf
- filter stop-words
- lemmatize
- …



Latent semantic analysis (LSA)

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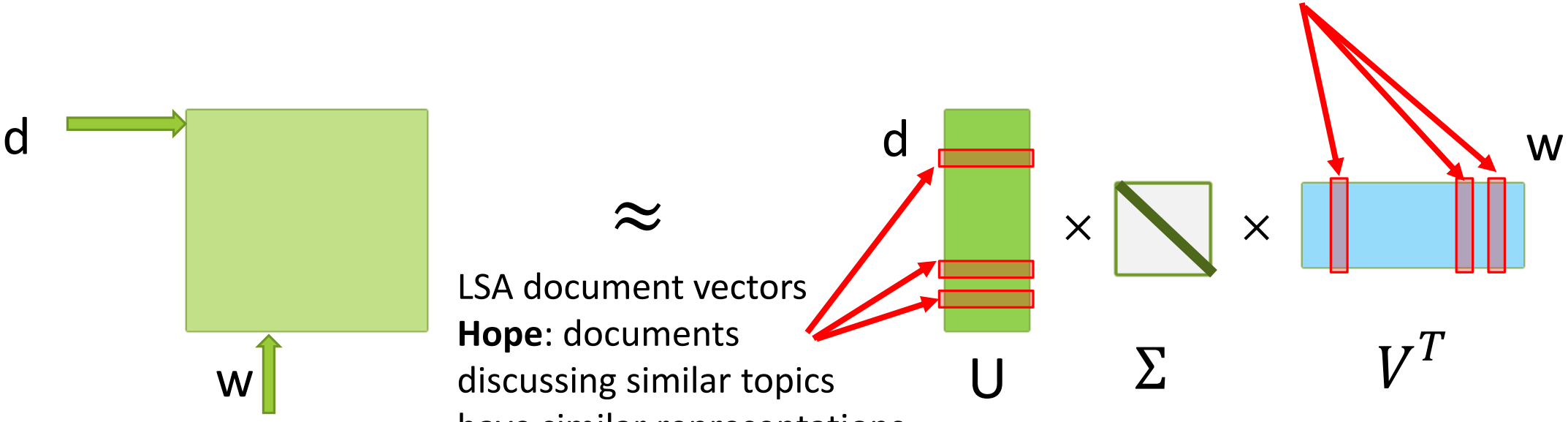


Latent semantic analysis (LSA)

X - document-term co-occurrence matrix

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LSA term vectors
Hope: term having common meaning are mapped to the same direction



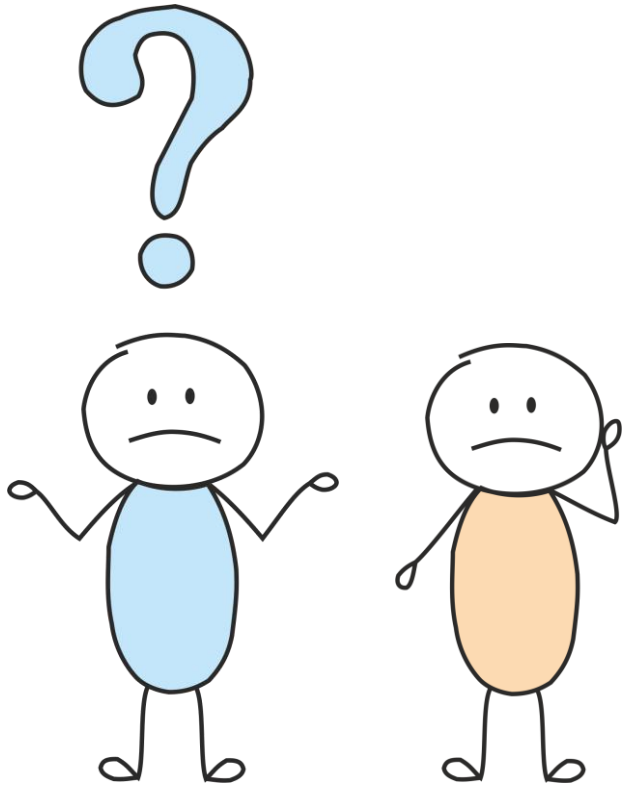
LSA document vectors
Hope: documents discussing similar topics have similar representations

Count-based methods

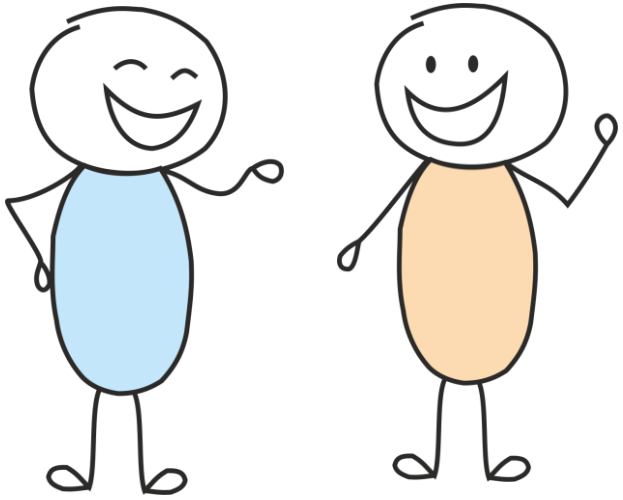
However, this is not the only way to induce distributional representations

(and not the best one)

Why not to learn distributed representations?



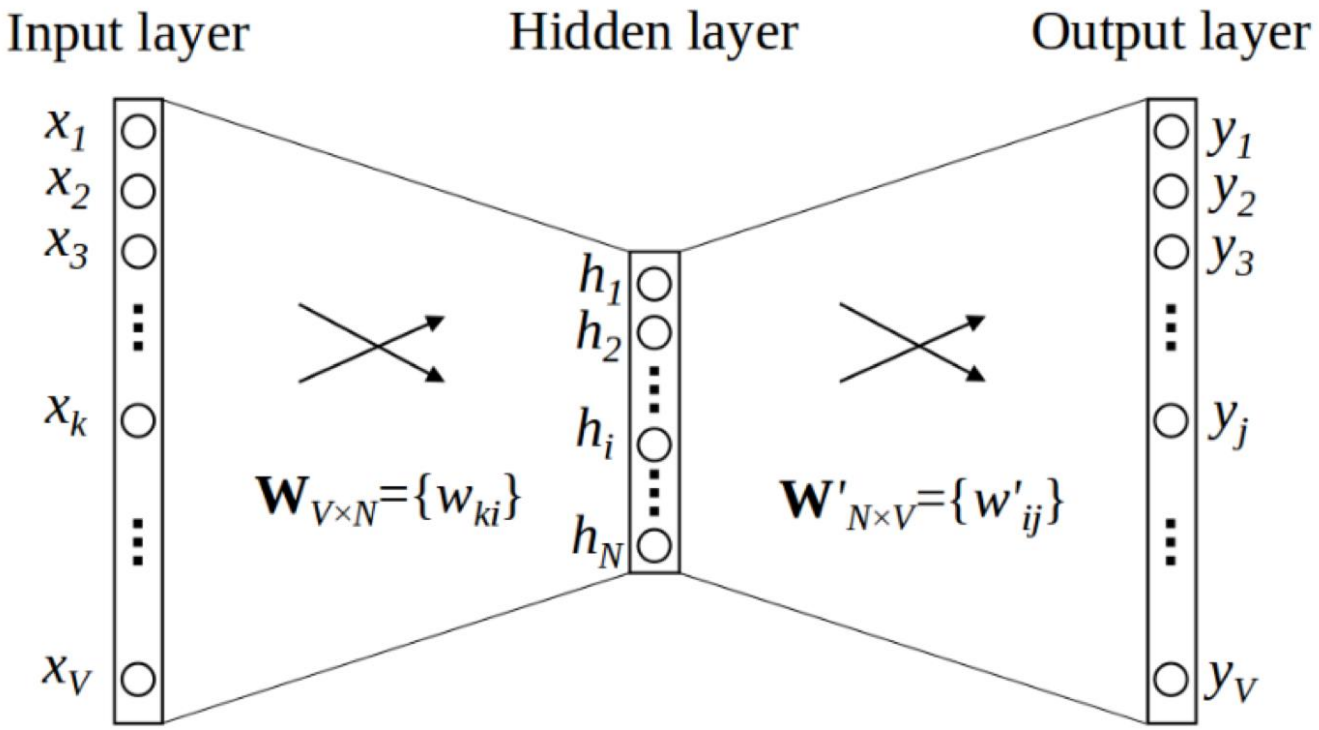
I mean, really, why?



We will learn a dense vector for each word, chosen so that it is similar to vectors of words that appear in similar contexts.

This **word vectors** are called **word embeddings** or word representations

Word2Vec

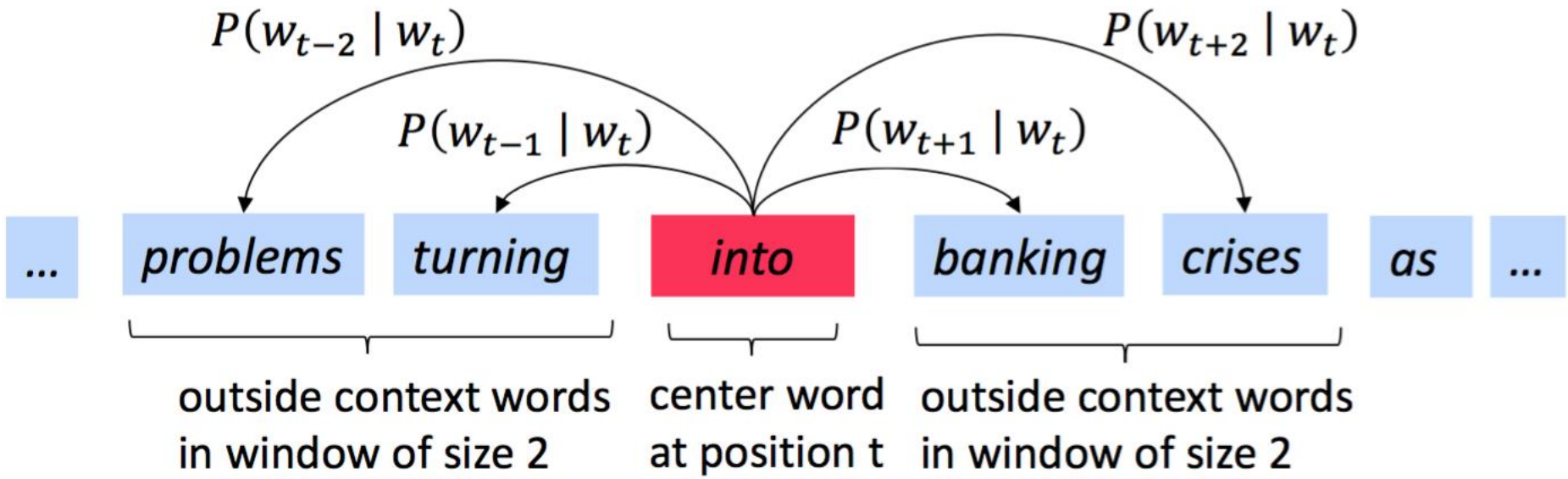


- a large corpus of text
- Every word in a fixed vocabulary is represented by a vector
- Go through each position t in the text, which has a center word c and context ("outside") words o
- Use the similarity of the word vectors for *c* and *o* to calculate the probability of *o* given *c* (or vice versa)
- Keep adjusting the word vectors to maximize this probability

Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

Word2Vec

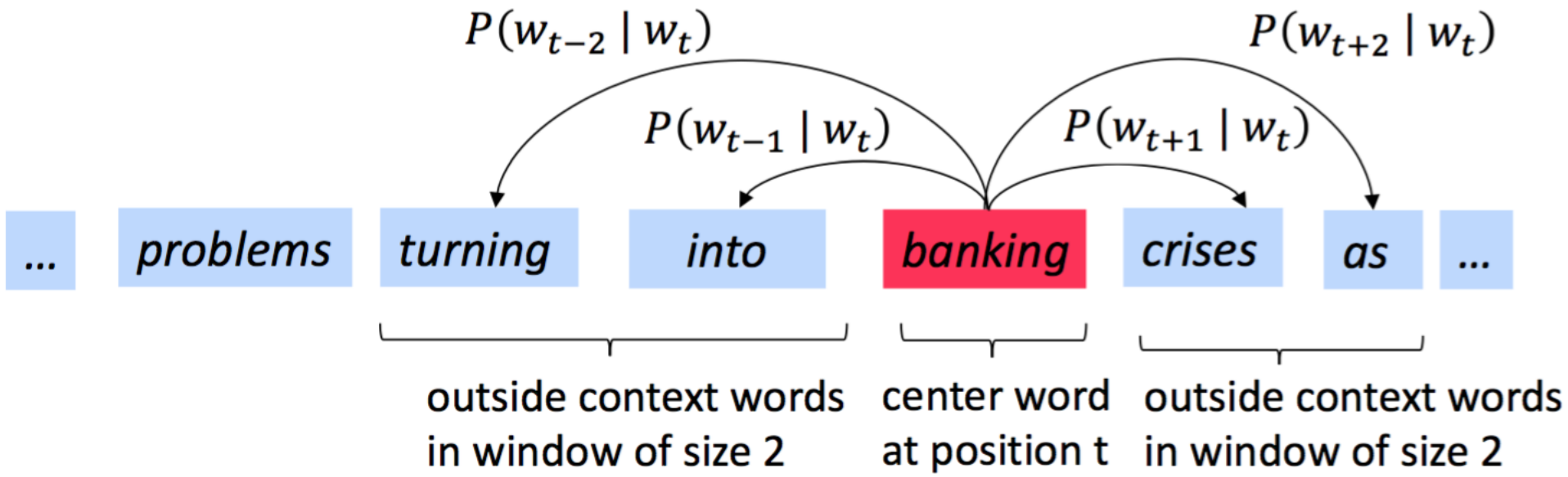
- Examples windows and and process for computing $P(w_{t+j}|w_j)$



http://web.stanford.edu/class/cs224n/syllabus.html

Word2Vec

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Word2Vec: objective function

For each position $t = 1, \ldots, T$, predict context words within a window of fixed size m, given center word w_t.

$$L(\theta) = \prod_{t=1}^{T} \prod_{\substack{-m \le j \le m \\ j \neq 0}} P\left(w_{t+j} \mid w_t; \theta\right)$$

Word2Vec: objective function

For each position $t = 1, \ldots, T$, predict context words within a window of fixed size m, given center word w_t.

Likelihood = $L(\theta) = \prod_{t=1}^{T} \prod_{\substack{-m \le j \le m \\ j \neq 0}} P(w_{t+j} \mid w_t; \theta)$

θ *is all variables to be optimized*

Word2Vec: objective function

The **objective function (or loss, or cost function)** $J(\theta)$ is the (average) negative log likelihood

$$J(\theta) = -\frac{1}{T}\log L(\theta) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{\substack{-m\leq j\leq m \\ j\neq 0}} \log P\left(w_{t+j} \mid w_t; \theta\right)$$

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Minimizing objective function ⟺ Maximizing predictive accuracy

Word2Vec: objective function

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- Question: **How to calculate $P(w_{t+j}|w_j, \theta)$?**

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- Answer: We will use two vectors per word *w*
  - v_w is a center word
  - u_w is a context word

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- Then for a center word **c** and a context word **o**:

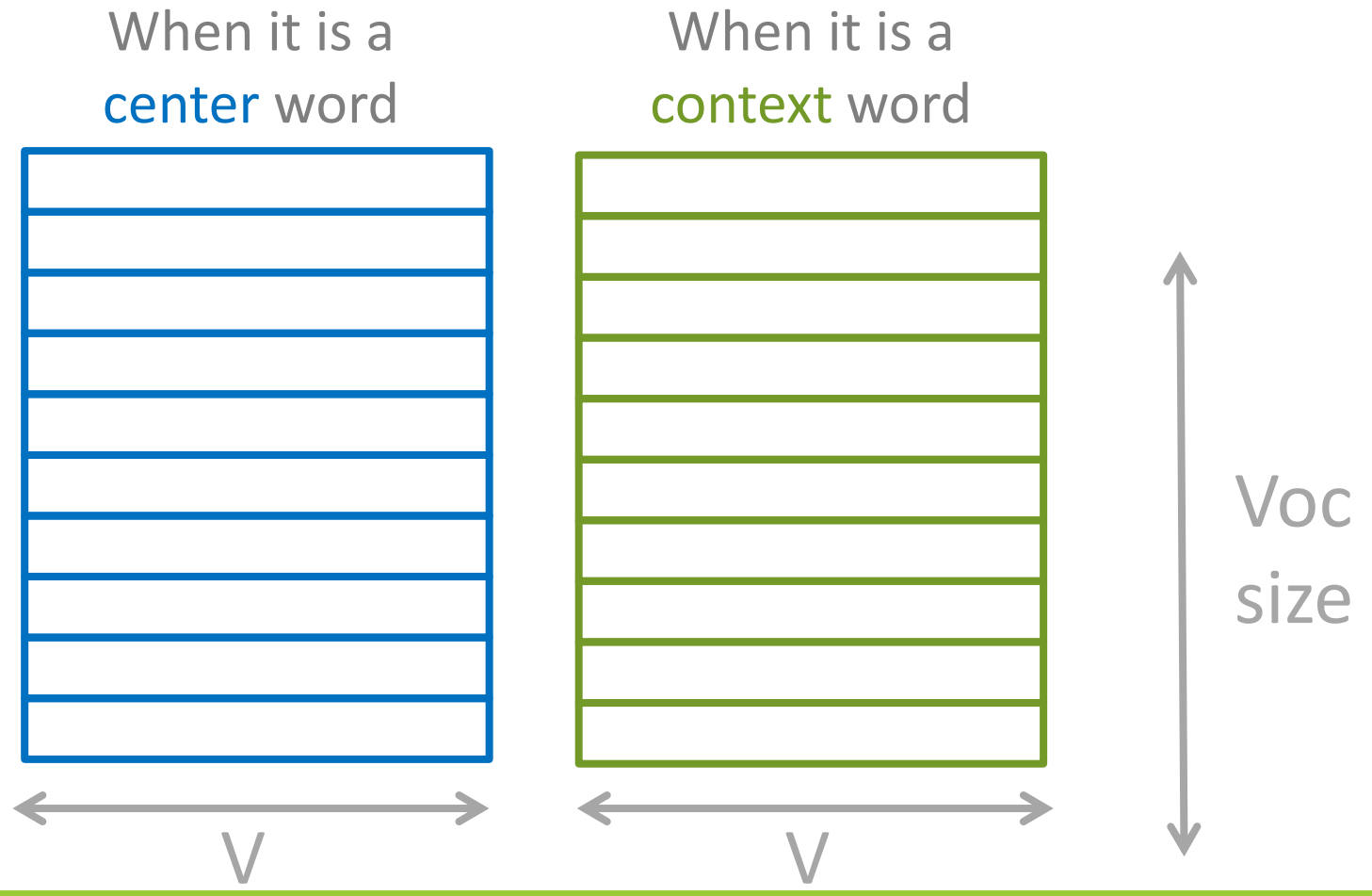
$$P(o|c) = \frac{\exp(u_o^T v_c)}{\sum_{w\in V}\exp(u_w^T v_c)}$$

Two vectors for each word

I saw a cat .

39 1592 10 2548 5

Token index in the vocabulary

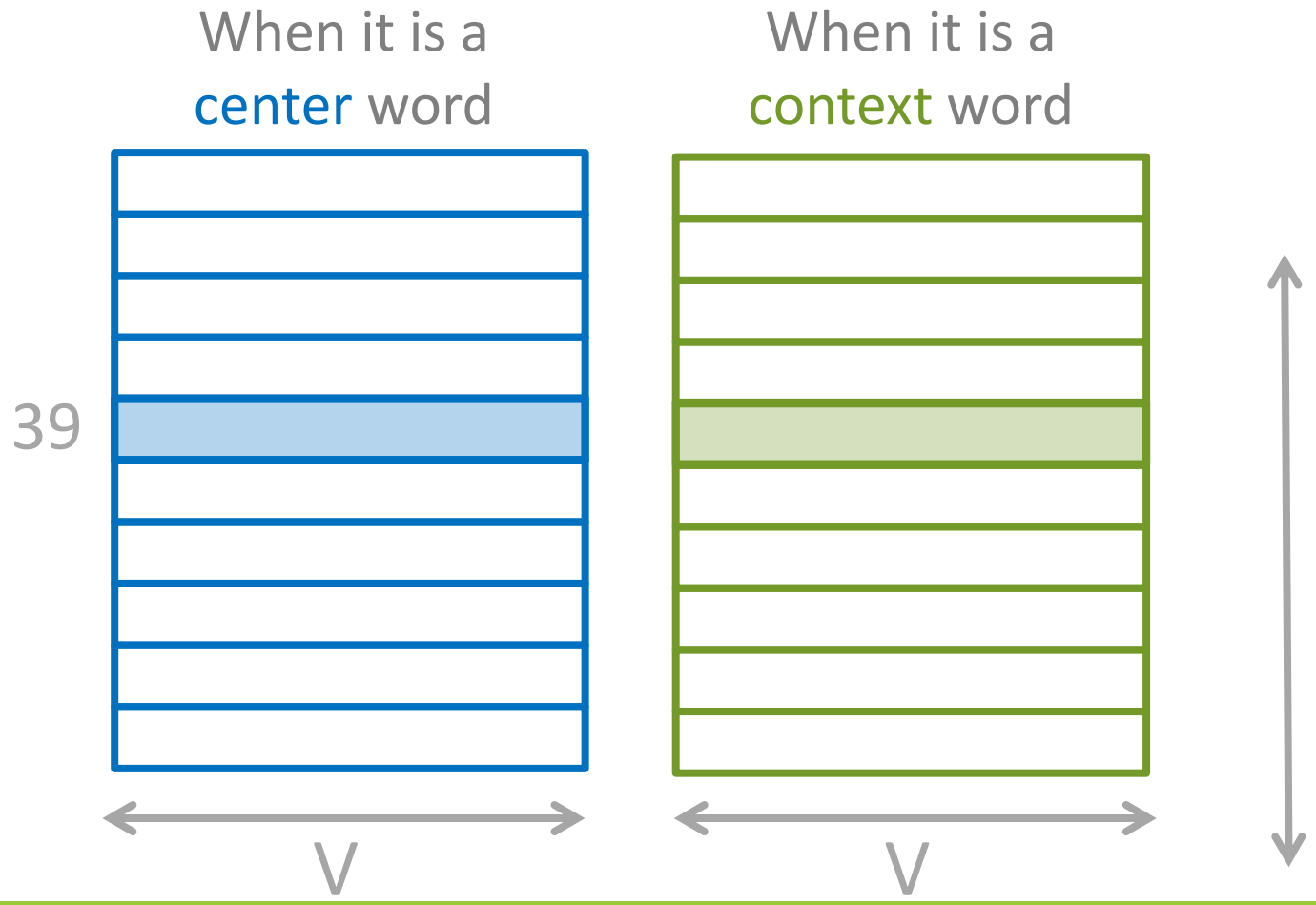


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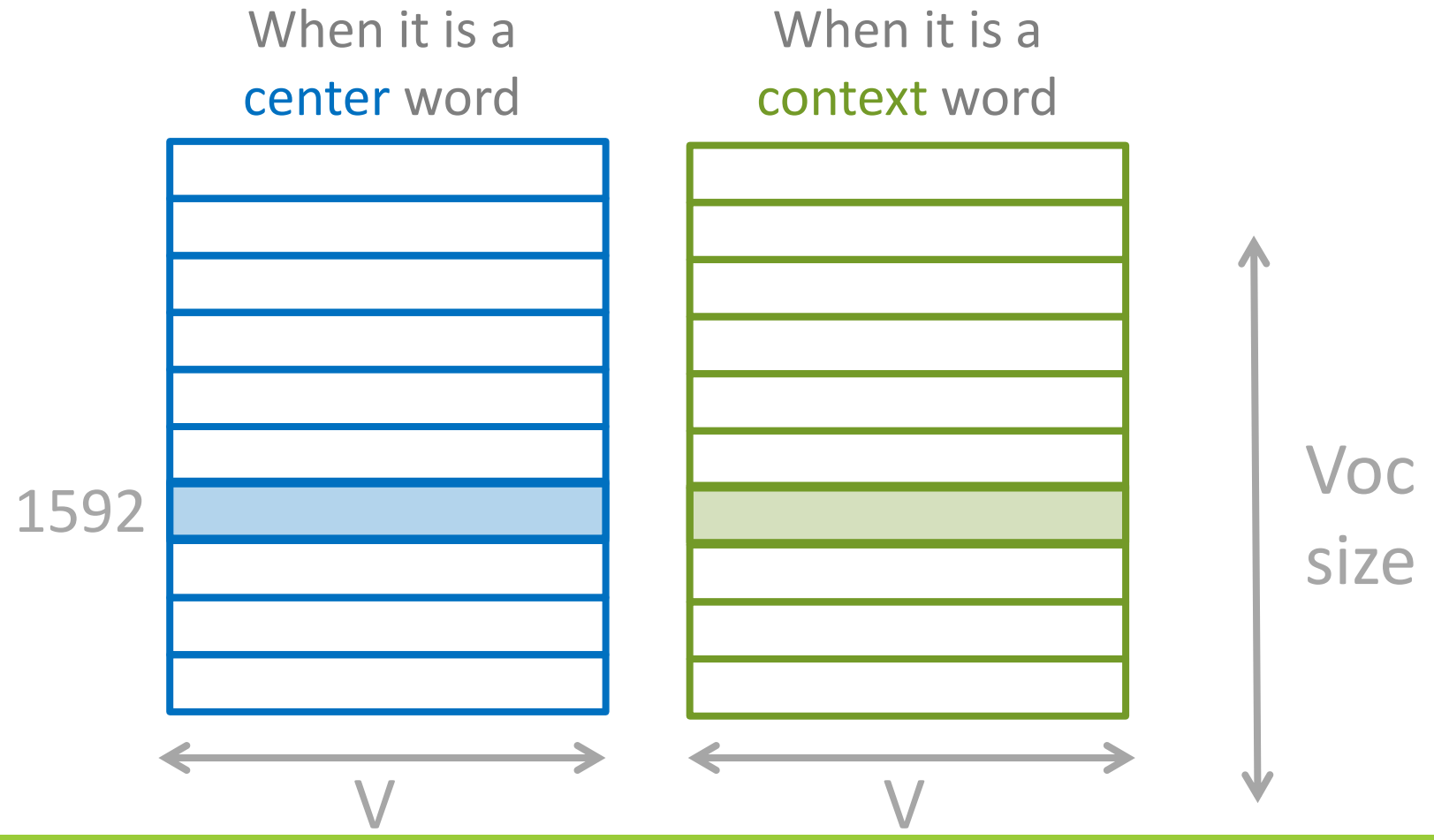


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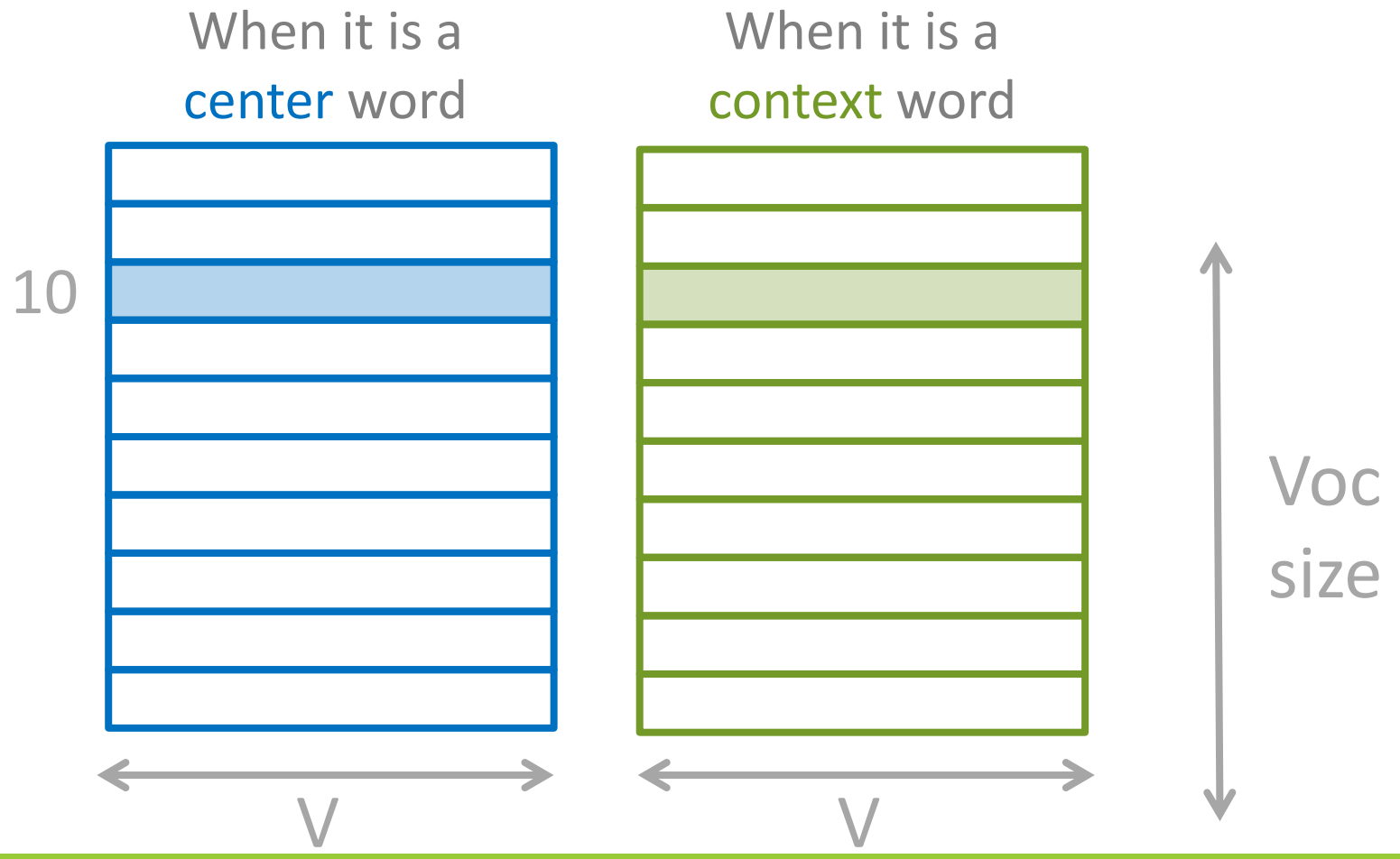


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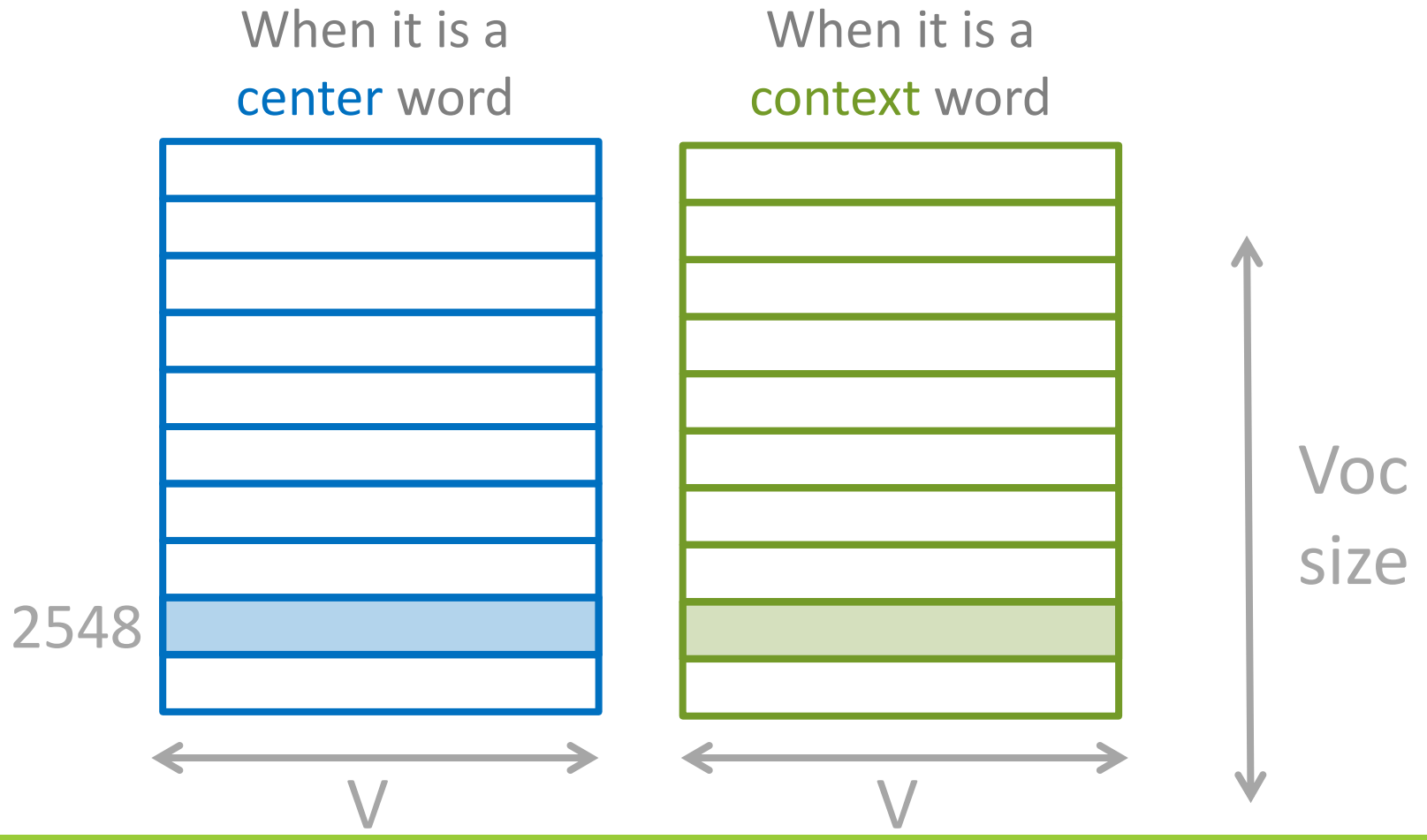


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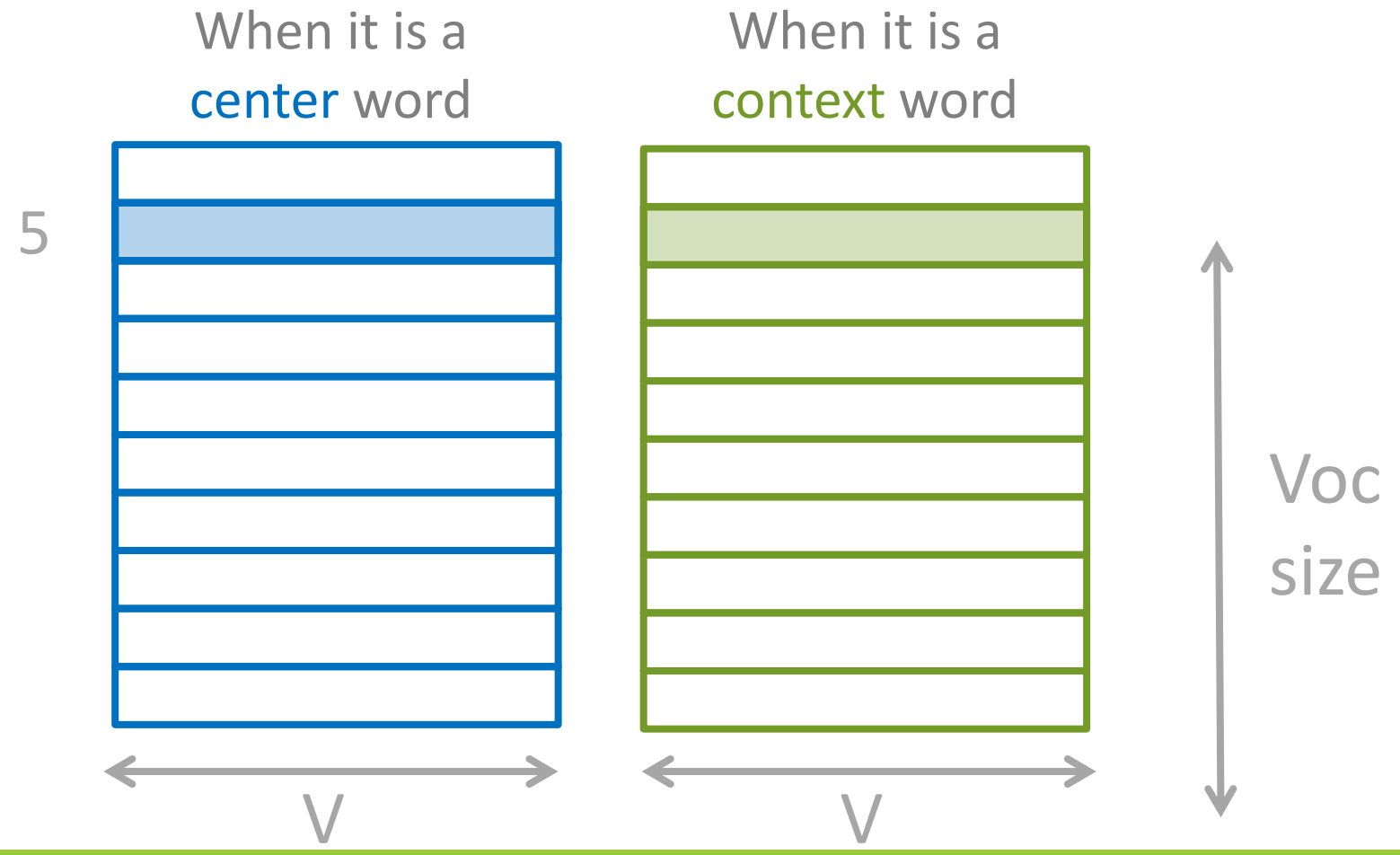


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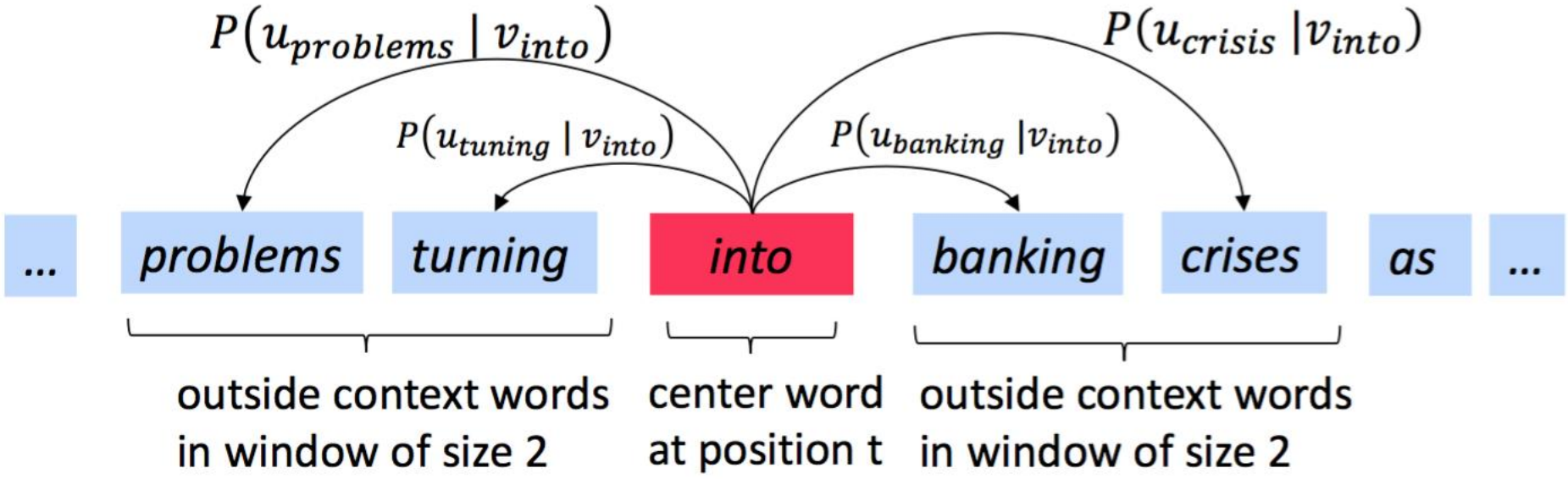
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Token index in the vocabulary



Word2Vec

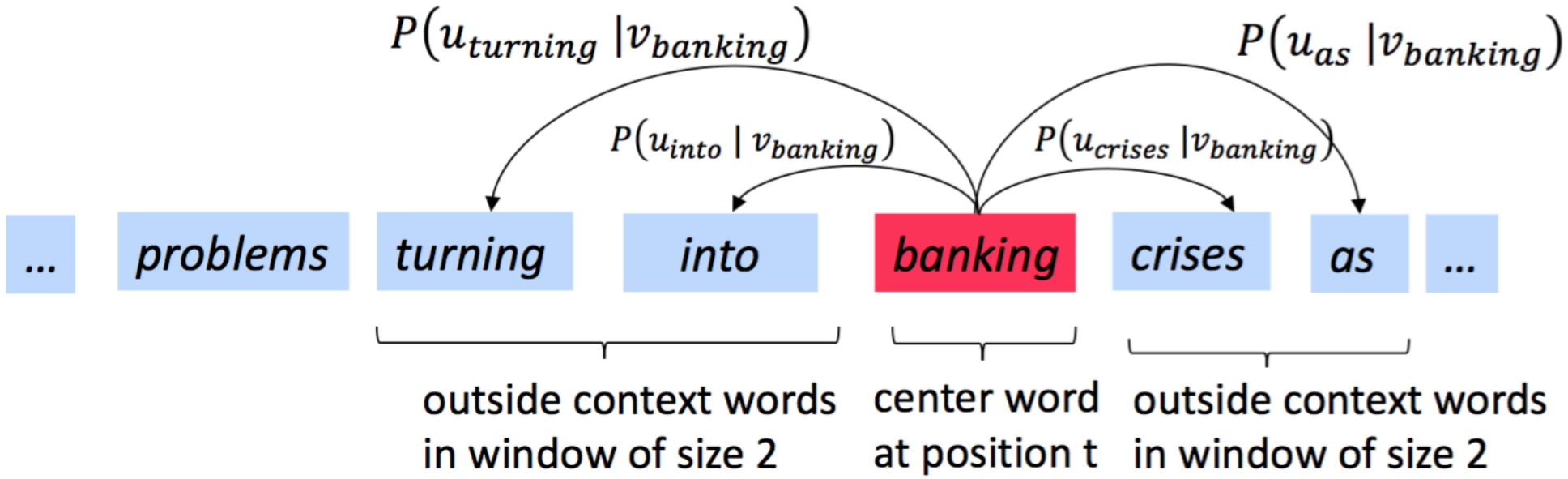
- Examples windows and and process for computing $P(w_{t+j}|w_j)$
- $P(u_{problems}|v_{into})$ is short for $P(problems|into;\ u_{problems}, v_{into}, \theta)$



http://web.stanford.edu/class/cs224n/syllabus.html

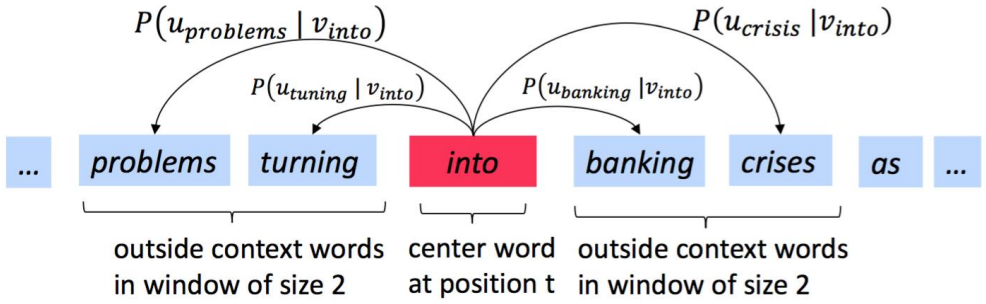
Word2Vec

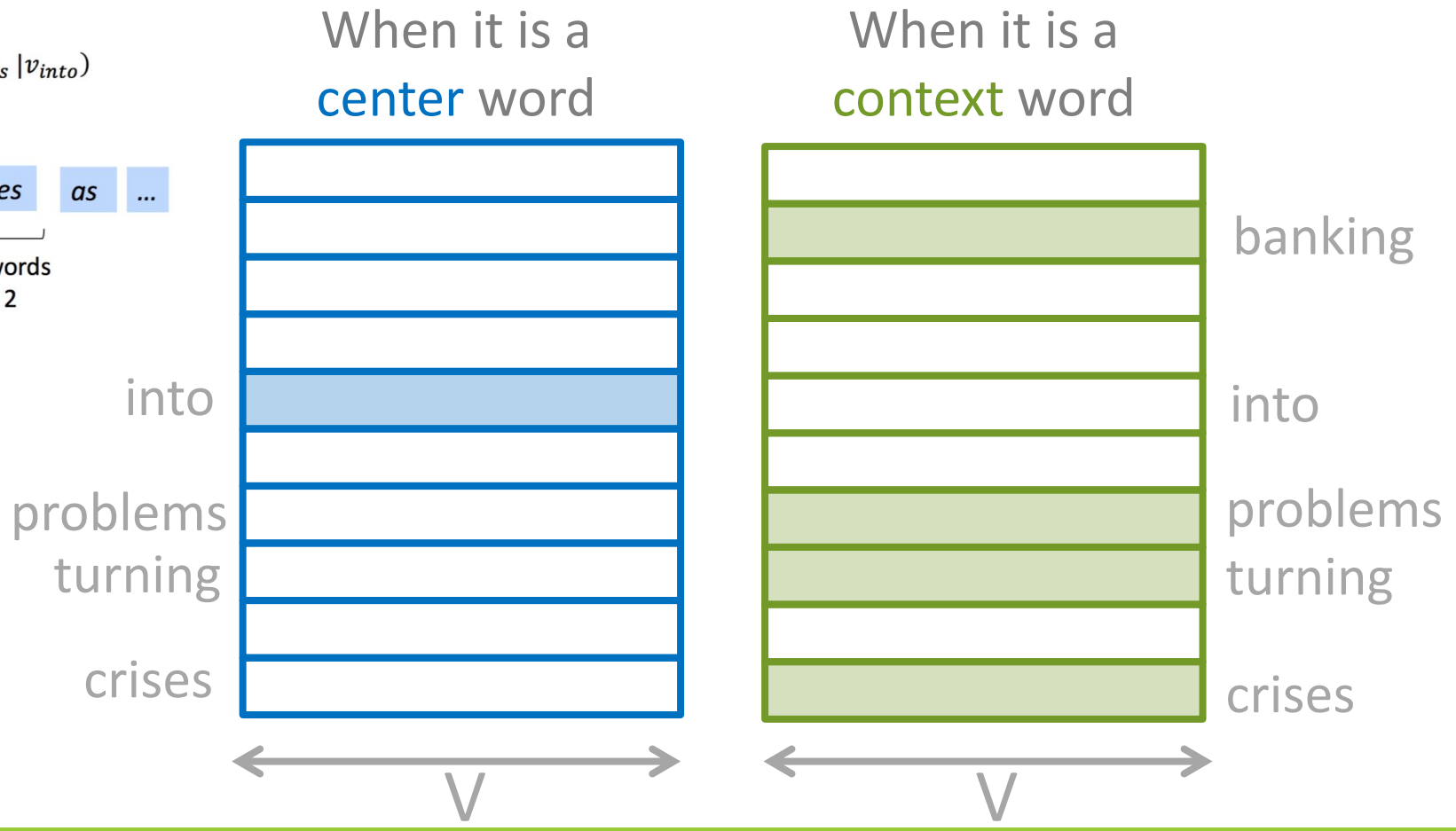
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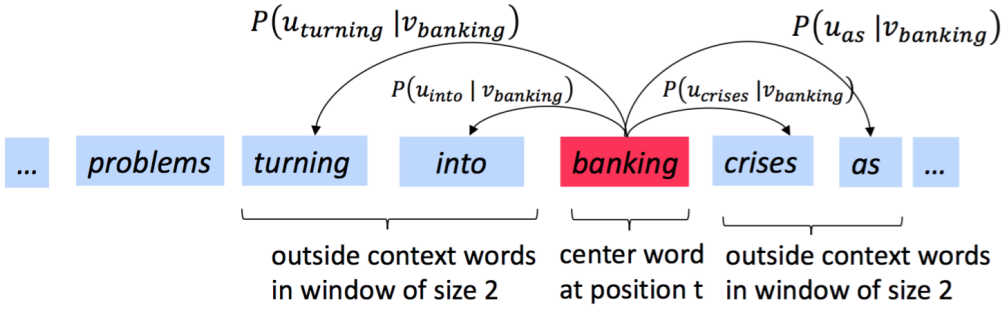
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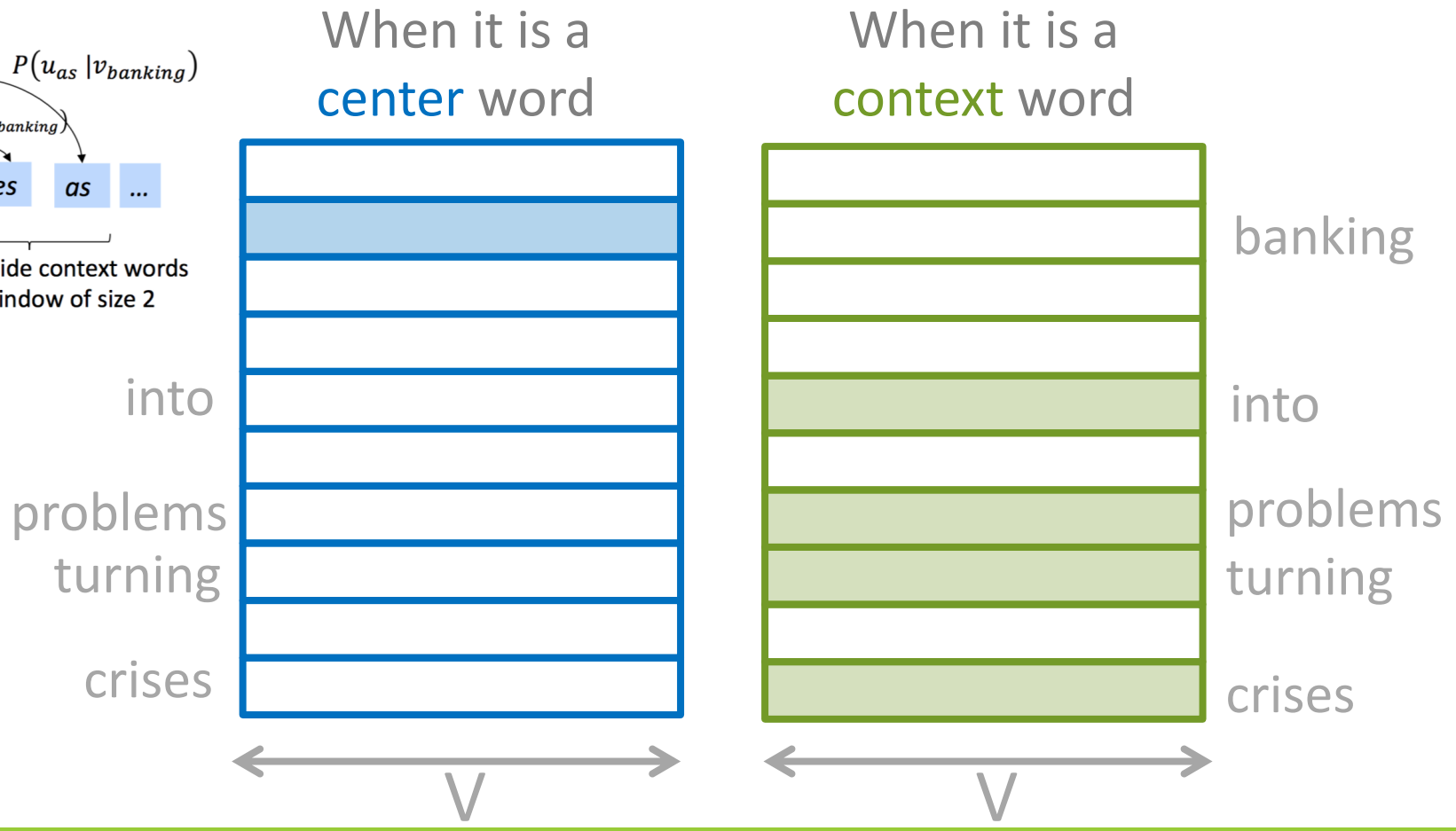
Two vectors for each word





Two vectors for each word





Word2Vec: prediction function

$$P(o|c) = \frac{\exp(u_o^T v_c)}{\sum_{w \in V} \exp(u_w^T v_c)}$$

Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

Word2Vec: prediction function

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Dot product measures similarity of ***o*** and ***c***
Larger dot product = larger probability

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After taking exponent, normalize over entire vocabulary

Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

This is softmax!

Softmax function $\mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$softmax(\boldsymbol{x})_i = \frac{\exp(x_i)}{\sum_{j=1}^{n} \exp(x_j)} = p_i$$

- maps arbitrary values x_i to a probability distribution p_i
- ”max” because amplifies probability of largest x_i
- “soft” because still assigns some probability to smaller x_i
- **often used in Deep Learning!**

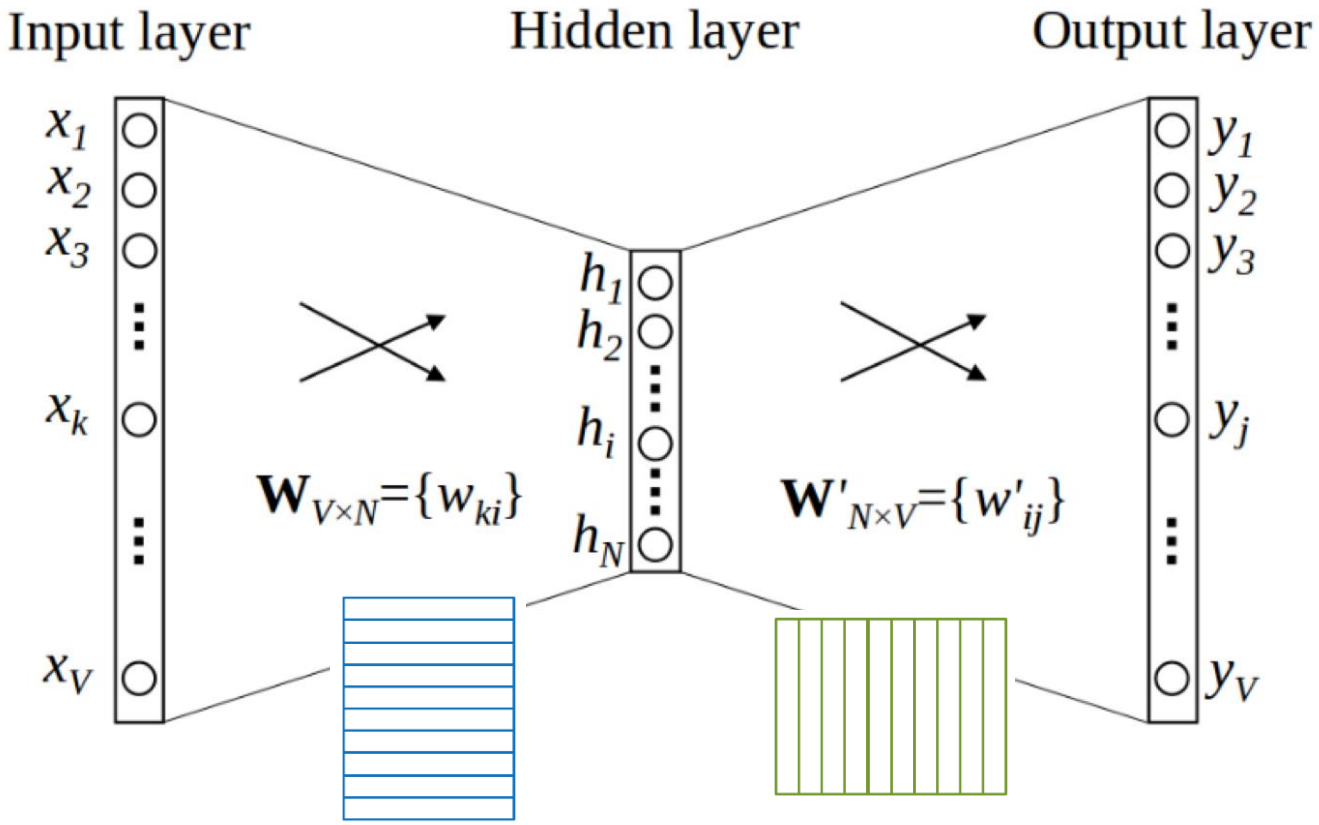
Where is $\boldsymbol{\theta}$?

$$\theta = \begin{bmatrix} v_{aardvark} \\ v_a \\ \vdots \\ v_{zebra} \\ u_{aardvark} \\ u_a \\ \vdots \\ u_{zebra} \end{bmatrix} \in \mathbb{R}^{2dV}$$

- θ - d-dimensional vectors for V words
- every word has two vectors!
- we optimize these parameters

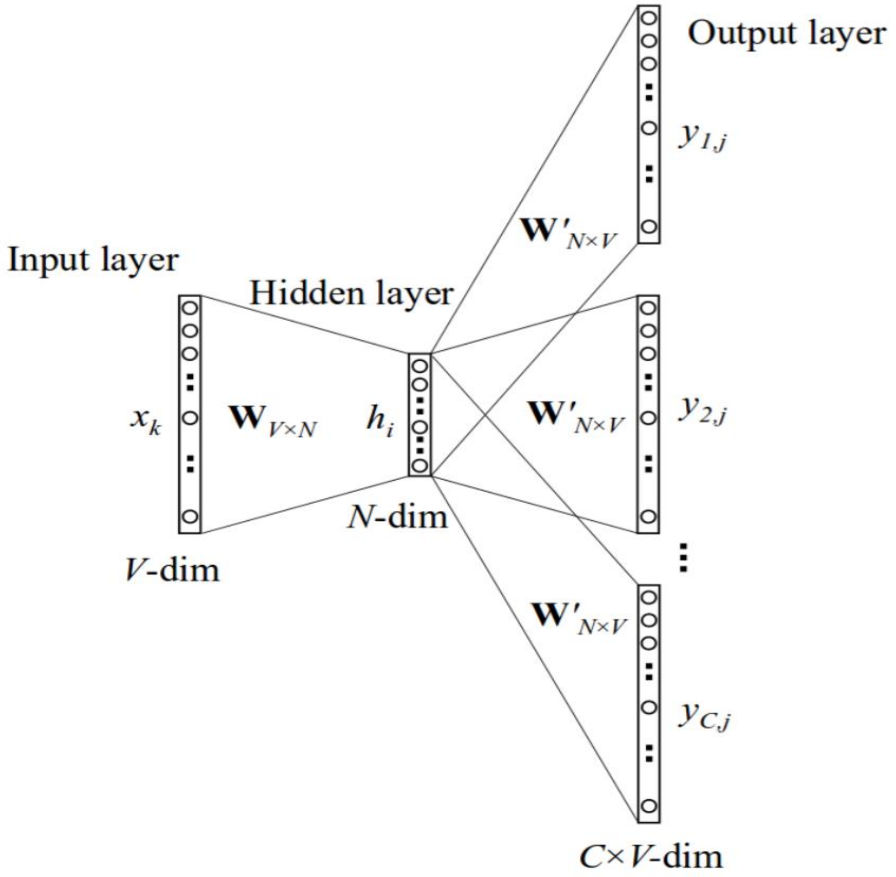
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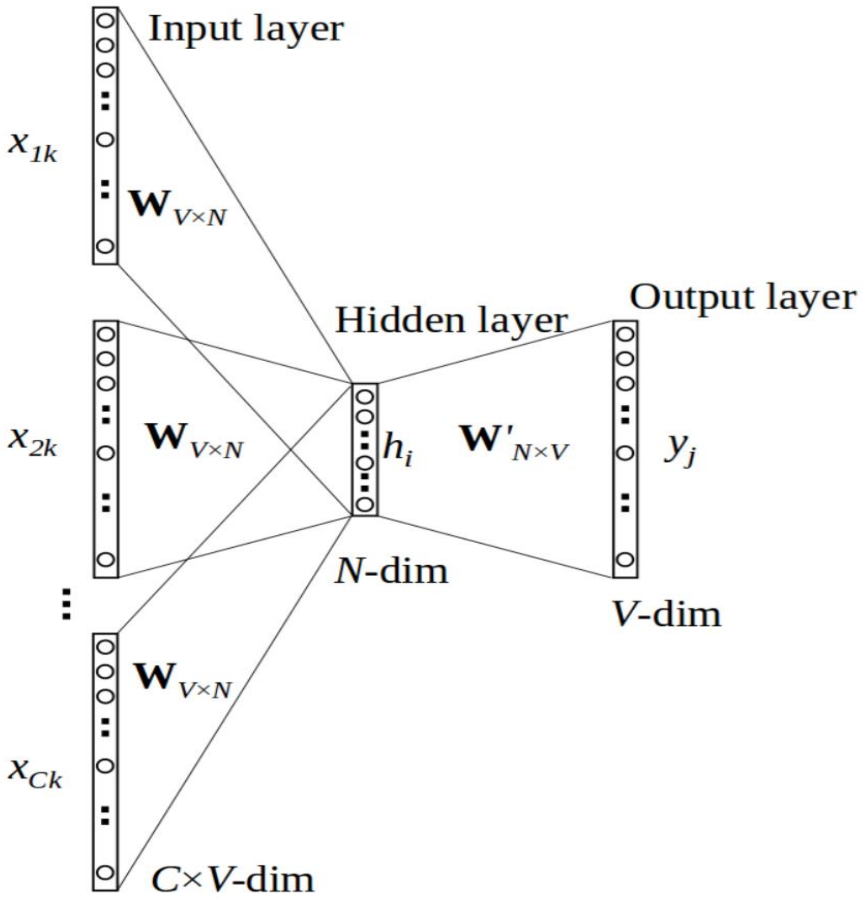
Word2Vec: Skip-gram (SG)

- Predict context (”outside”) words (position independent) given center word



Word2Vec: Continuous Bag of Words (CBOW)

- Predict center word from (bag of) context words



Word2Vec: Additional efficiency in training

$$P(o|c) = \frac{\exp(u_o^T v_c)}{\sum_{w \in V} \exp(u_w^T v_c)}$$



Huge sum! Time for calculating gradients is proportional to |V|

Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

Word2Vec: Additional efficiency in training

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Possible solutions:

- Hierarchical softmax
- Negative sampling

Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

Word2Vec: Additional efficiency in training

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Huge sum! Time for calculating gradients is proportional to |V|

Possible solutions:

- Hierarchical softmax
- Negative sampling

$$\sum_{w \in V} \exp(u_w^T v_c) \longrightarrow \sum_{w \in \{o\} \cup S_k} \exp(u_w^T v_c)$$

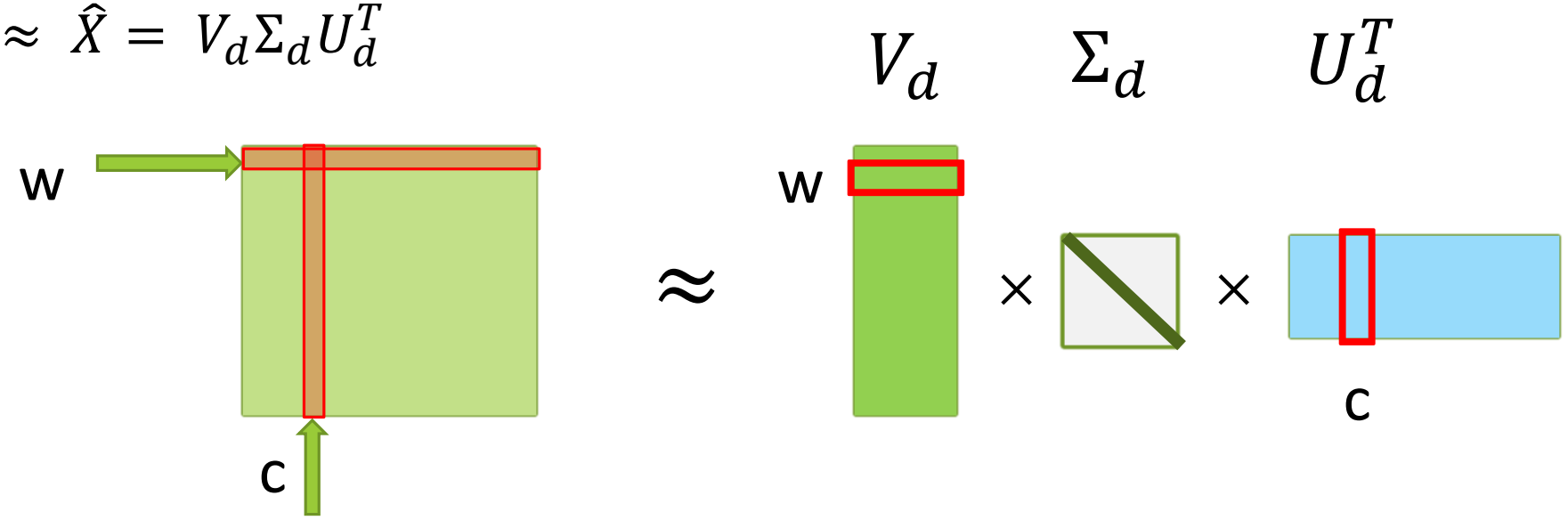
Sum over a small subset: *negative sample,* $|S_k| = k$

Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

Word2Vec:
(Near) equivalence to matrix factorization

$$PMI(w,c) = \log \frac{N(w,c) \times |V|}{N(w)N(c)}$$

$$PMI = X \approx \hat{X} = V_d \Sigma_d U_d^T$$

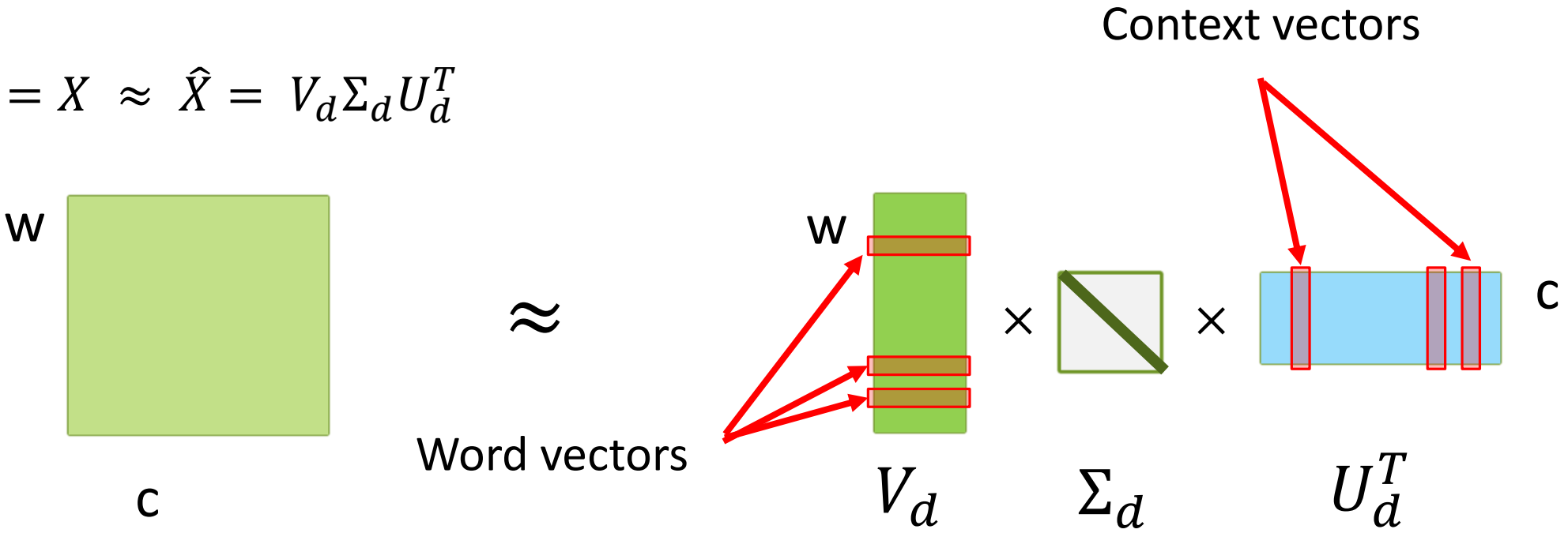


Levy et al, TACL 2015 http://www.aclweb.org/anthology/Q15-1016

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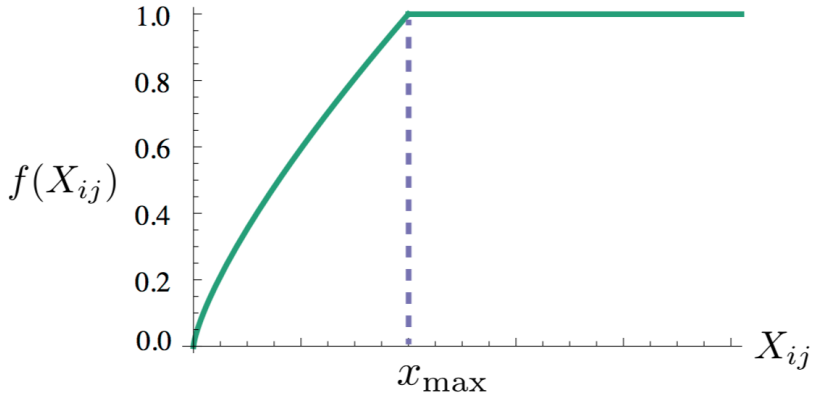


Levy et al, TACL 2015 http://www.aclweb.org/anthology/Q15-1016

GloVe: combine count-based and direct prediction methods

$$J(\theta) = \frac{1}{2} \sum_{i,j=1}^{W} f(P_{ij})(u_i^T v_j - \log P_{ij})^2$$

$$X_{final} = U + V$$

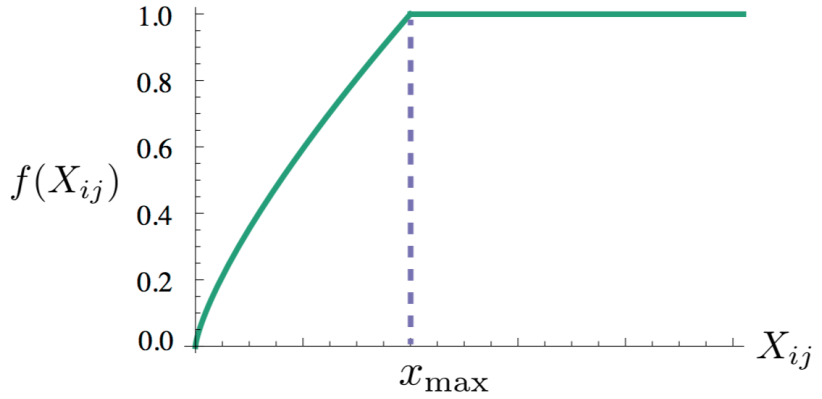


Pennington et al., EMNLP 2014, https://www.aclweb.org/anthology/D14-1162

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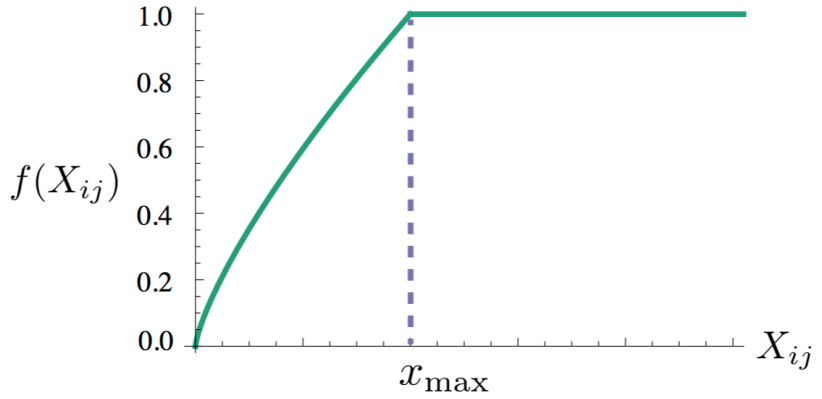
probability that word *j* appears in the context of word *i*

Pennington et al., EMNLP 2014, https://www.aclweb.org/anthology/D14-1162

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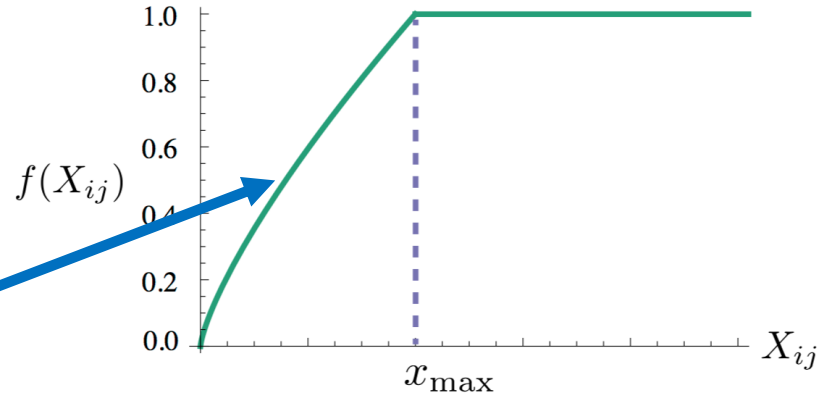
the idea is close to factorizing the log of the co-occurrence matrix (closely related to LSA)

Pennington et al., EMNLP 2014, https://www.aclweb.org/anthology/D14-1162

GloVe: combine count-based and direct prediction methods

$$J(\theta) = \frac{1}{2} \sum_{i,j=1}^{W} f(P_{ij}) (u_i^T v_j - \log P_{ij})^2$$

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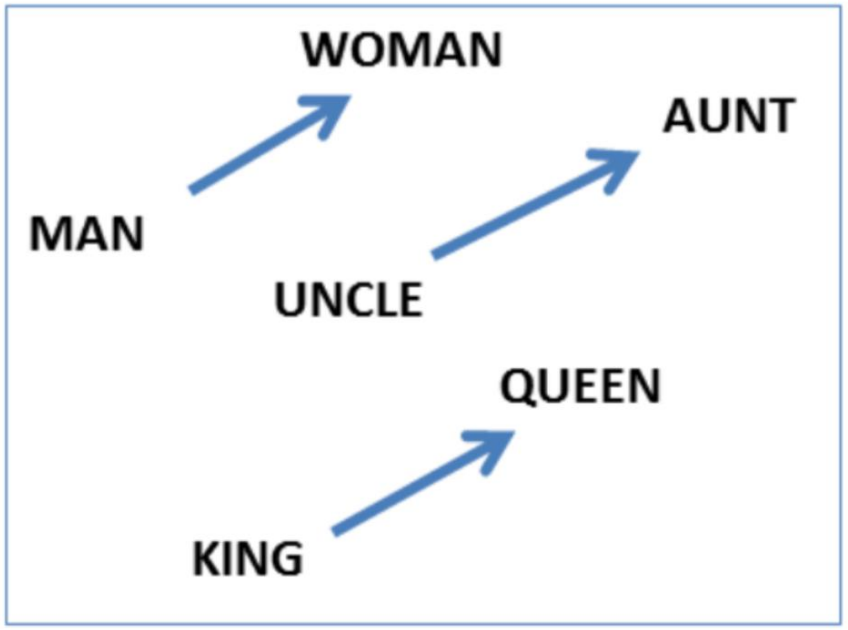
discard rare noisy co-occurrences

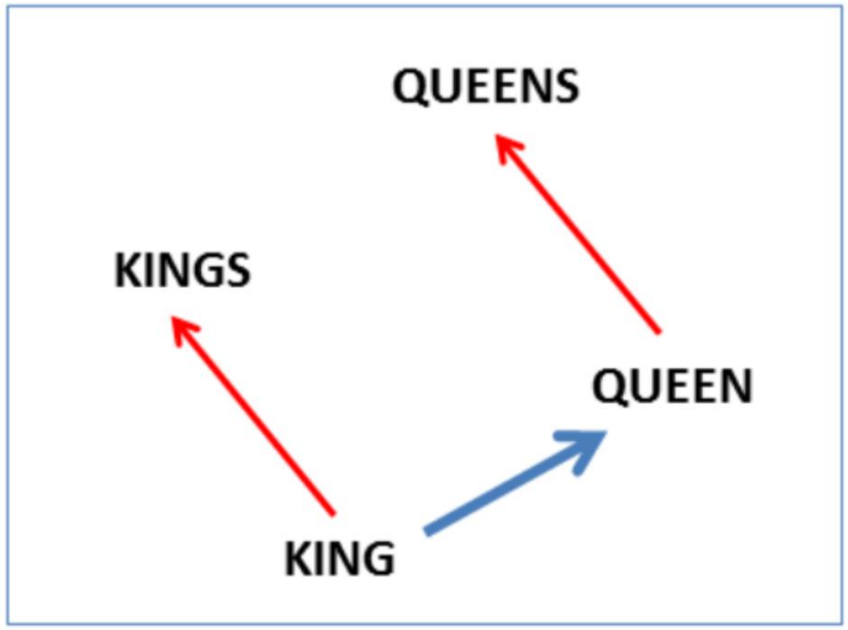
Pennington et al., EMNLP 2014, https://www.aclweb.org/anthology/D14-1162

And now the fun starts!

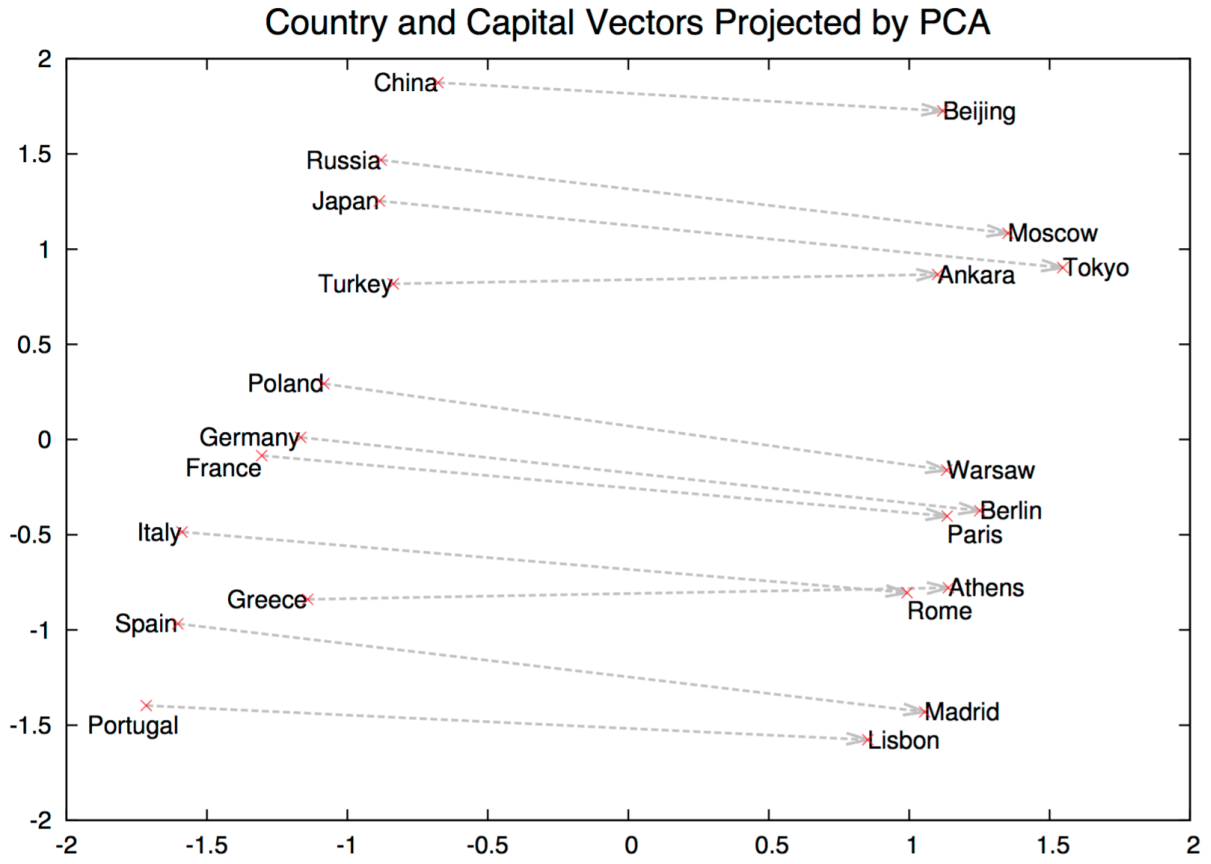
Word2Vec: what are relations between vectors?

$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$



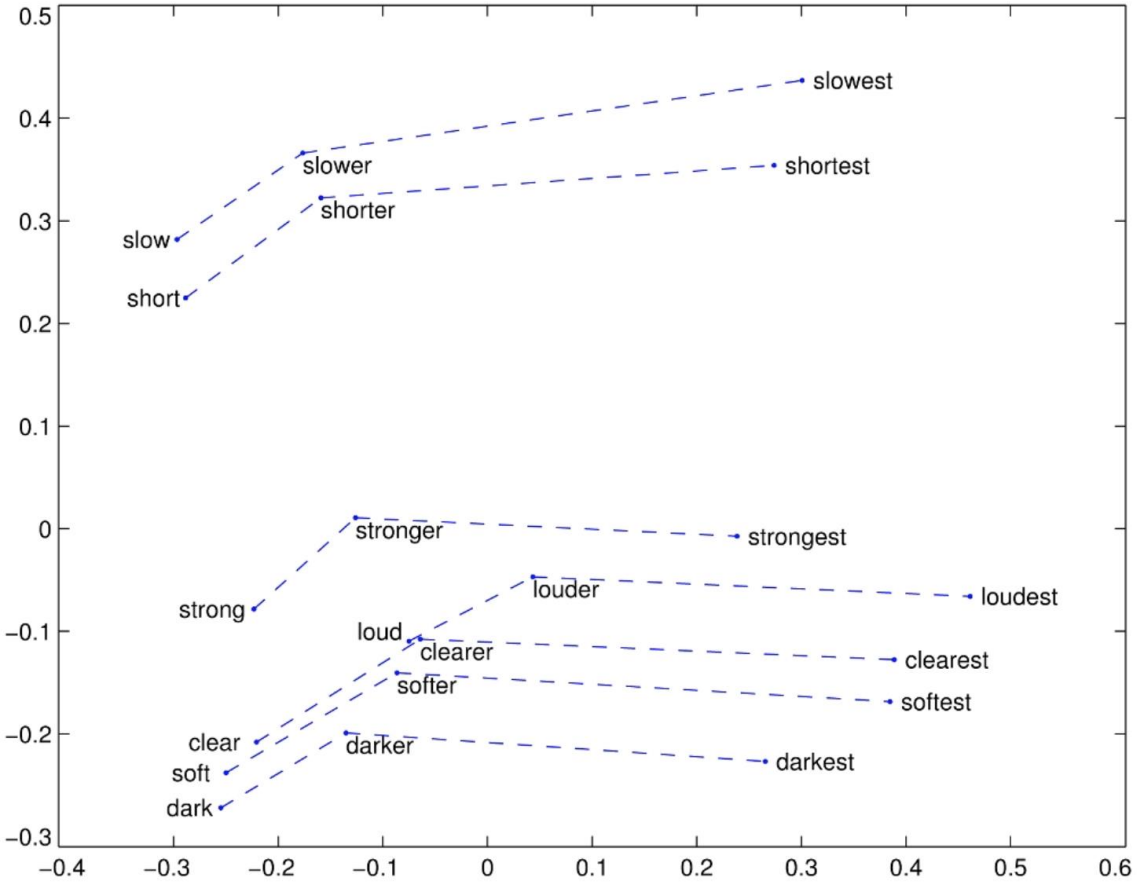


What are relations between vectors?



Mikolov et al, 2013, https://arxiv.org/pdf/1310.4546.pdf

What are relations between vectors?



(http://nlp.stanford.edu/projects/glove/)

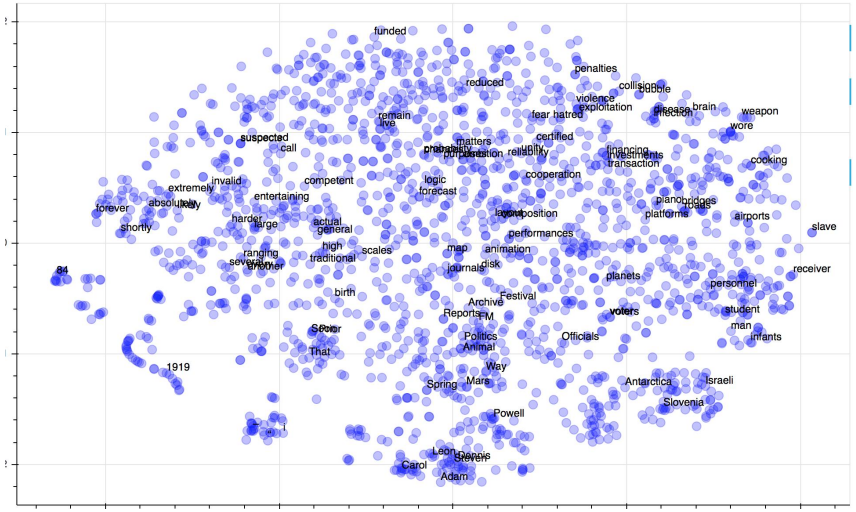
What are relations between vectors?

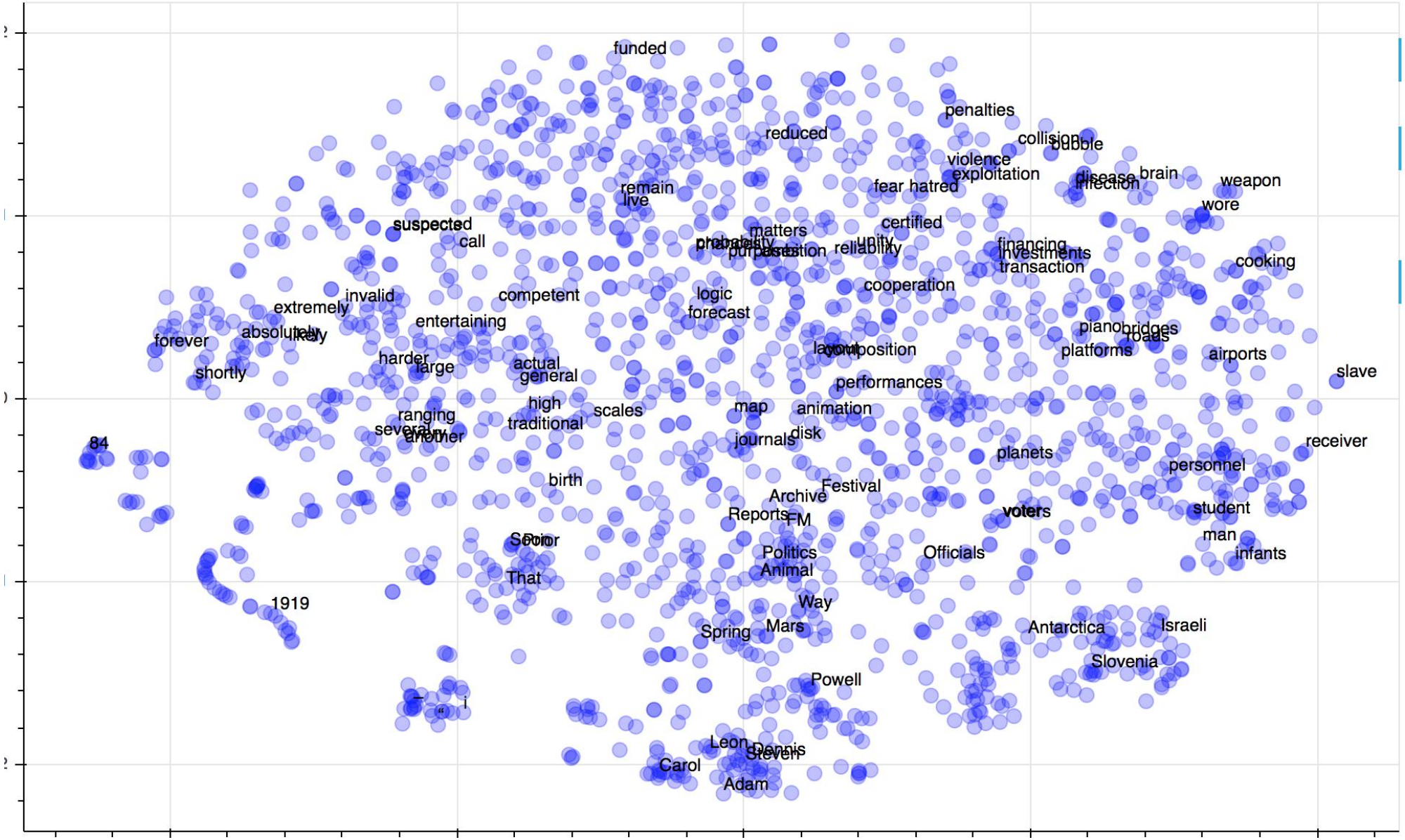
nearest neighbors of *frog*	Litoria	Leptodactylidae	Rana	Eleutherodactylus
Pictures				

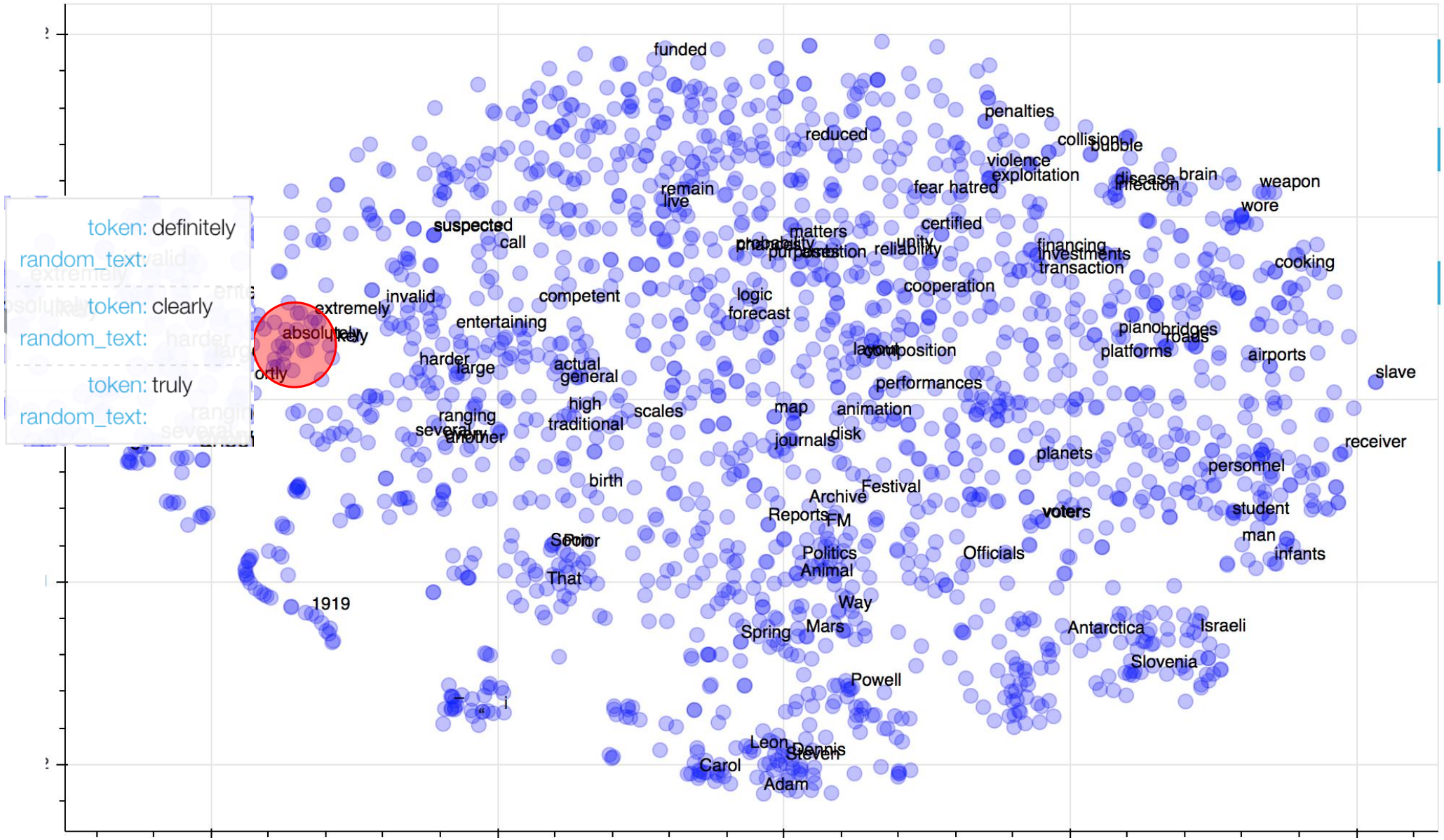
(http://nlp.stanford.edu/projects/glove/)

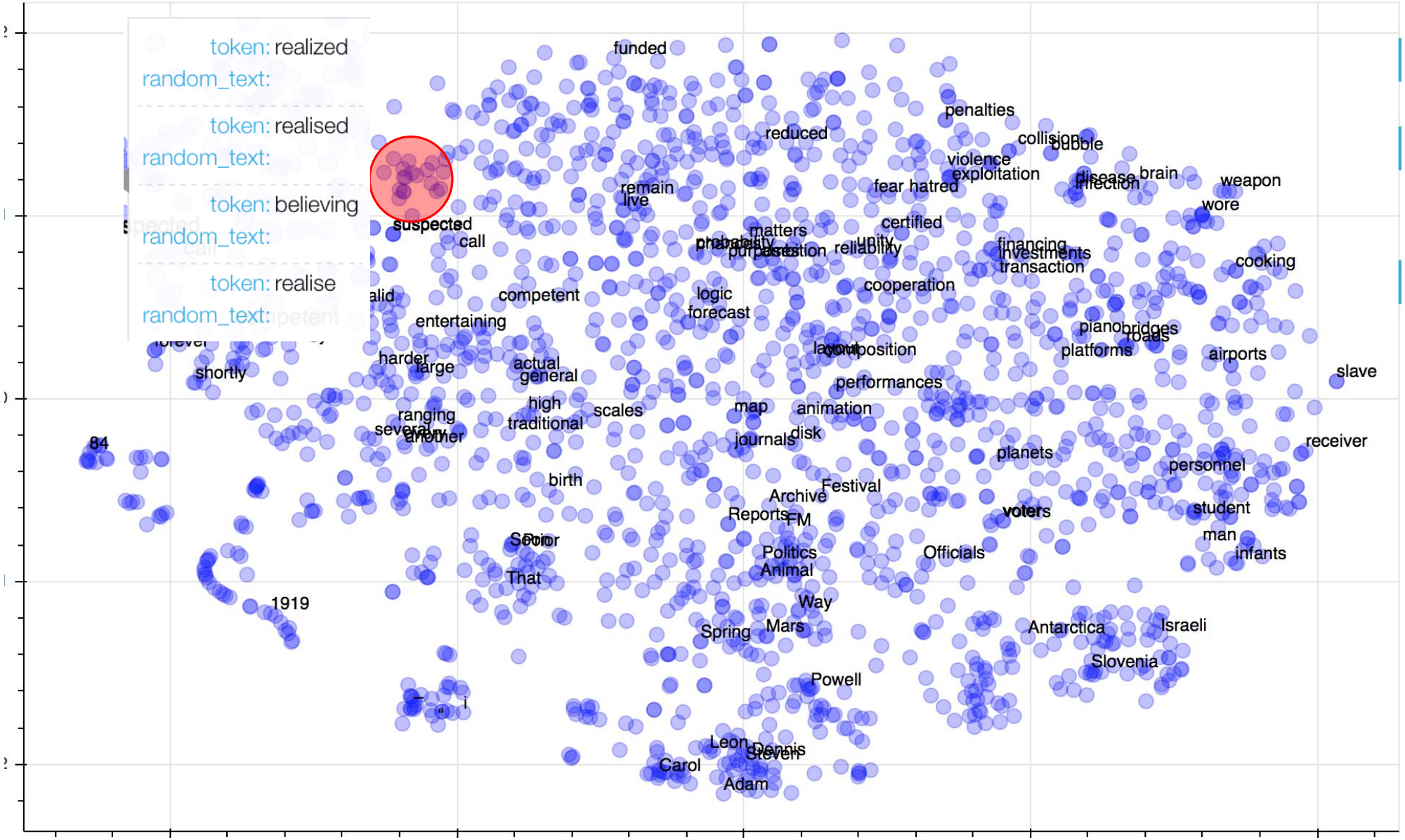
Let’s walk through space…

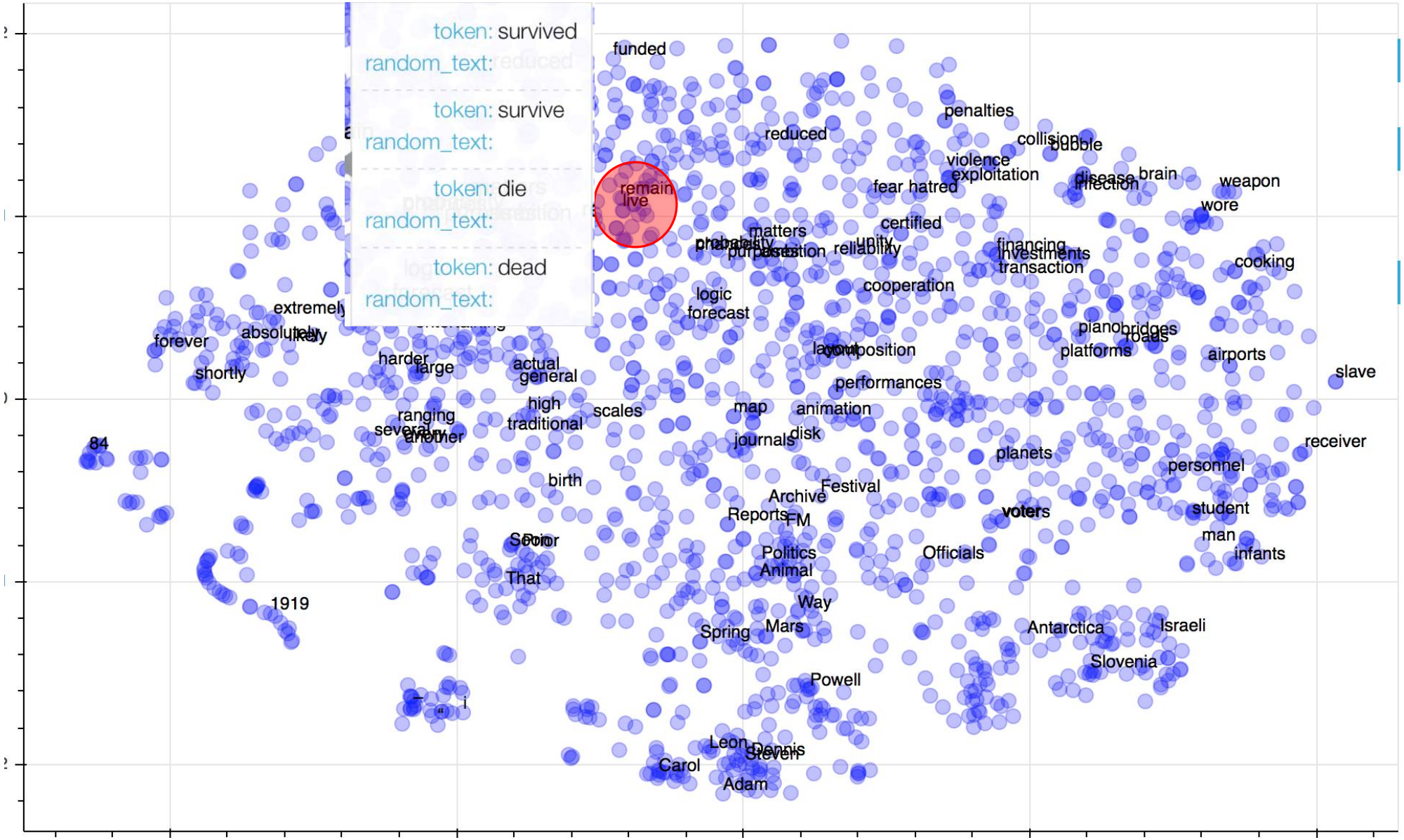
Let’s walk through space…
Semantic space!

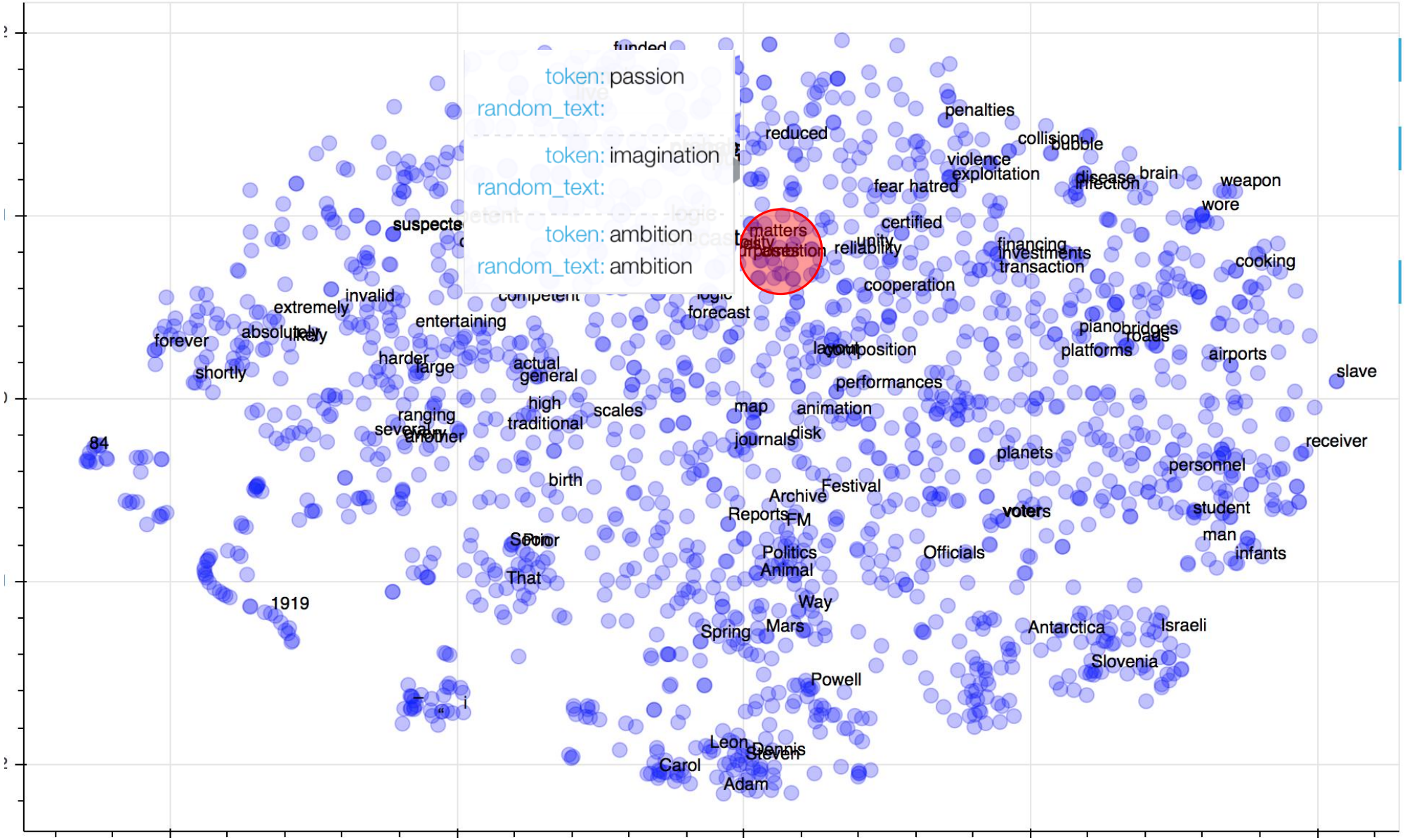


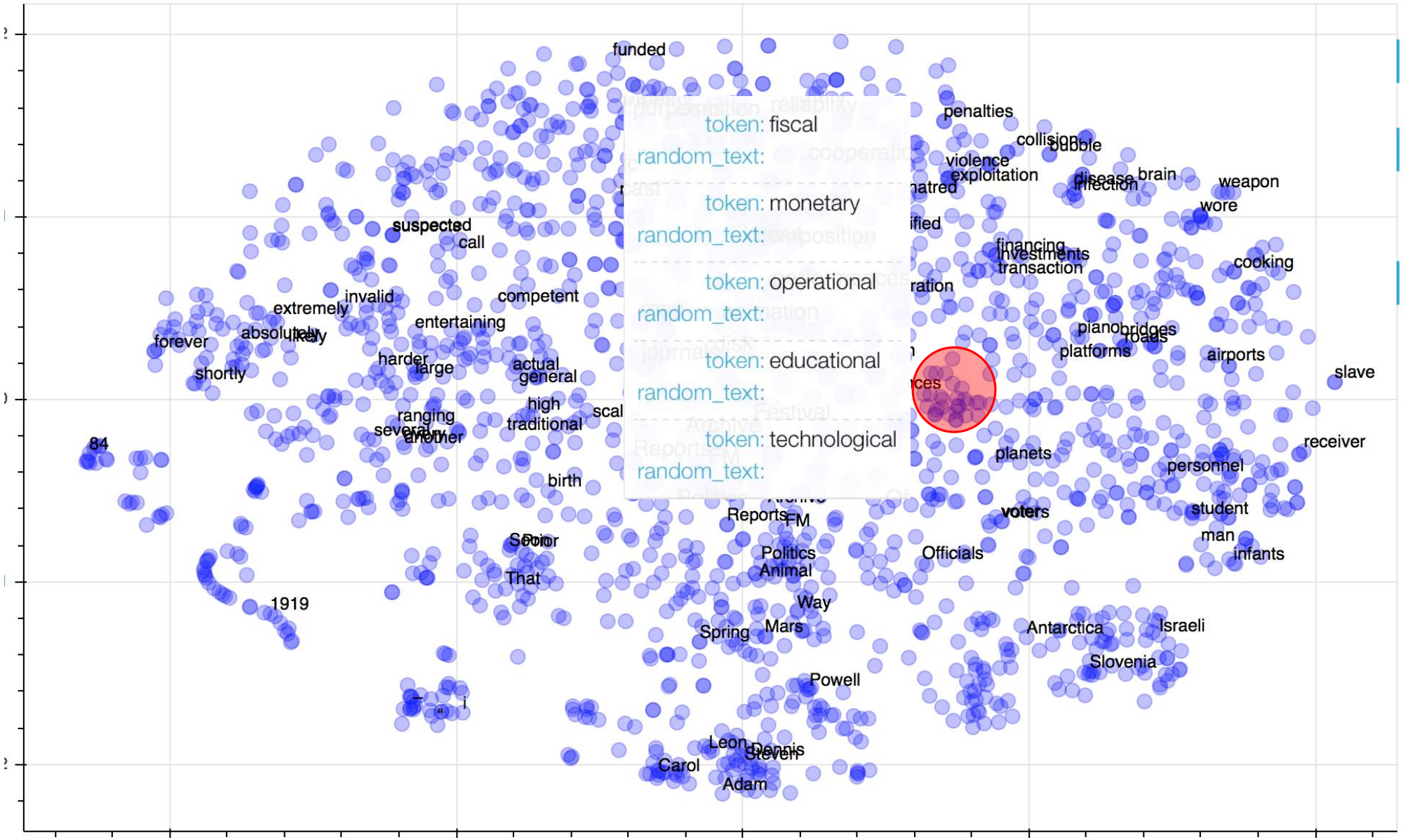

funded
penalties
reduced
collision
bubble
violence
exploitation
disease
brain
weapon
infection
wore
remain
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fear
hatred
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call
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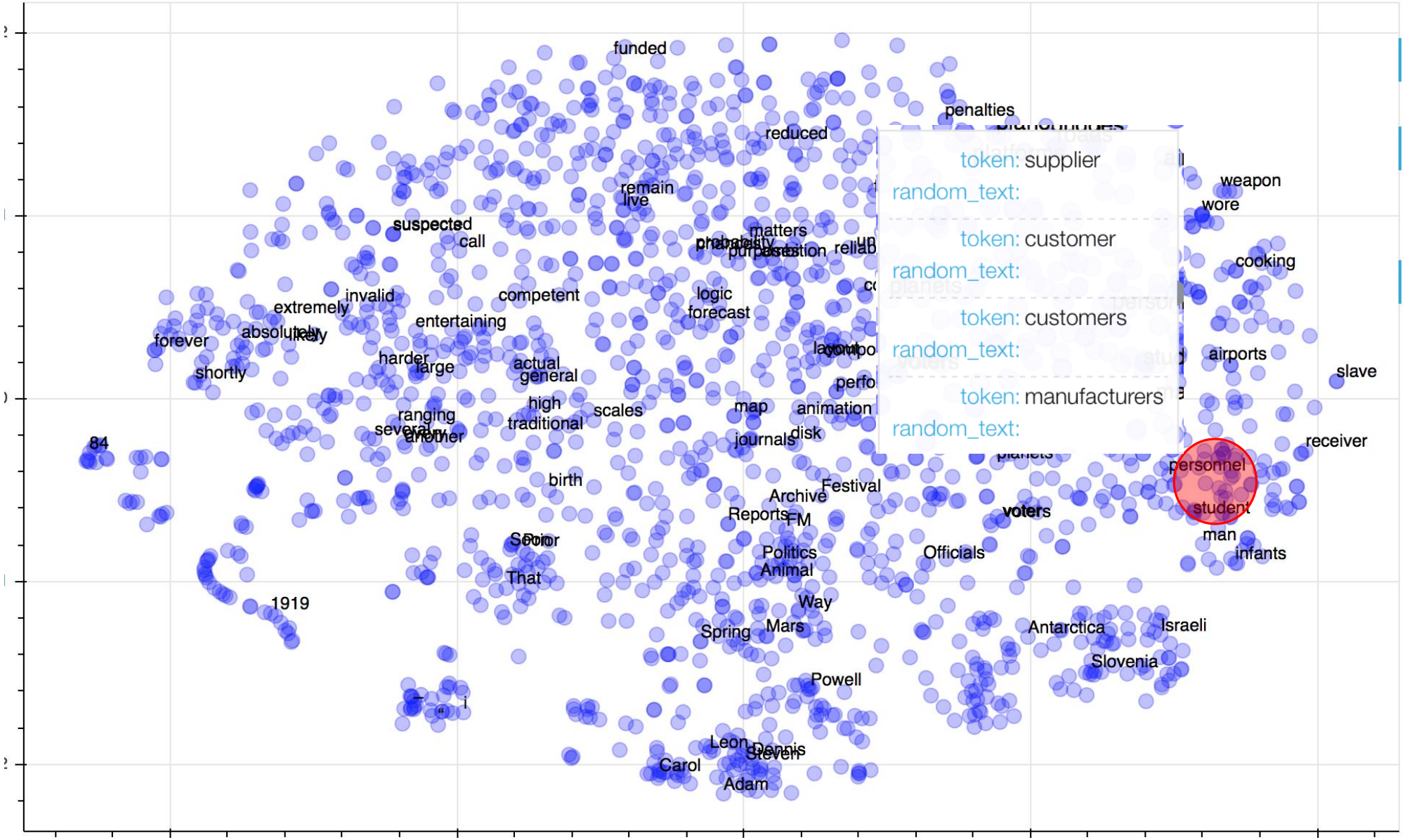

token: definitely
random_text:
token: clearly
random_text:
token: truly
random_text:
funded
penalties
reduced
collision
bubble
violence
exploitation
disease
brain
weapon
wore
remain
live
fear
hatred
suspected
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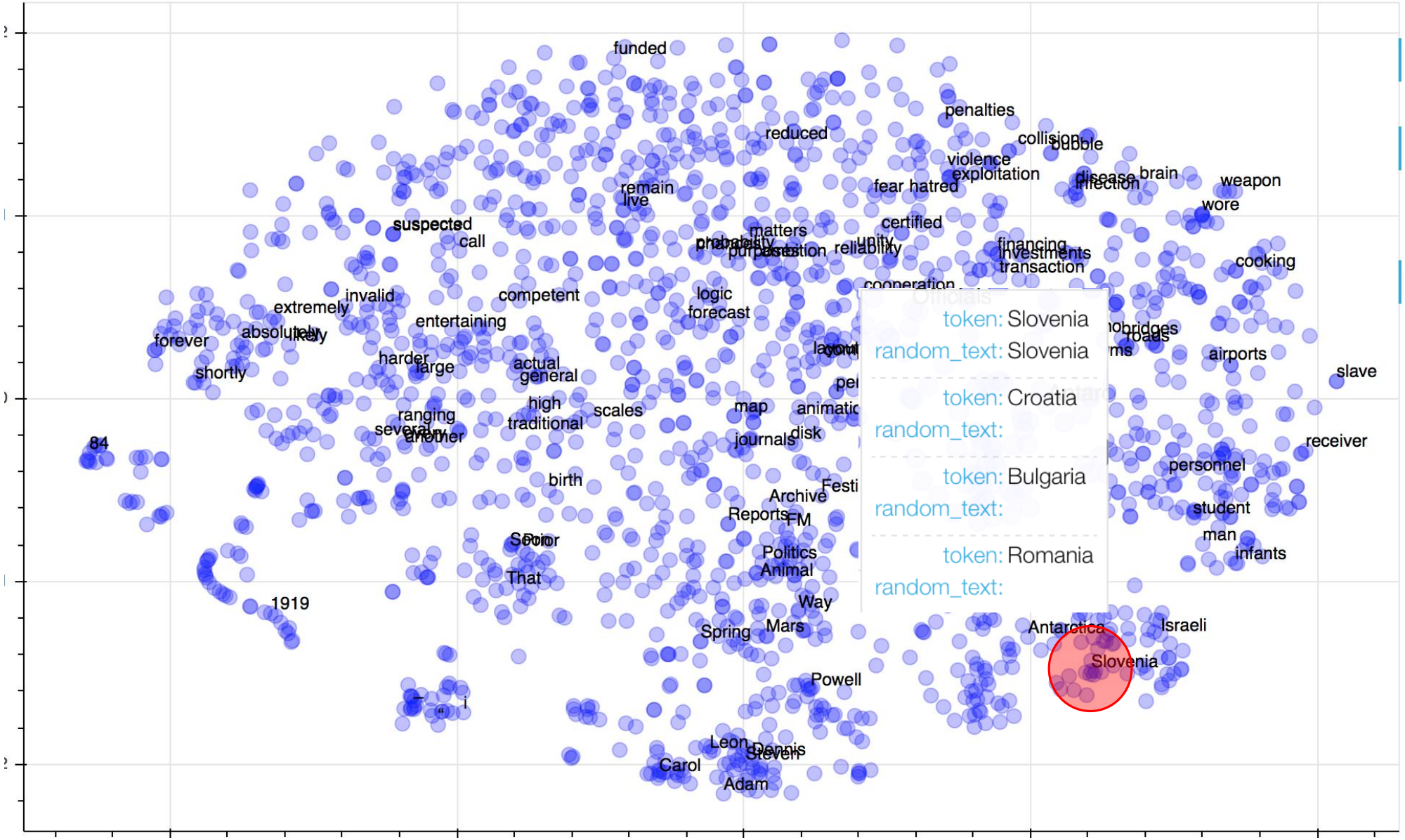

token: realized
random_text:
token: realised
random_text:
token: believing
random_text:
token: realise
random_text:
funded
penalties
reduced
collision
bubble
violence
exploitation
disease
brain
weapon
wore
fear
hatred
remain
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suspected
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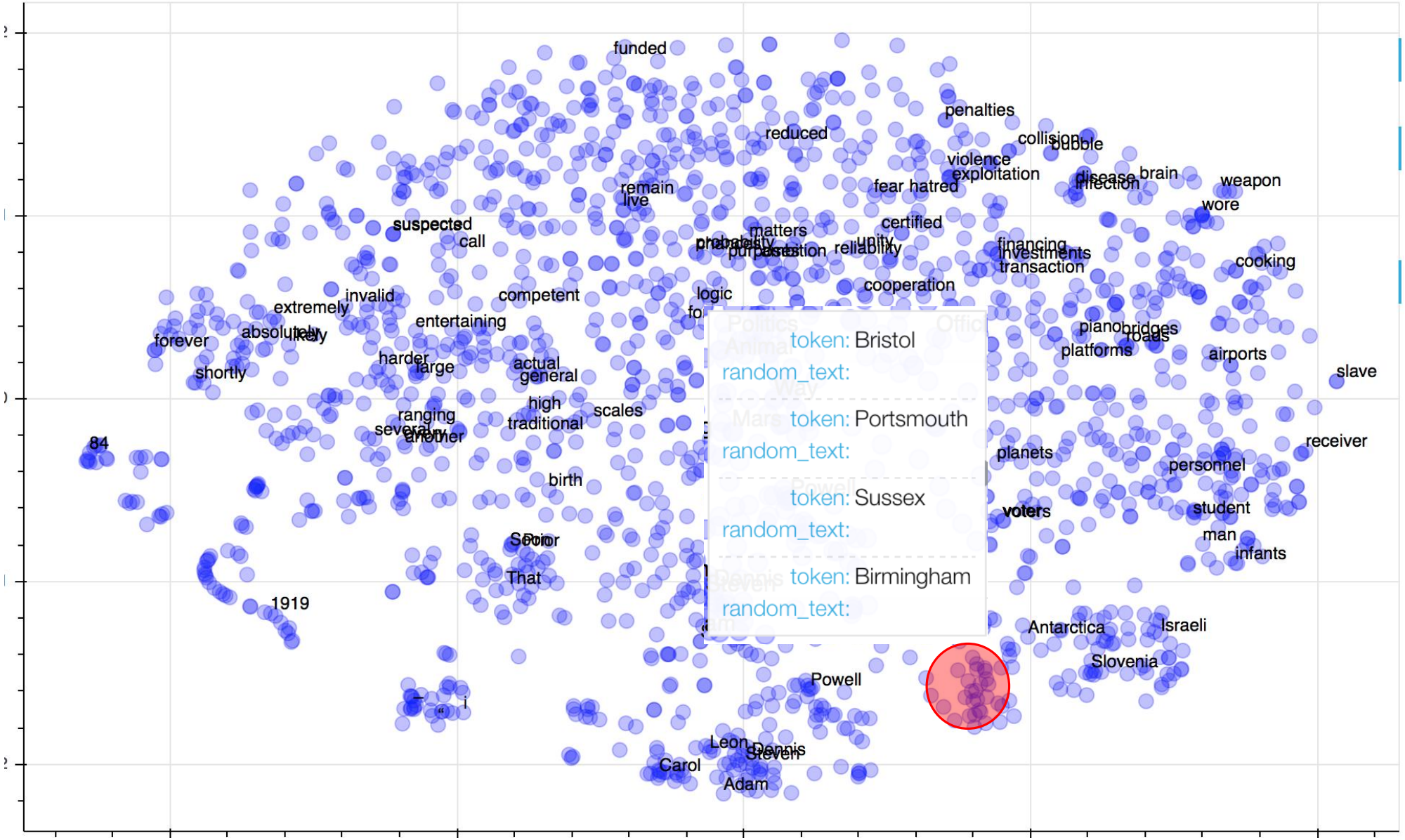

token: survived
random_text:
token: survive
random_text:
token: die
random_text:
token: dead
random_text:
funded
penalties
reduced
collision
bubble
violence
exploitation
disease
brain
weapon
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fear
hatred
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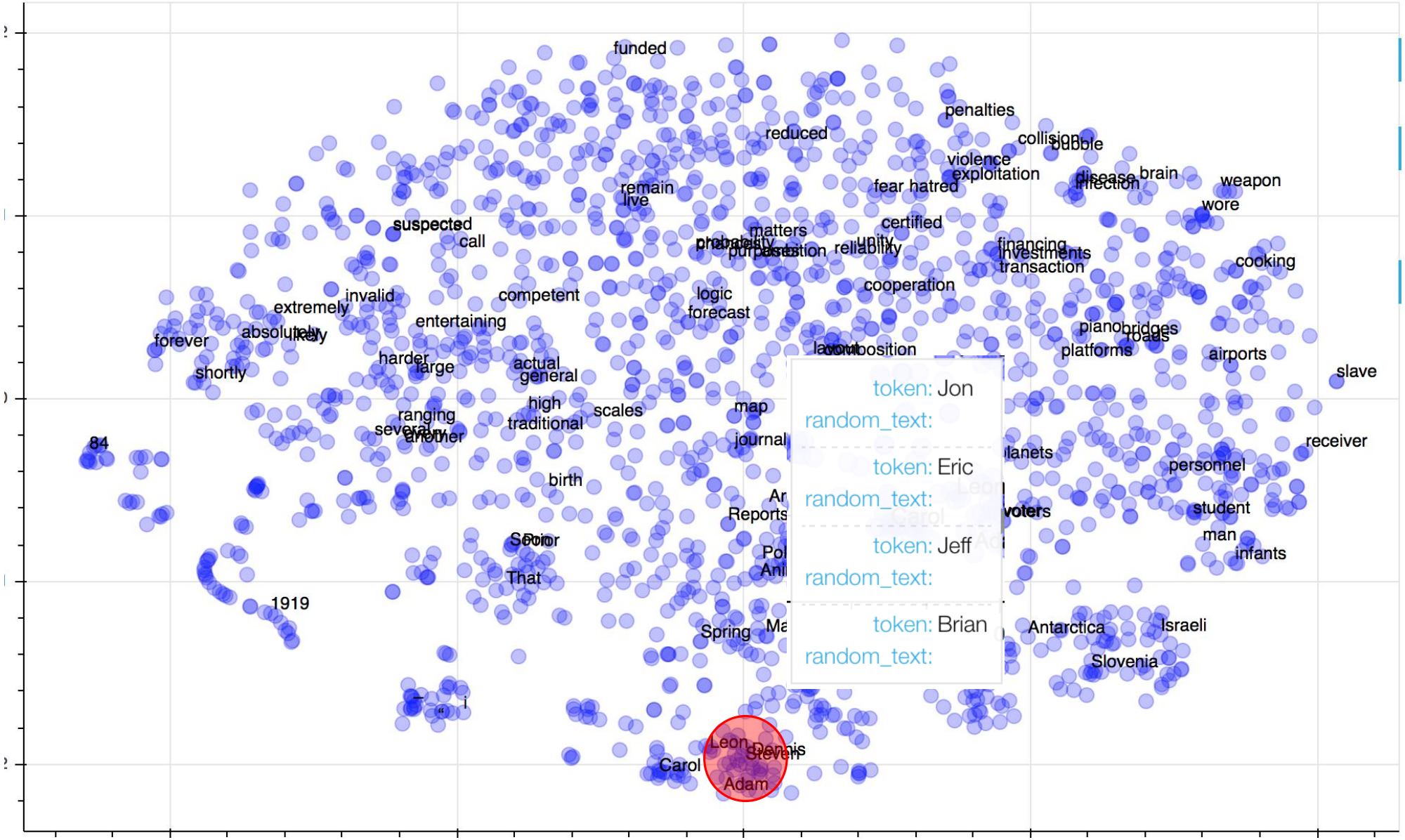

token: passion
random_text:
token: imagination
random_text:
token: ambition
random_text: ambition
penalties
reduced
collision
bubble
violence
exploitation
disease
brain
weapon
wore
fear hatred
infection
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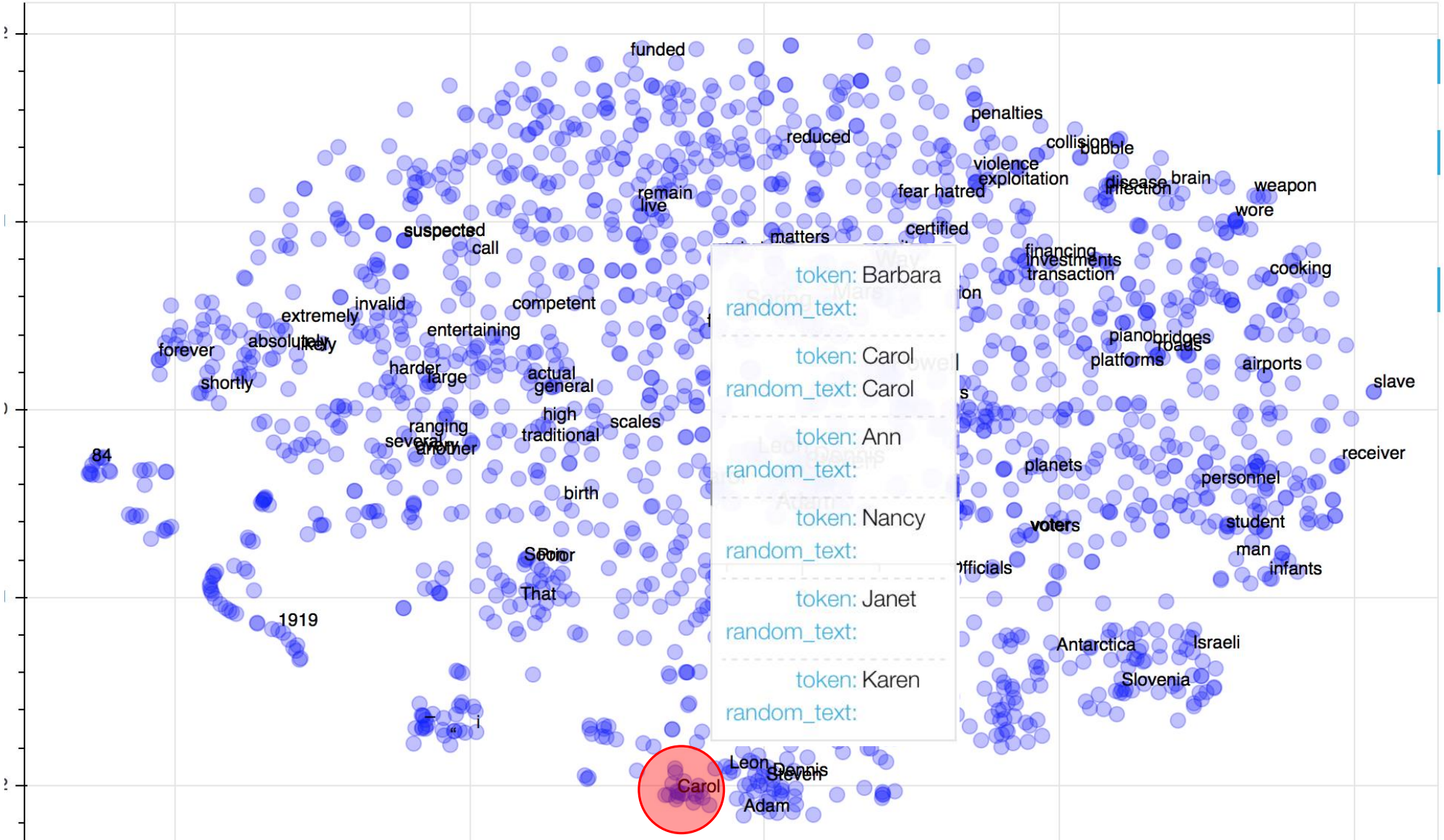

token: fiscal
random_text:
token: monetary
random_text:
token: operational
random_text:
token: educational
random_text:
token: technological
random_text:
funded
penalties
collision
bubble
violence
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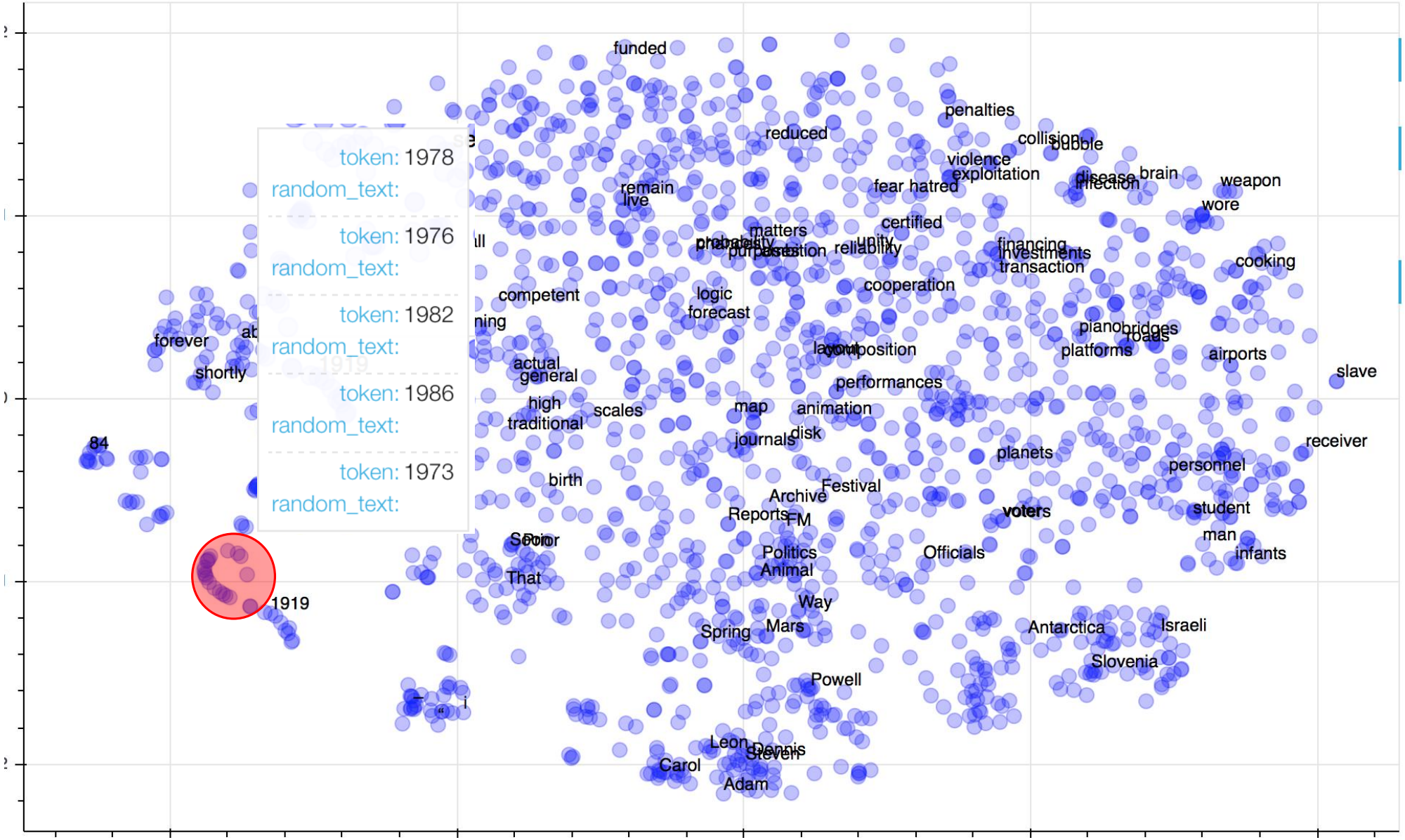

token: supplier
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token: customer
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token: customers
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token: manufacturers
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Dennis
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token: Bulgaria
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token: Romania
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token: Bristol
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token: Sussex
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token: Eric
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token: Brian
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Slovenia
Powell
Leon
Carol
Adam

How to evaluate embeddings

Intrinsic: evaluation on a specific/intermediate subtask

- word analogies: "*a* is to *b* as *c* is to ___?"
- word similarity: correlation of the rankings
- …

Extrinsic: evaluation on a real task

- take some task (MT, NER, coreference resolution, …) or several tasks
- train with different pretrained word embeddings
- if the task quality is better -> win!

What if..

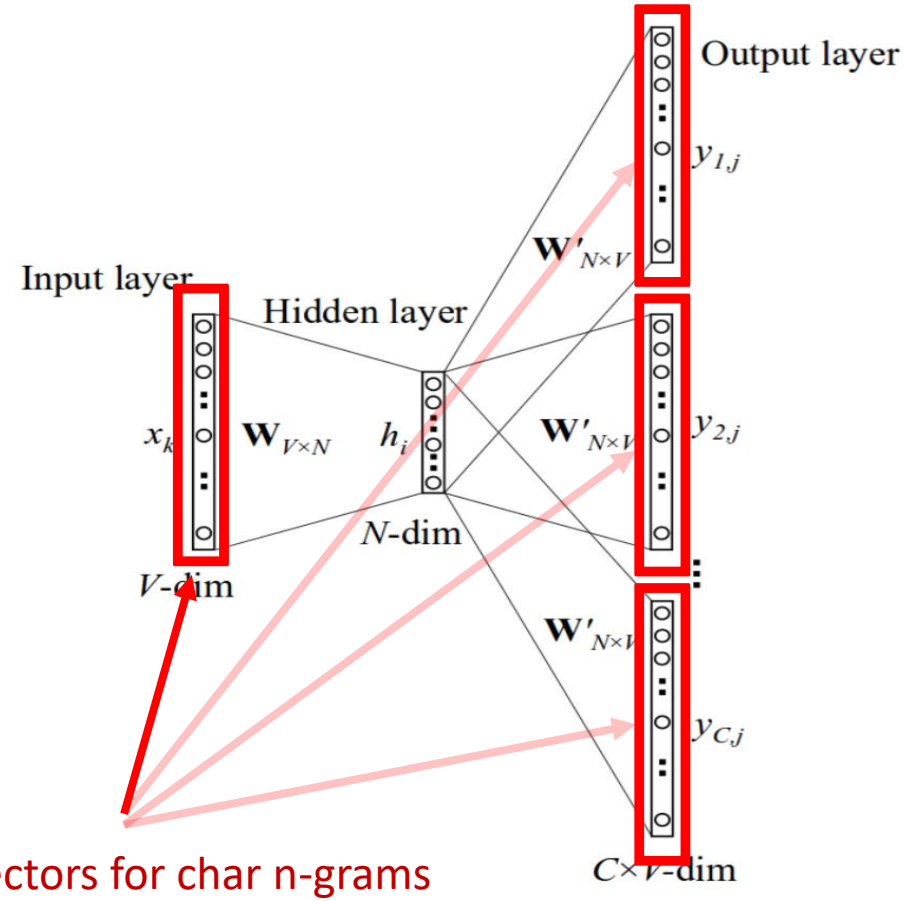
We want to use subword information?

Adding subword information: FastText

Model: SG-NS (skip-gram with negative sampling)

Change the way word vectors are formed:

- each word represented as a bag of character n-gram `<wh, whe, her, ere, re> <where>`
- associate a vector representation to each n-gram
- represent a word by the sum of the vector representations of its n-grams



Sum of vectors for char n-grams

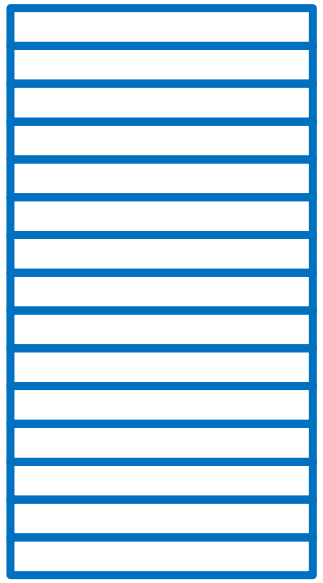
Bojanovsky et al, TACL 2017 http://aclweb.org/anthology/Q17-1010

Word2Vec vs FastText

Word2Vec

where

FastText

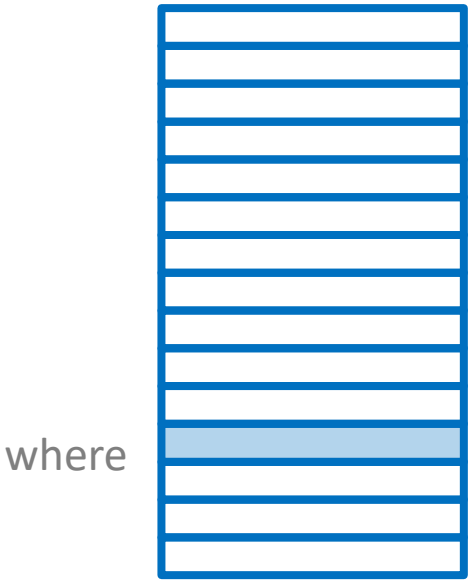


Word2Vec vs FastText

Word2Vec

where

FastText

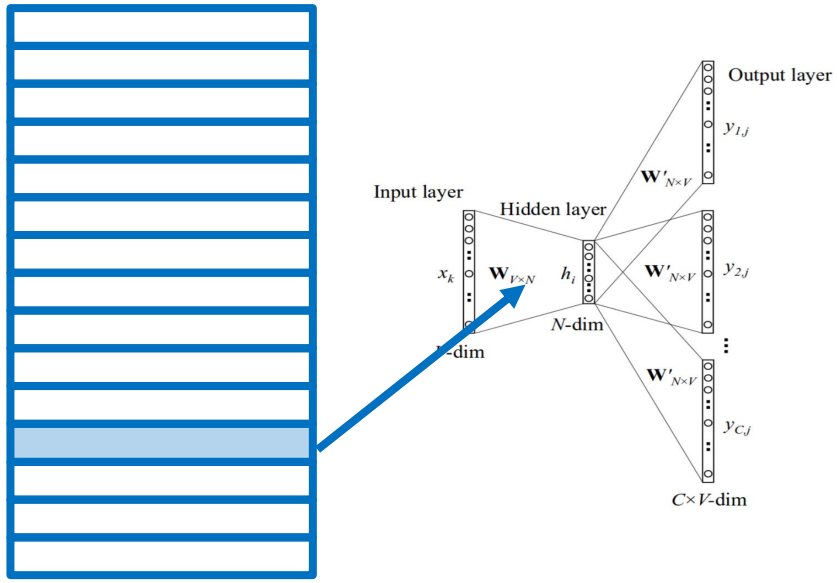


Word2Vec vs FastText

Word2Vec

where

FastText



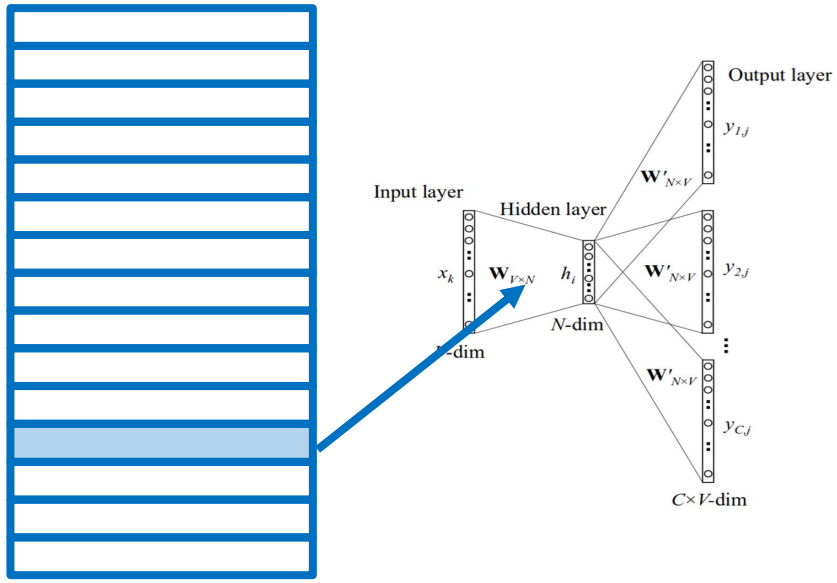
Word2Vec vs FastText

Word2Vec

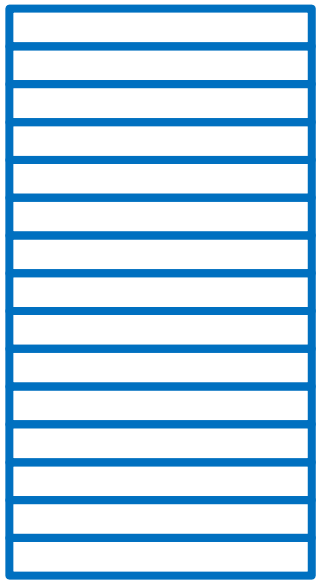
where

FastText

<wh, whe, her, ere, re> <where>



where

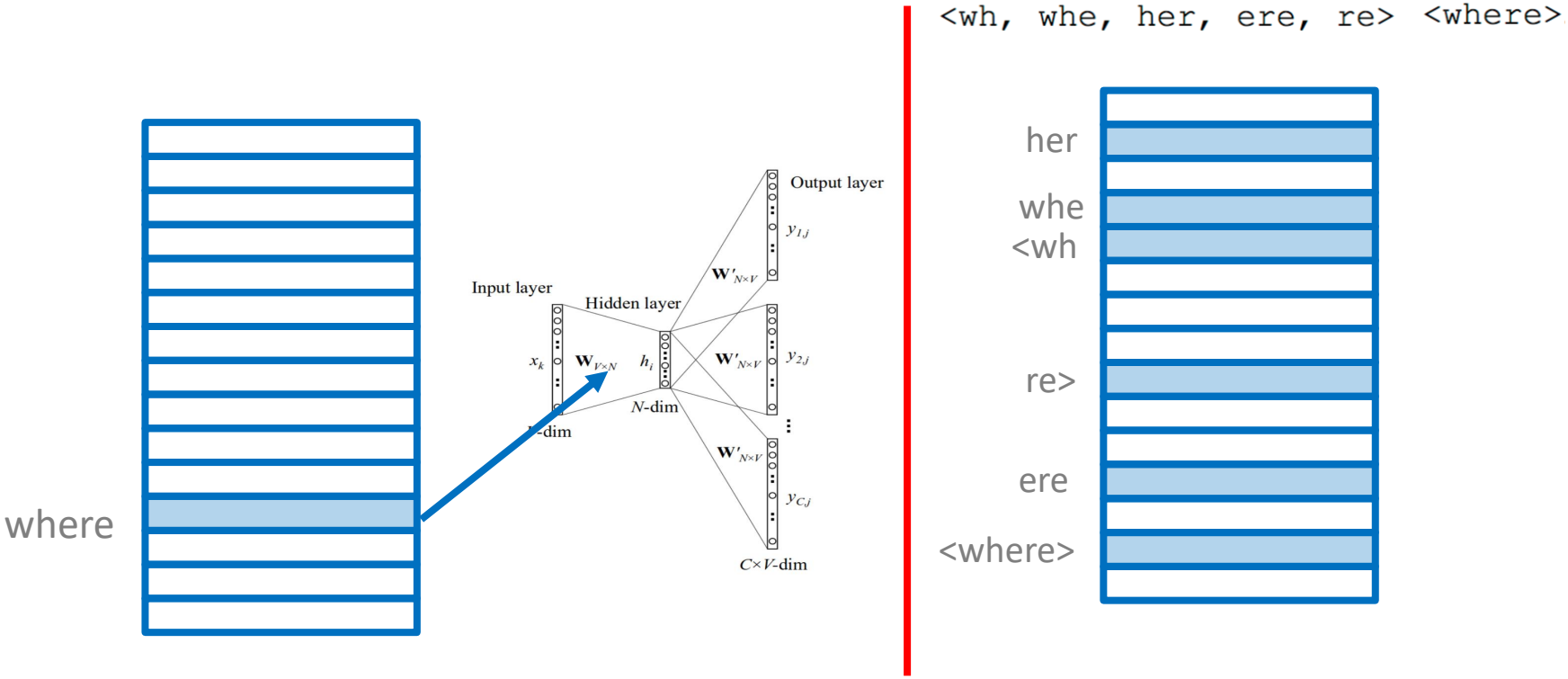


Word2Vec vs FastText

Word2Vec

where

FastText

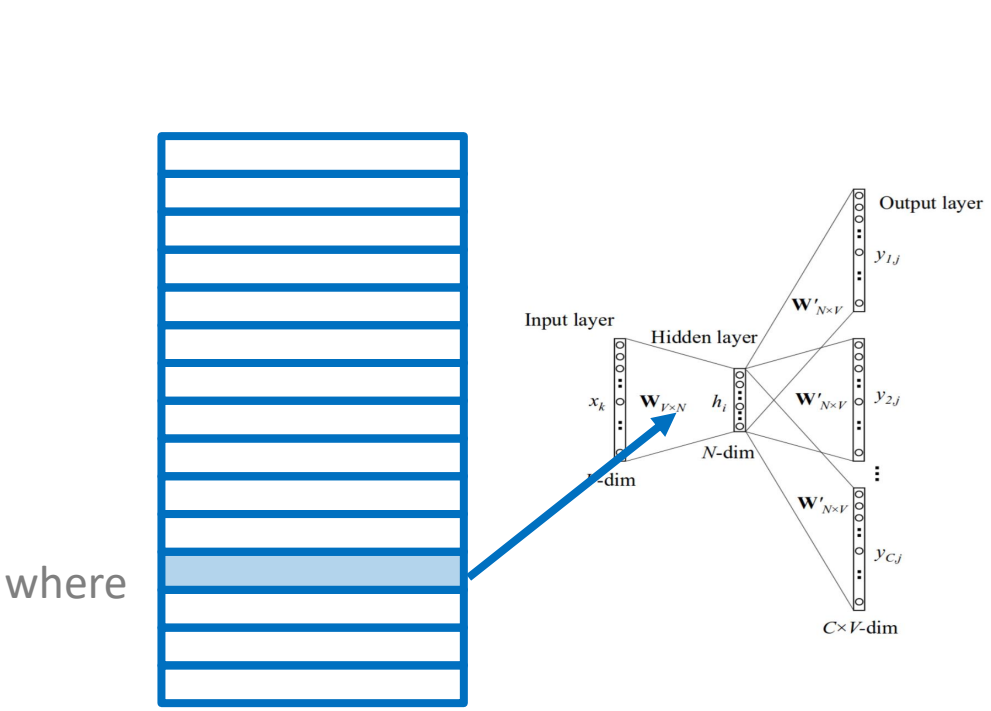


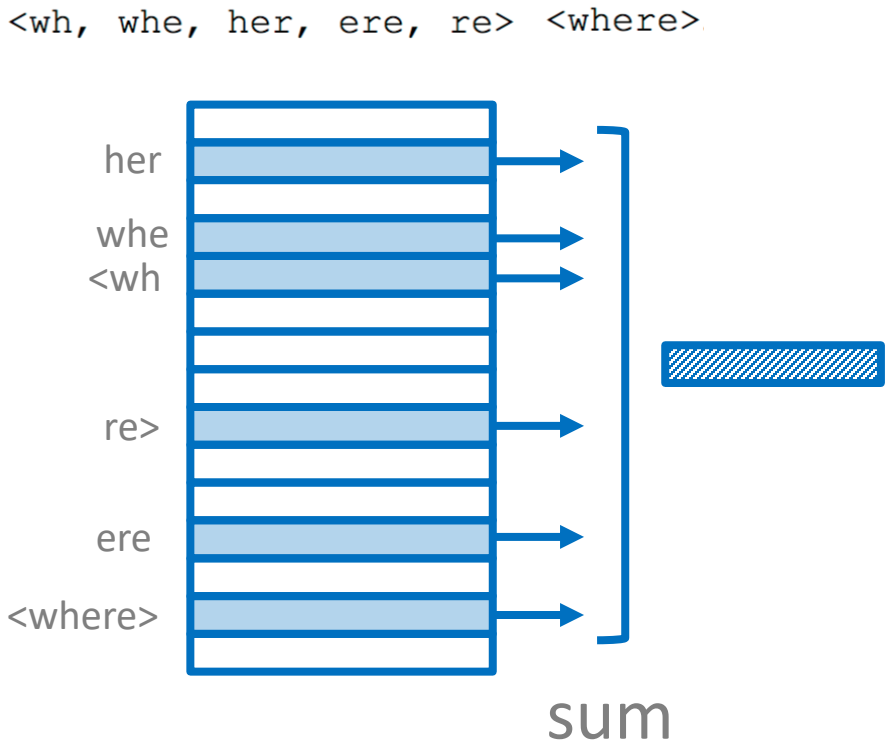
Word2Vec vs FastText

Word2Vec

where

FastText

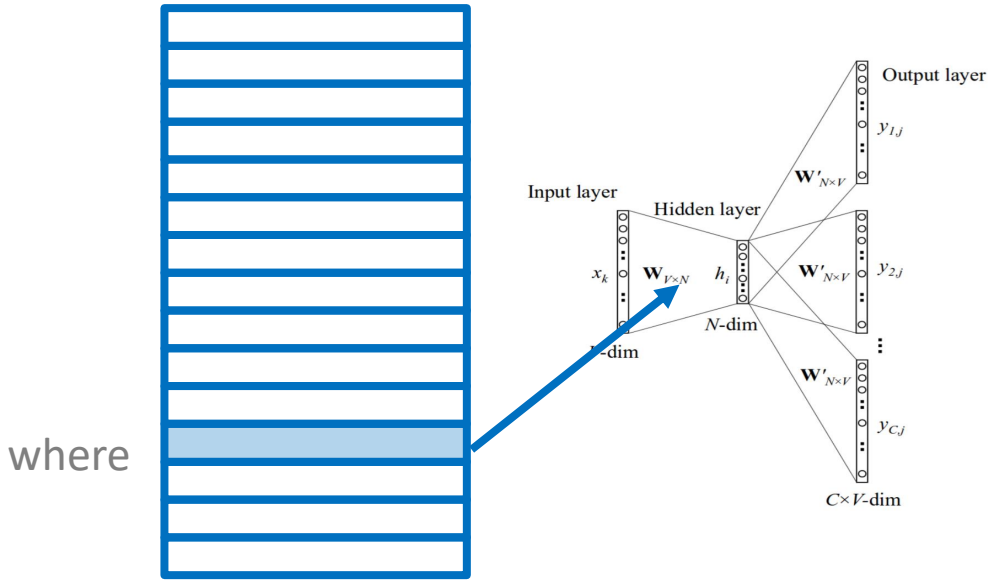




Word2Vec vs FastText

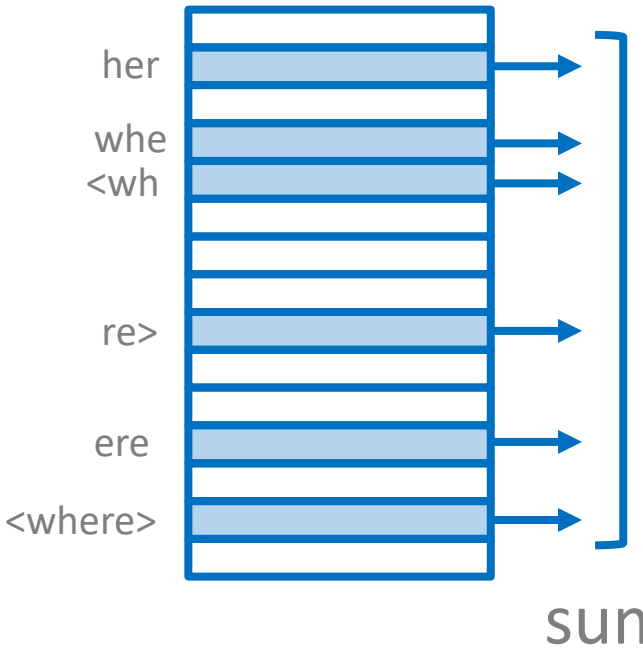
Word2Vec

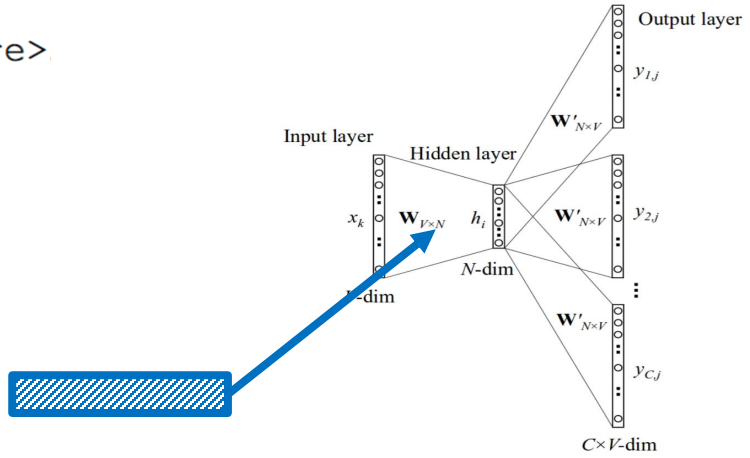
where



FastText

<wh, whe, her, ere, re> <where>





What if..

We abstract the skip-gram model to the sentence level?

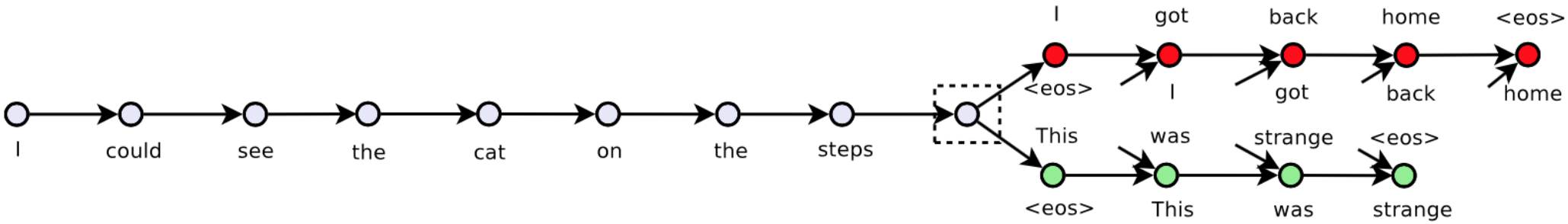
Skip-Thought Vectors

Before:

- use a word to predict its surrounding context

Now:

- encode a sentence to predict the sentences around it



Kiros et al., NIPS 2015 https://papers.nips.cc/paper/5950-skip-thought-vectors.pdf

What if..

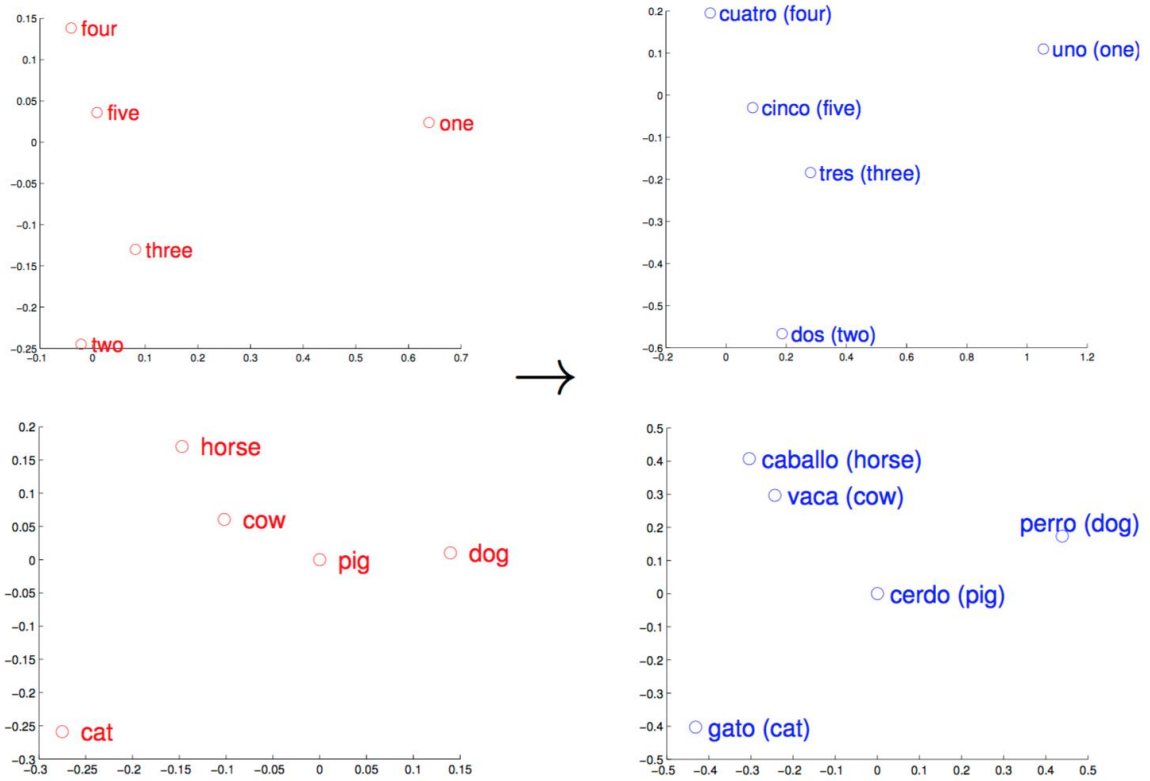
We exploit the structure of semantic space?

Exploiting Similarities among Languages for Machine Translation

- we are given a set of word pairs and their associated vector representations
- find a transformation matrix W

$$\min_W \sum_{i=1}^{n} \|Wx_i - z_i\|^2$$

- for any given new word we can map it to the other language space



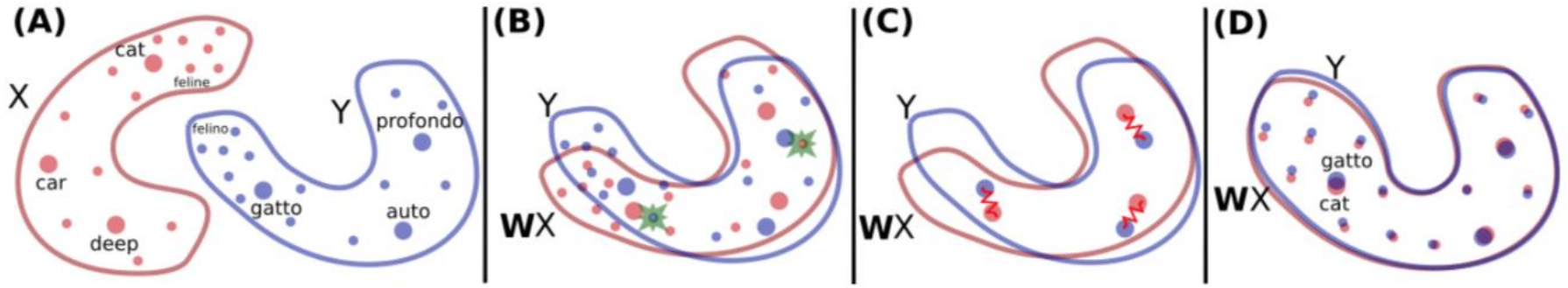
Mikolov et al, 2013, https://arxiv.org/pdf/1309.4168.pdf

Here we supposed that we already know some word pairs.

But what if we know nothing about the languages?

Word Translation Without Parallel Data

Map semantic spaces so that their samples are indistinguishable



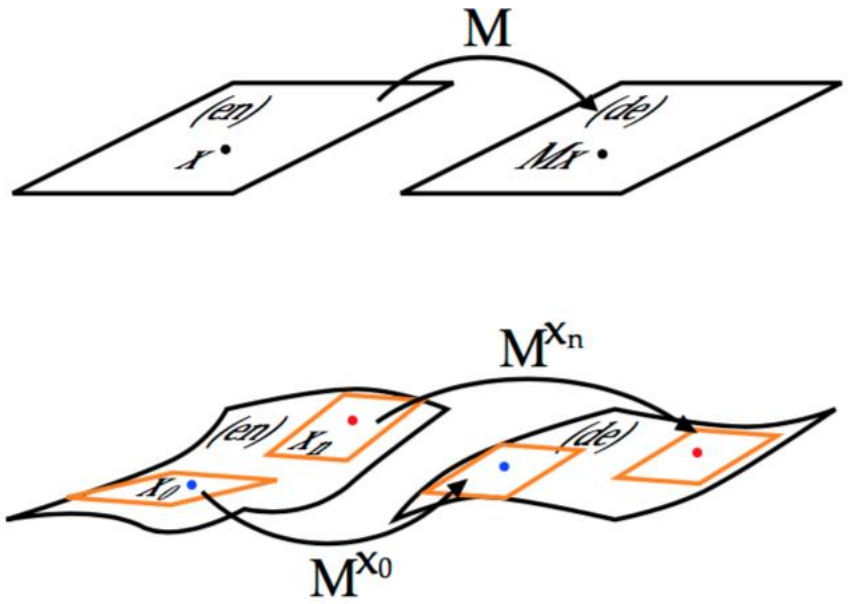
Conneau et al, ICLR 2018, https://arxiv.org/pdf/1710.04087.pdf

Are the underlying maps really linear?

Look at the local linear approximations:

- If they're identical, then the mapping is indeed linear
- If they are not, probably not

  (actually, they're not)

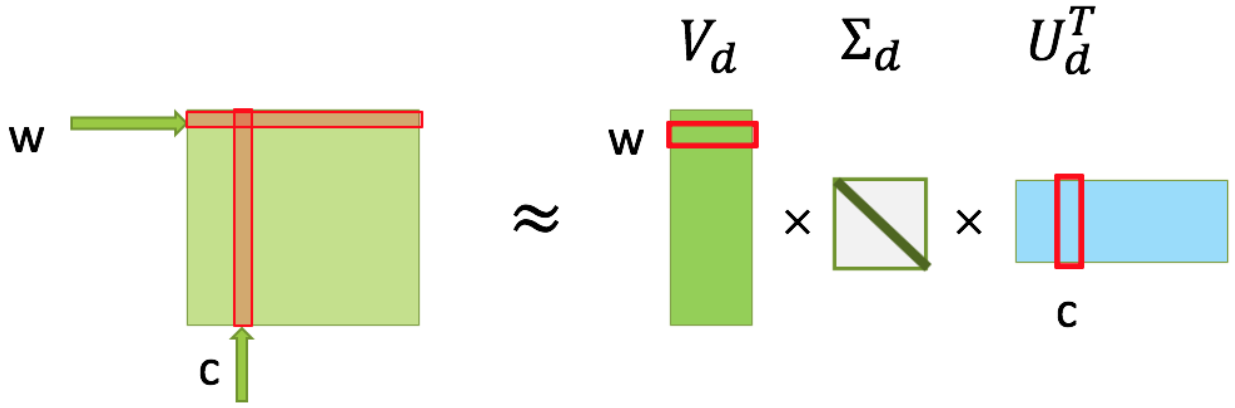


Nakashole & Flauger, ACL 2018, http://aclweb.org/anthology/P18-2036

What happed over the last year

On the Dimensionality of Word Embedding

- Word embeddings are obtained by factorizing PMI matrix
- Dimensionality – how many largest singular values are taken



Yin & Shen, NIPS 2018, https://arxiv.org/pdf/1812.04224.pdf

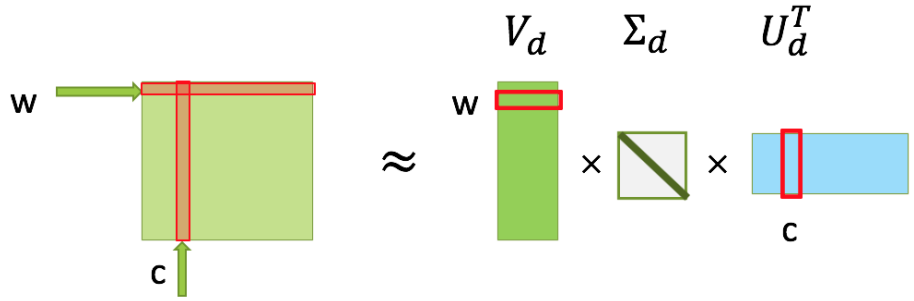
On the Dimensionality of Word Embedding

M– true signal matrix ← you DO NOT have it

$\widetilde{M}$ - estimated signal matrix ← this is what you have

The Bias-Variance Trade-off:

- Small dimension – throw away signal
- Large dimension – take noise



Yin & Shen, NIPS 2018, https://arxiv.org/pdf/1812.04224.pdf

Analogies Explained: Towards Understanding Word Embeddings

"man is to king as woman is to queen"

⇕

man *transforms to* king as woman *transforms to* queen

⇕

{woman, king} *paraphrases* {man, queen}

⇓

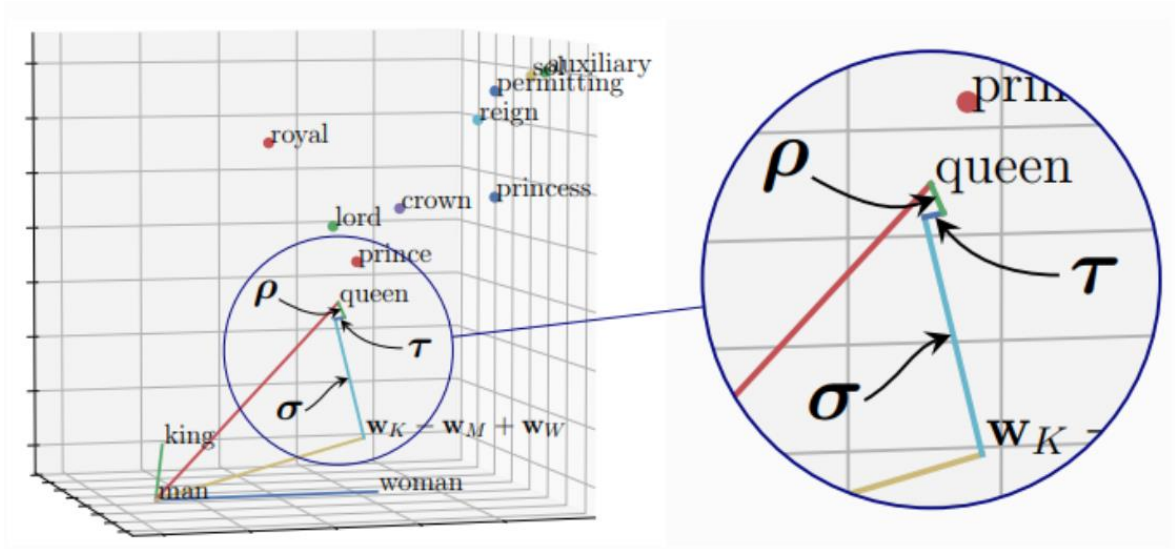
$PMI_{king} - PMI_{man} + PMI_{woman} \approx PMI_{queen}$

⇓

$w_{king} - w_{man} + w_{woman} \approx w_{queen}$

semantic

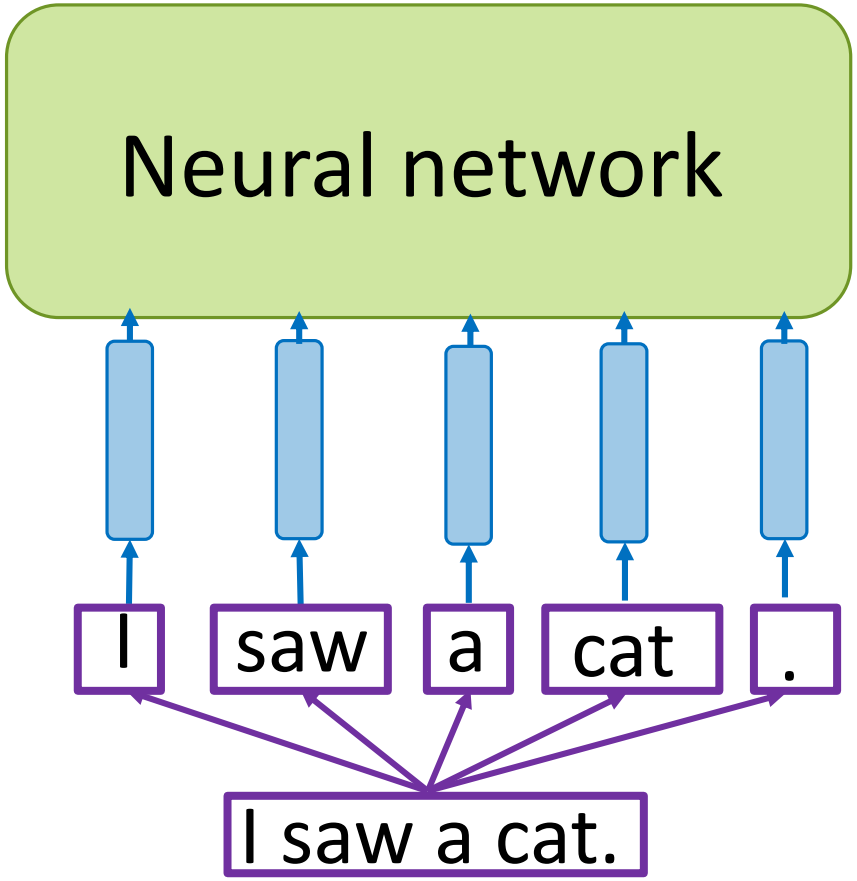
geometric



Allen & Hospedales, ICML 2019, *Best Paper Honourable Mention,* https://arxiv.org/abs/1901.09813
Official blog: https://carl-allen.github.io/nlp/2019/07/01/explaining-analogies-explained.html

A piece of practice:
When we really need to learn word representations?

Why do we need word representation?



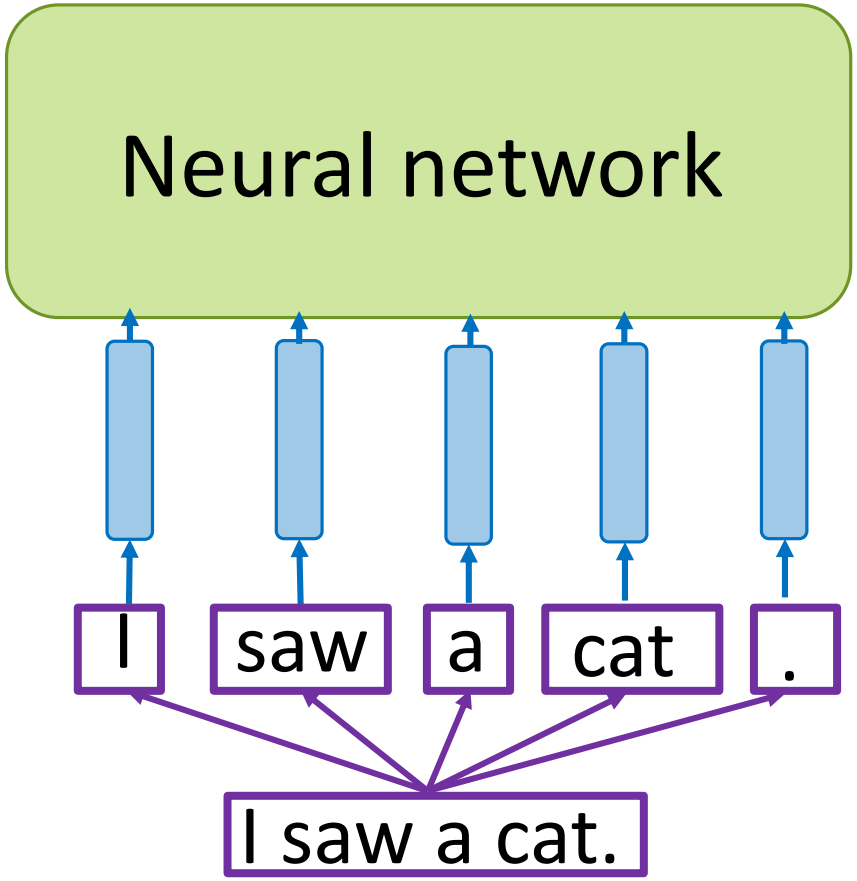
Any NN for solving any task

Word representation - vector (word embedding)

Sequence of tokens

Text

Do we REALLY need to learn word representations in advance?



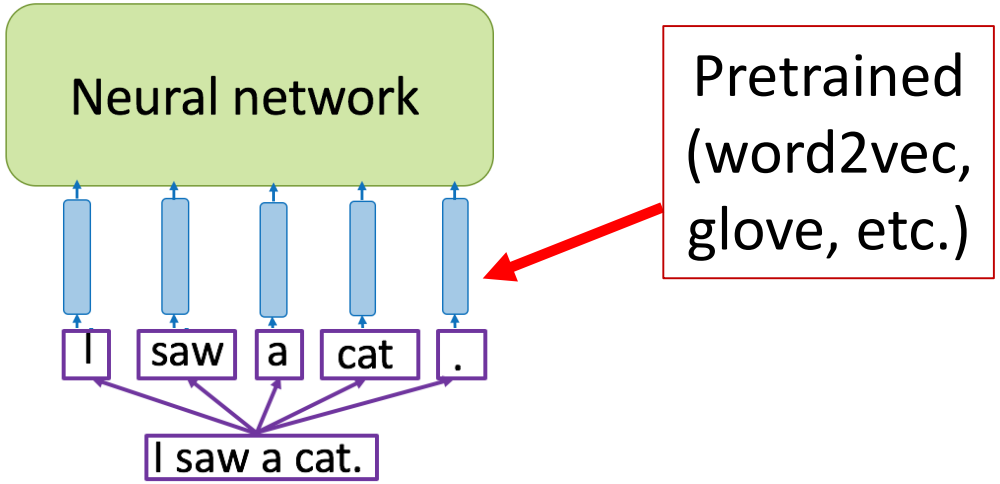
Ok, but if we **already have an NN** for our task, why do we have to learn parameters for word embeddings using **some other NN**?

When to use pretrained embeddings?

Not enough data or the task is too simple



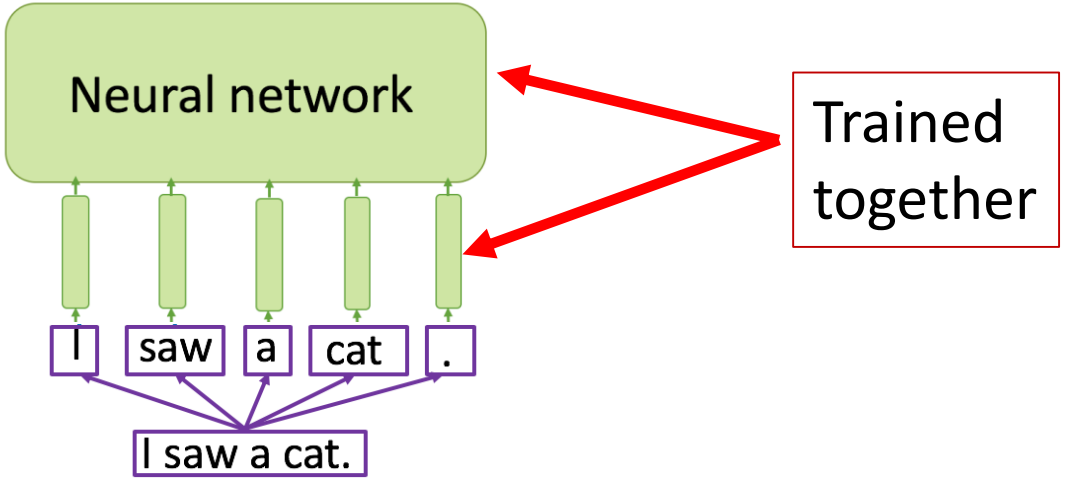
Use pretrained on the other task



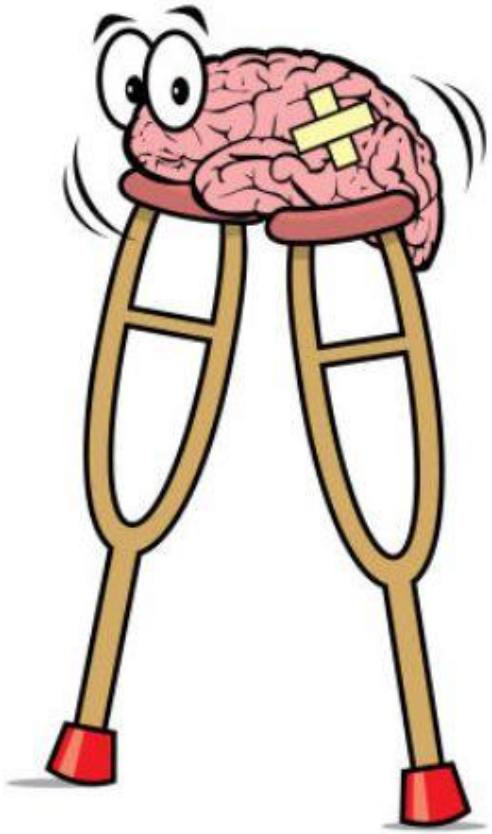
Enough data and a hard task (LM, MT, …)



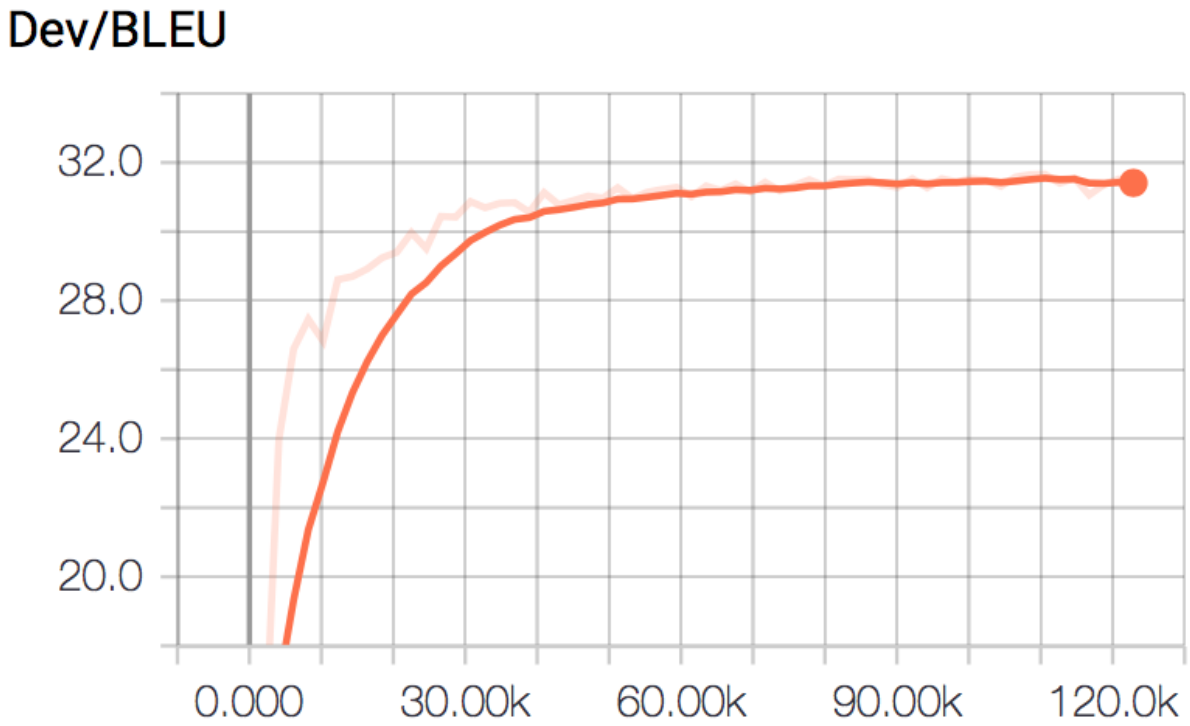
Train with the model

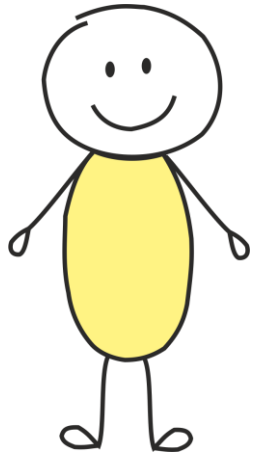


Hack of the day

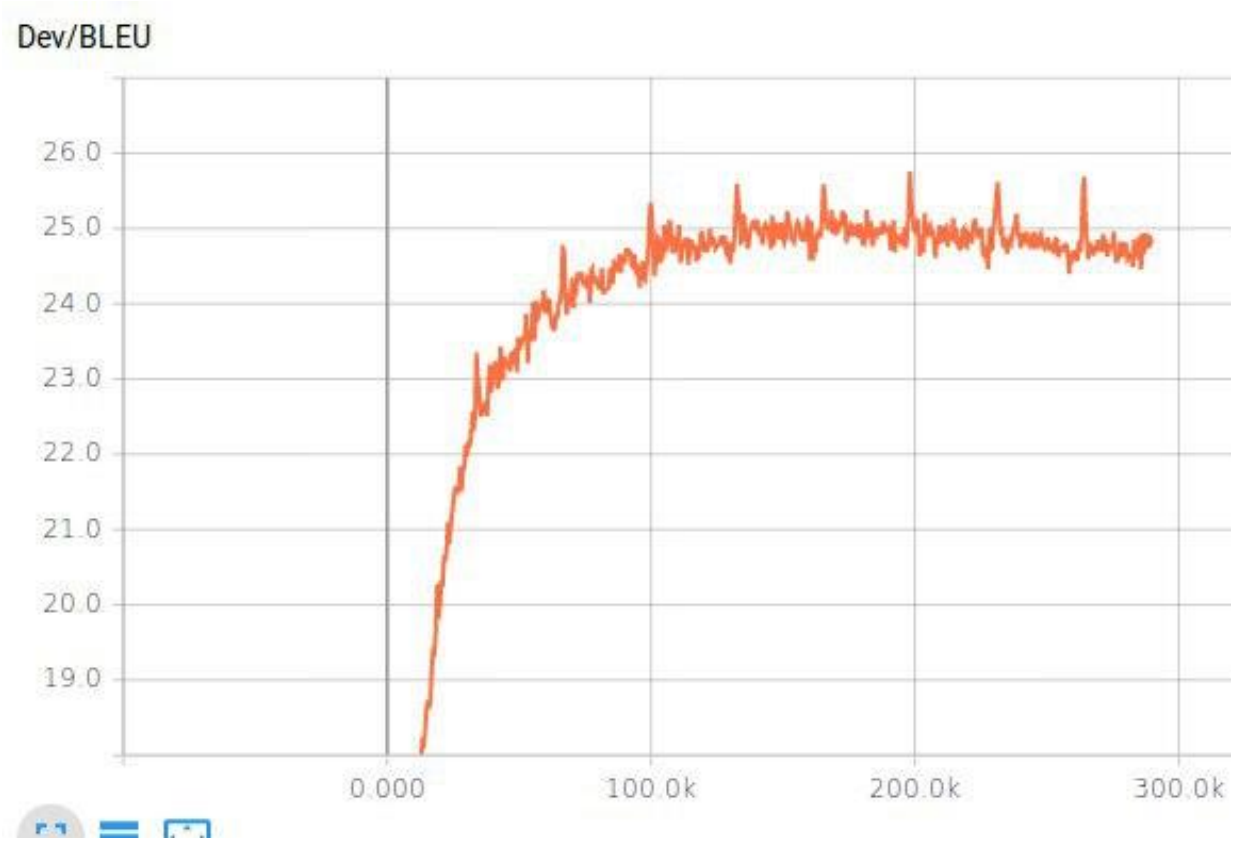


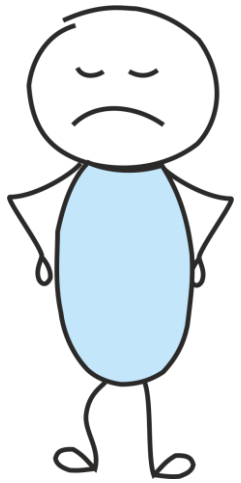
Tensorboard of a healthy man



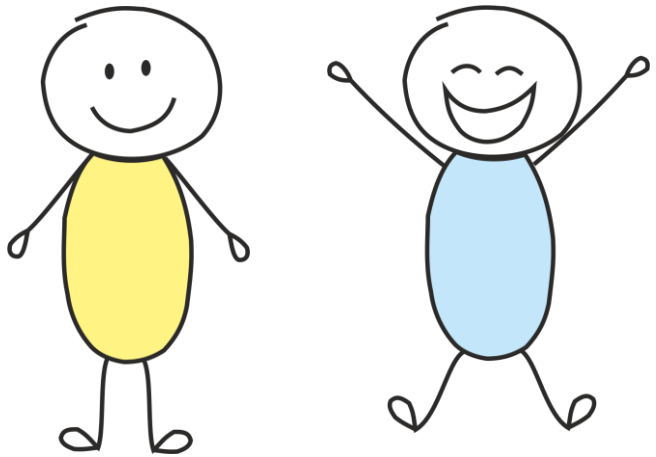


Tensorboard of a man who doesn't shuffle his data





Shuffle your data!



Congratulations, you've just survived the first NLP lecture!

Looking forward to the next week's episode…

Sincerely yours,

Yandex Research